Class 8
Physics

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1. Basic Mathematics for Physics

LEVEL-I

1. **Answer :** C) 90°

Solution : In the sexagesimal system of measurement, a single right angle is uniformly divided into 90 equal parts, which are defined as degrees. Therefore, one right angle is exactly equal to 90°.

2. **Answer :** C) 1 radian

Solution : The radian measure of an angle subtended at the center of a circle is mathematically defined by the formula $\theta = \frac{s}{r}$, where s is the arc length and r is the radius. When the arc length is equal to the radius ($s = r$), the angle is $\frac{r}{r} = 1$ radian.

3. **Answer :** C) $\frac{hypotenuse}{opposite}$

Solution : In a right-angled triangle, the cosecant (csc) of an angle is a trigonometric ratio defined as the length of the hypotenuse divided by the length of the opposite side.

4. **Answer :** B) $-\frac{12}{13}$

Solution : According to the fundamental Pythagorean identity, we know that $\sin^2\theta + \cos^2\theta = 1$. Substituting the given value, we get $\left(\frac{5}{13}\right)^2 + \cos^2\theta = 1$, which simplifies to $\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Taking the square root gives $\cos\theta = \pm\frac{12}{13}$. Because the angle θ lies in Quadrant II where cosine is negative, the correct value is $-\frac{12}{13}$.

5. **Answer :** B) $\frac{16}{25}$

Solution : Using the provided trigonometric identity $\cos^2\theta + \sin^2\theta = 1$, we can substitute the given value $\sin\theta = \frac{3}{5}$ into the equation. This results in $\cos^2\theta + \left(\frac{3}{5}\right)^2 = 1$, yielding $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Note that this exact fraction is missing from the provided options, but it is the mathematically correct derivation.

6. **Answer :** B) $\frac{1}{2}F(2x+3) + C$

Solution : We use the method of integration by substitution. Let $u = 2x + 3$. Differentiating both sides with respect to x gives $du = 2dx$, which means $dx = \frac{du}{2}$. Substituting this into the integral gives $\int f(u)\frac{du}{2} = \frac{1}{2}\int f(u)du$. Since we are given that $\int f(x)dx = F(x) + C$, this evaluates to $\frac{1}{2}F(u) + C$, or $\frac{1}{2}F(2x+3) + C$.

7. **Answer :** A) $x^3 + x^2 + C$

Solution : We apply the power rule for integration, $\int x^n\,dx = \frac{x^{n+1}}{n+1}$, to each term separately. The integral of $3x^2$ is $3\left(\frac{x^3}{3}\right) = x^3$, and the integral of $2x$ is $2\left(\frac{x^2}{2}\right) = x^2$. Adding the constant of integration yields the final result $x^3 + x^2 + C$.

8. **Answer :** A) 1

Solution : Before differentiating, we can simplify the given function algebraically. The function $y = \frac{x^2}{x}$ simplifies to $y = x$ for all $x \neq 0$. The derivative of x with respect to x is logically and mathematically 1.

9. **Answer :** C) $\sin x + x\cos x$

Solution : To differentiate $y = x\sin x$, we must apply the product rule: $\frac{d}{dx}(uv) =$

$u\frac{dv}{dx} + v\frac{du}{dx}$. Let $u = x$ and $v = \sin x$. The derivative is $x\left(\frac{d}{dx}\sin x\right) + \sin x\left(\frac{d}{dx}x\right)$. This simplifies directly to $x\cos x + \sin x\,(1) = \sin x + x\cos x$.

10. **Answer :** B) $15x^2$

Solution : Using the power rule for differentiation, we multiply the coefficient by the exponent and decrease the exponent by one. For $y = 5x^3$, the derivative is $5 \times 3x^{3-1} = 15x^2$.

11. **Answer :** C) 10

Solution : The Law of Sines states the proportional relationship $\frac{a}{\sin A} = \frac{b}{\sin B}$ for any triangle ABC. Substituting the known values, we have $\frac{5}{1/2} = \frac{b}{\sin B}$. Evaluating the left side gives $5 \times 2 = 10$, so the value of $\frac{b}{\sin B}$ is exactly 10.

12. **Answer :** $\frac{2}{3}$ / A) $\frac{2}{7}$ (cos C)

Solution : To find $\cos A$, we use the cosine rule formula: $\cos A = \frac{b^2+c^2-a^2}{2bc}$. Substituting the given side lengths $a = 7$, $b = 8$, and $c = 9$ yields $\cos A = \frac{8^2+9^2-7^2}{2(8)(9)}$. This calculates to $\frac{64+81-49}{144} = \frac{96}{144}$, which mathematically simplifies to $\frac{2}{3}$. In the question it is asking for cos C instead of cos A then the answer would be $\frac{2}{7}$.

13. **Answer :** C) Not defined

Solution : Given that $\tan\theta = 1$, we know the principal angle θ is 45°. The problem asks for the value of $\tan(45° + \theta)$, which substitutes to $\tan(45° + 45°) = \tan(90°)$. The tangent function approaches infinity and is strictly mathematically undefined at 90°.

14. **Answer :** B) $-\frac{1}{2}$

Solution : By applying the standard trigonometric reduction identity $\sin(180° + x) = -\sin x$, the expression $\sin(180° + 30°)$ simplifies to $-\sin(30°)$. Since the standard value of $\sin(30°)$ is $\frac{1}{2}$, the final evaluated answer is $-\frac{1}{2}$.

15. **Answer :** A) 1

Solution : We first simplify the terms using complementary angle identities, where $\sin(90° - \theta) = \cos\theta$ and $\cos(90° - \theta) = \sin\theta$. Substituting these into the original expression yields $\cos\theta \times \cos\theta + \sin\theta \times \sin\theta$, which simplifies to $\cos^2\theta + \sin^2\theta$. According to the Pythagorean trigonometric identity, this sum is always equal to 1.

16. **Answer :** A) 2x

Solution : We are given that $\tan\theta = 1$, which is a constant numeric value. Substituting this into the function gives us $\frac{d}{dx}(1 \cdot x^2)$, or simply $\frac{d}{dx}(x^2)$. Using the basic power rule for differentiation, the result is exactly $2x$.

17. **Answer :** C) Not defined

Solution : We apply the provided compound angle formula for tangent. By substituting $\theta_1 = 45°$ and $\theta_2 = 45°$, the formula yields $\frac{\tan 45° + \tan 45°}{1 - \tan 45° \tan 45°}$. Knowing that $\tan 45° = 1$, the denominator becomes $1 - (1)(1) = 0$. Since division by zero is strictly prohibited in real number arithmetic, the expression is not defined.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** B) $\frac{5}{12}$

Solution : Given $\cos A = \frac{12}{13}$, we know the ratio of the adjacent side to the hypotenuse is $\frac{12}{13}$. Using the Pythagorean theorem, the opposite side is $\sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$. The tangent function is the ratio of opposite to adjacent, which gives $\tan A = \frac{5}{12}$.

2. **Answer :** A) $\frac{4}{5}$

Solution : The given equation $3\tan A = 4$ can be rearranged algebraically to $\tan A = \frac{4}{3}$, representing the ratio of the opposite side to the adjacent side. By applying the Pythagorean theorem, the hypotenuse is calculated as $\sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. The sine ratio is opposite over hypotenuse, resulting in $\sin A = \frac{4}{5}$.

3. **Answer :** A) 1

Solution : The given trigonometric expression $\sin A \cos C + \cos A \sin C$ is the standard expansion formula for $\sin(A + C)$. In right-angled triangle ABC with the right angle at B, the sum of the remaining two acute angles A and C must be 90°. Therefore, the expression simplifies to sin(90°), which exactly equals 1.

4. **Answer :** A) 0

Solution : The function $y = \sin(2\pi)$ represents the sine of a specific constant angle. The numerical value of $\sin(2\pi)$ is 0. Since the derivative of any mathematical constant with respect to a variable is identically zero, $\frac{dy}{dx} = 0$.

5. **Answer :** C) $6xe^{3x^2}$

Solution : To find the derivative of an exponential function with a complex exponent, we must apply the chain rule formula $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$. Here, the inner function is $u = 3x^2$, and its derivative is $\frac{du}{dx} = 6x$. Multiplying the outer derivative by the inner derivative yields $e^{3x^2} \times 6x = 6xe^{3x^2}$.

6. **Answer :** C) $-6x$

Solution : We start by finding the first derivative of the function $y = -x^3$, which by the power rule is $\frac{dy}{dx} = -3x^2$. To find the second derivative, we differentiate the first derivative once more with respect to x. This yields $\frac{d^2y}{dx^2} = -3(2x) = -6x$.

7. **Answer :** B) $\frac{x|x|}{2} + c$

Solution : The integral of the absolute value function $|x|$ requires a generalized formula that correctly handles both positive and negative domains of x. The expression $\frac{x|x|}{2} + c$ perfectly satisfies this condition, yielding $\frac{x^2}{2}$ when $x > 0$ and $-\frac{x^2}{2}$ when $x < 0$, which aligns with the piecewise integration of $|x|$.

8. **Answer :** B) $\tan x - x + c$

Solution : Direct integration of $\tan^2 x$ is not standard, so we use the Pythagorean trigonometric identity $\tan^2 x = \sec^2 x - 1$ to rewrite the integrand. The integral becomes $\int (\sec^2 x - 1)\, dx$. Integrating term by term, the antiderivative of $\sec^2 x$ is $\tan x$, and the antiderivative of 1 is x, giving a final answer of $\tan x - x + c$.

9. **Answer :** C) $\tan x - \sec x + C$

Solution : To evaluate the integral $\int \frac{1}{1+\sin x} dx$, we multiply the numerator and the denominator by the conjugate $(1 - \sin x)$. This changes the integrand to $\frac{1-\sin x}{1-\sin^2 x}$, which simplifies to $\frac{1-\sin x}{\cos^2 x}$ using trigonometric identities. Splitting the fraction gives $\int (\sec^2 x - \sec x \tan x)\, dx$. Integrating these standard forms results directly in $\tan x - \sec x + C$.

10. **Answer :** C) $x^4 + c$

Solution : We evaluate the integral $\int 4x^3 dx$ by directly applying the power rule for integration. We increase the exponent by one and divide by the new exponent, which mathematically gives $4\left(\frac{x^4}{4}\right)$. The fours cancel out, leaving $x^4 + c$.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. **Answer :** B) $\cot 90° = 0$, C) $\sec 45° = \sqrt{2}$

Solution : Evaluating each trigonometric option manually: $\tan 90°$ is mathematically undefined (making A incorrect), $\cot 90° = \frac{\cos 90°}{\sin 90°} = \frac{0}{1} = 0$ (making B correct) , $\sec 45° = \frac{1}{\cos 45°} = \frac{1}{1/\sqrt{2}} = \sqrt{2}$ (making C correct), and finally $\sin 30° = \frac{1}{2}$ (making D incorrect).

12. **Answer :** D) $A = 48°$

Solution : We are given the equation $\cos A = \sin(A - 6°)$. By using the complementary trigonometric identity $\cos A = \sin(90° - A)$, we can equate the angles: $90° - A = A - 6°$. Solving for A algebraically, we get $2A = 96°$, which strictly implies $A = 48°$.

13. **Answer :** B) $x + C$

Solution : The integrand contains the term $e^{\log x}$. By the fundamental property of logarithms and exponentials (assuming base e), $e^{\log x}$ simplifies perfectly to just x. The integral then becomes $\int \frac{x}{x} dx$, which simplifies further to $\int 1\, dx$. The standard antiderivative of 1 with respect to x is $x + C$.

14. **Answer :** B) 0

Solution : The mathematical expression 4π consists entirely of constants with no variables present. In calculus, the derivative of any fixed constant value with respect to a variable is fundamentally 0.

ASSERTION AND REASON TYPE

15. **Answer :** A) Both A and R are true, and R is the correct explanation of A **Solution :** The assertion correctly identifies the sexagesimal system as measuring angles in degrees, minutes, and seconds. The reason is also factually correct and acts as the precise definitional framework for the assertion, correctly stating that one degree is divided into 60 minutes, and one minute into 60 seconds.

16. **Answer :** A) Both A and R are true, and R is the correct explanation of A **Solution :** The assertion accurately states that the second derivative of displacement yields acceleration. The reason accurately notes that velocity is the first derivative (rate of change) of displacement. Since acceleration is the rate of change of velocity, and velocity is the first derivative of displacement, acceleration becomes the second derivative of displacement velocity.

17. **Answer :** A) Both A and R are true, and R is the correct explanation of A **Solution :** The assertion that the derivative of a constant is zero is a fundamental rule in calculus. The reason flawlessly defines the conceptual basis for this rule: because a

derivative measures the rate of change, and a constant value by definition never changes regardless of the independent variable, its rate of change must physically and mathematically be zero.

18. **Answer :** C) A is true, but R is false

Solution : The assertion is mathematically correct; the instantaneous rate of change at a specific point on a curve is exactly the slope of the tangent line at that specific point. However, the reason is fundamentally false because the average rate of change is conceptually represented by the slope of a secant line connecting two points, not a tangent line.

MATRIX MATCHING TYPE

19. **Answer :** a → R($4x^{-3}+c$), b → P($\frac{x^{n+1}}{n+1}+c$), c → S($2\log x-\frac{3}{x}-\frac{x^{-4}}{4}+c$)(there is error in option), d → Q ($\frac{x^3}{3}+c$)

Solution : For part a, applying the power rule to integrate $-12x^{-4}$ yields $-12\left(\frac{x^{-3}}{-3}\right)+c=4x^{-3}+c$, firmly matching R. For part b, the standard formula for integrating x^n is $\frac{x^{n+1}}{n+1}+c$, matching P. For part c, integrating term by term results in $2\log x-\frac{3}{x}-\frac{x^{-4}}{4}+c$, which conceptually corresponds to S For part d, integrating x^2 directly yields $\frac{x^3}{3}+c$, matching Q.

20. **Answer :** a → S, b → P, c → Q, d → T

Solution : For part a, differentiating a linear term kx leaves only the constant coefficient k, matching S. For part b, rewriting $\frac{1}{x}$ as x^{-1} and applying the power rule yields $-1x^{-2}$ or $\frac{-1}{x^2}$, matching P. For part c, the fundamental derivative of the tangent function $\tan x$ is standardly known to be $\sec^2 x$, matching Q. For part d, the fundamental derivative of the natural logarithm $\log_e x$ is definitively $\frac{1}{x}$, matching T.

COMPREHENSION TYPE

Passage-I

21. **Answer :** A) $\frac{41}{29}$

Solution : Using the provided right-angled triangle dimensions where hypotenuse $AC=29$, adjacent $AB=20$, and opposite $BC=21$ relative to angle A, we define the basic ratios. Here, $\sin A=\frac{21}{29}$ and $\cos A=\frac{20}{29}$. Summing these two fractions yields $\frac{21+20}{29}=\frac{41}{29}$.

22. **Answer :** B) $\frac{\cos A}{\sin A}$

Solution : According to fundamental trigonometric identities, the cotangent function (cot) is mathematically defined as the ratio of the cosine of an angle to the sine of that same angle.

23. **Answer :** D) $\frac{20}{29}\times\frac{20}{29}$

Solution : We start by algebraically simplifying the given mathematical expression $\frac{\sin A\times\cos A}{\tan A}$. By replacing $\tan A$ with its equivalent ratio $\frac{\sin A}{\cos A}$, the expression structurally becomes $\frac{\sin A\times\cos A}{\sin A/\cos A}$. The sine terms cancel out, leaving $\cos^2 A$. From earlier calculations, $\cos A=\frac{20}{29}$, so squaring it results exactly in $\frac{20}{29}\times\frac{20}{29}$, or $\frac{400}{841}$.

Passage-II

24. **Answer :** A) $\frac{1}{2}t^2$

Solution : The displacement function is given as $x = \sin(90° - \theta)\, t^2$. Substituting the explicit condition $\theta = 60°$ into the formula produces $x = \sin(90° - 60°)\, t^2 = \sin(30°)\, t^2$. Because the standard value of $\sin(30°)$ is $\frac{1}{2}$, the final displacement equation correctly simplifies to $\frac{1}{2}t^2$.

25. **Answer :** A) $2t\cos\theta$

Solution : First, simplify the given displacement equation using a complementary identity: $x = \sin(90° - \theta)\, t^2 = \cos\theta \cdot t^2$. To locate the velocity, we mathematically differentiate displacement with respect to time t. Treating $\cos\theta$ as a constant multiplier, $\frac{dx}{dt} = \cos\theta\,(2t) = 2t\cos\theta$.

26. **Answer :** A) $2\cos\theta$

Solution : Acceleration is defined as the first derivative of velocity with respect to time. Using the velocity expression from the preceding $v = (2\cos\theta)t$, we differentiate this linearly with respect to t. The variable t drops out, yielding the constant acceleration mathematically equal to $2\cos\theta$.

27. **Answer :** 1

Solution : This question relies entirely on the primary Pythagorean identity in fundamental trigonometry. For absolutely any real angle θ, the sum of the square of its sine and the square of its cosine is eternally constrained to logically equal 1.

28. **Answer :** 0

Solution : The mathematical operation requests the derivative of 21193 strictly with respect to x. Because 21193 is a strict numerical constant devoid of any variables, its rate of change is inherently non-existent, making its derivative 0.

29. **Answer :** 6

Solution : To evaluate this definite integral, we first meticulously find the general antiderivative of the polynomial polynomial: $\int (4x^3 - 3x^2 + 2x + 5)\, dx = x^4 - x^3 + x^2 + 5x$. We then evaluate this expression at the upper boundary limit (1) and strictly subtract the evaluation at the lower boundary limit (0). This results in $\big(1^4 - 1^3 + 1^2 + 5(1)\big) - (0) = 1 - 1 + 1 + 5 = 6$.

2. Units And Dimensions

Units and Measurements

LEVEL-I

1. **Answer :** C) 1000
The metric prefix "kilo" represents a factor of 1000. Therefore, 1 kilometre is equal to 1000 meters.

2. **Answer :** B) Ampere
According to the International System of Units (SI), the fundamental unit for measuring electric current is the Ampere.

3. **Answer :** B) 10^{-6}
A micron, also known as a micrometer, is a submultiple of the meter. 1 micron is equal to one-millionth of a meter, which is mathematically represented as 10^{-6} m.

4. **Answer :** D) Astronomical unit (A.U)
By standard astronomical definition, the mean distance from the center of the Earth to the center of the Sun is called one Astronomical Unit (A.U).

5. **Answer :** A) Light year
A light year is a unit of astronomical distance defined as the total distance that light travels in a vacuum over the duration of one Earth year.

6. **Answer :** B) 1.5×10^{11}
One Astronomical Unit (A.U) is the average distance between the Earth and the Sun. In metric terms, it is approximately equal to 150 million kilometers, which is 1.5×10^{11} meters.

7. **Answer :** A) 9.5×10^{15}
The distance light travels in one year is calculated by multiplying the speed of light (3×10^8 m/s) by the number of seconds in a year (31,536,000 s). This results in approximately 9.46×10^{15} meters, which rounds to 9.5×10^{15} meters.

8. **Answer :** D) 1609
The mile is a unit of linear measure in the imperial system. 1 mile is exactly equal to 1.609344 kilometers, which is approximately 1609 meters.

9. **Answer :** B) 10^{-15}
A Fermi (or femtometer) is a unit of length used primarily in nuclear physics to measure the diameter of atomic nuclei. 1 Fermi is defined as 10^{-15} meters.

10. **Answer :** D) 10^{-8}
An angstrom is a unit of length used for measuring atomic and molecular dimensions. It is officially defined as 10^{-8} centimeters.

11. **Answer :** C) 10^{-8}
In nuclear engineering and astrophysics, a shake is an informal unit of time equal to 10 nanoseconds, or 10^{-8} seconds.

12. **Answer :** C) 1.67×10^{-27}
One atomic mass unit (a.m.u) is defined as one-twelfth the mass of a carbon-12 atom. In kilograms, this standard reference mass is approximately 1.67×10^{-27} kg.

13. **Answer :** C) 1000
A metric ton (or tonne) is a non-SI metric unit of mass. 1 metric ton is exactly equal to 1000 kilograms.

14. **Answer :** B) 14.59 kg
The slug is an English unit of mass defined by Newton's second law ($F = ma$). It is

the mass that accelerates by 1 ft/s² when a force of 1 pound-force is applied. 1 slug is approximately equal to 14.59 kilograms.

15. **Answer :** C) 454 grams
A pound is a unit of mass in the imperial system. The international standard pound is exactly 0.45359237 kilograms, which is approximately 454 grams.

16. **Answer :** B) $10^5 N/m^2$
One standard atmosphere (atm) is the approximate average atmospheric pressure at sea level. In standard SI units (Pascals or N/m²), 1 atm is equal to 101,325 N/m², which is roughly approximated as $10^5 N/m^2$.

17. **Answer :** B) 10^5 Pa
A bar is a metric unit of pressure defined as exactly 100,000 Pascals (Pa). Therefore, 1 bar equals 10^5 Pa.

18. **Answer :** C) 746 W
Horsepower (H.P) is a unit of power often used for engines and motors. One mechanical horsepower is equivalent to approximately 746 watts.

19. **Answer :** D) $1.6 \times 10^{-19} J$
An electron volt (eV) is a unit of energy equal to the work done on an electron in accelerating it through a potential difference of one volt. The charge of an electron is 1.6×10^{-19} Coulombs, making 1 eV = 1.6×10^{-19} Joules.

20. **Answer :** B) 10^8
The weber is the SI unit of magnetic flux, whereas the maxwell is the CGS unit. 1 weber is equal to 10^8 maxwells. (In the question it should be maxwell instead of gauss).

21. **Answer :** A) 10^4
Tesla is the SI unit for magnetic flux density, and Gauss is the corresponding CGS unit. The conversion factor is 1 Tesla = 10,000 Gauss, or 10^4 Gauss.

22. **Answer :** B) $\frac{10^3}{4\pi}$
The oersted is the CGS unit of magnetic field strength (H-field). When converting to the SI unit of Amperes per meter (A/m), 1 oersted is equal to $\frac{1000}{4\pi}$ A/m.

23. **Answer :** D) Poise
The coefficient of viscosity measures a fluid's resistance to flow. In the centimeter-gram-second (CGS) system, the designated unit for dynamic viscosity is the Poise.

24. **Answer :** C) 365
A solar year, also known as a tropical year, is the time it takes for the Earth to complete one revolution around the Sun. In standard calendar tracking, it is represented broadly as 365 days.

25. **Answer :** C) 4
A leap year in the Gregorian calendar usually occurs every 4 years. If a year is evenly divisible by 4 (with exceptions for century years not divisible by 400), it is classified as a leap year.

26. **Answer :** B) 30.48
The foot is a unit of length in the imperial system. One inch is officially 2.54 cm, and since there are 12 inches in a foot, 1 foot equals $12 \times 2.54 = 30.48$ cm.

27. **Answer :** B) Lunar month
A lunar month, also known as a synodic month, is defined as the time it takes the Moon to complete one full cycle of its phases (e.g., from new moon to new moon) as it orbits the Earth.

LEVEL-II

(SINGLE CORRECT OPTION TYPE)

1. **Answer :** D) Shake
Parsec, Light year, and Angstrom are all units of spatial distance or length. A shake is a unit of time equal to 10 nanoseconds, so it is not a unit of length.

2. **Answer :** A) 4 min
A mean solar day (24 hours) is the time for the Earth to rotate relative to the Sun. A sidereal day (approx 23 hours 56 minutes) is the rotation relative to distant stars. The difference is approximately 4 minutes.

3. **Answer :** D) Fermi
Nuclear radii are extremely small, on the order of 10^{-15} meters. The Fermi (femtometer) is exactly 10^{-15} meters, making it the proper specific unit for measuring subatomic and nuclear sizes.

4. **Answer :** A) S.I
The C.G.S, M.K.S, and F.P.S systems are purely mechanical systems based entirely on three fundamental units: mass, length, and time. The S.I system extends this base to seven fundamental units (adding temperature, current, luminous intensity, and amount of substance).

5. **Answer :** B) Bar
Fermi, Micron, and Light year are standard units used to measure length or distance. A bar is a unit used strictly to measure fluid and atmospheric pressure.

6. **Answer :** B) Kilogram
In the SI system, mass is a fundamental physical quantity, and its fundamental unit is the kilogram. Newton and Acceleration are derived units/quantities, and Length is a quantity, not a unit.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

7. **Answer :** A) Mean solar day, B) Lunar month, C) Leap year
Mean solar day, lunar month, and leap year are all measurements of time intervals. A light year is a measurement of distance, not time.

8. **Answer :** A) Steradian is unit of solid angle, B) Mole is the unit of amount of substance, C) Candela is the unit of luminous intensity
Options A, B, and C correctly match physical quantities to their standard SI units. Option D is incorrect because the kilogram is the unit of mass, not length.

9. **Answer :** A) Pound, B) Slug, C) A.M.U, D) Quintal
All of the listed options are units of mass. The pound and slug are imperial units, A.M.U is used at the atomic scale, and the quintal is a metric unit representing 100 kilograms.

10. **Answer :** A) 100 kg, B) 10^5 grams
A quintal is defined as 100 kilograms. Because 1 kilogram equals 1000 grams, 100 kg is equal to 100,000 grams (or 10^5 grams). Therefore, both A and B are mathematically equivalent and correct.

ASSERTION AND REASON TYPE

11. **Answer :** A) Both A and R are true, and R is correct explanation of A
The historical SI definition of the Ampere was based on the force between two parallel current-carrying wires. This definition exists entirely because electric current inherently produces a magnetic field that exerts measurable interaction forces on other conductors.

12. **Answer :** B) Both A and R are true, but R is NOT correct explanation of A
It is a true fact that the Kelvin scale is anchored to the triple point of water (Assertion is true). It is also true that temperature is a fundamental SI quantity (Reason is true). However, being a fundamental quantity is not the scientific reason why the triple point of water is used as the calibration standard for the Kelvin unit.

COMPREHENSION TYPE (Passage-I)

13. **Answer :** A) $10^{-1}m$
The metric prefix "deci" signifies one-tenth of a base unit. Therefore, 1 decimeter equals 10^{-1} meters.

14. **Answer :** C) 10^{-3}
The metric prefix "milli" signifies one-thousandth of a base unit. Therefore, 1 millisecond equals 10^{-3} seconds.

15. **Answer :** B) 10^{-12}
The metric prefix "pico" signifies one-trillionth of a base unit. Therefore, a physical unit factored by a pico prefix is equal to 10^{-12} of the base value.

COMPREHENSION TYPE (Passage-II)

16. **Answer :** C) Area
Mass, temperature, and length are fundamentally independent physical quantities in standard systems. Area is a derived quantity calculated by multiplying lengths together ($Area = length \times width$).

17. **Answer :** D) Electric current
Electric current is an independent, fundamental physical quantity. Candela, Ampere, and Mole are fundamentally defined units of measurement, not physical quantities themselves.

18. **Answer :** C) 1 kelvin
The Kelvin is the SI fundamental unit of thermodynamic temperature. It was historically defined exactly as the fraction $1/273.16$ of the thermodynamic temperature of the triple point of pure water.

MATRIX MATCHING TYPE

19. **Answer :** a) → R, T ; b) → S ; c) → P ; d) → Q
An angstrom (a) equals 10^{-10} meters (R) and 10^{-8} centimeters (T). A mile (b) is approximately equal to 1.6 kilometers (S). A light year (c) equates to approximately 9.5×10^{12} kilometers (P). A barn (d) is an area unit used in nuclear physics defined as 10^{-28} square meters (Q).

20. **Answer :** a) → R ; b) → P ; c) → S ; d) → Q, T
One kilometre (a) is defined as 1000 meters (R). One standard mean solar day (b) contains exactly 86,400 seconds (P). One metric ton (c) is 1000 kg, which equals 10 quintals (S), since a quintal is 100 kg. A standard year (d) encompasses 365 days (Q) and consists of 12 months (T).

INTEGER TYPE

21. **Answer :** 4
We start by converting both units to meters. 1 micrometer (μm) is equal to 10^{-6} m. 1 Angstrom (A°) is equal to 10^{-10} m. Dividing the micrometer value by the Angstrom value yields $10^{-6}/10^{-10} = 10^{4}$. Therefore, the integer x is 4.

22. **Answer :** 9
The prefix "nano" denotes a factor of one-billionth. Consequently, 1 nanometer (nm) is equal to 10^{-9} meters. Matching this to the formula 10^{-x} meters, we determine that x is exactly 9.

23. **Answer :** 5
The prompt provides a list of terms: Length, mass, time, temperature, and force. All five of these terms represent independent or derived physical properties that can be measured, making all 5 examples of physical quantities.

Dimensions

LEVEL-I

1. **Answer :** B) Pascal
Pressure is defined as force per unit area. In the International System of Units (SI), the unit of force is the Newton (N) and the unit of area is the square meter (m^2). Therefore, the SI unit of pressure is N/m^2, which is formally named the Pascal (Pa).

2. **Answer :** C) Thrust
Thrust is the macroscopic force applied perpendicularly to a surface, meaning it is fundamentally a type of force. Therefore, thrust and force share the exact same dimensional formula, which is $[MLT^{-2}]$.

3. **Answer :** B) Work and Energy
Work is defined as force multiplied by displacement, yielding dimensions of $[ML^2T^{-2}]$. Energy is the capacity to do work and shares the exact same dimensional formula.

4. **Answer :** A) $[ML^2T^{-2}]$
A calorie is a unit of heat, which is a form of energy. All forms of energy and work have the dimensional formula $[ML^2T^{-2}]$.

5. **Answer :** C) Gravitational constant
We can determine the dimensions of the universal gravitational constant (G) using Newton's law of universal gravitation, $F = G\frac{m_1m_2}{r^2}$. Rearranging for G gives $G = \frac{Fr^2}{m_1m_2}$. Substituting the dimensions for force ($[MLT^{-2}]$), radius squared ($[L^2]$), and mass squared ($[M^2]$) yields $\frac{[MLT^{-2}][L^2]}{[M^2]}$, which simplifies to $[M^{-1}L^3T^{-2}]$.

6. **Answer :** C) Time
The dimensional formula for impulse (force × time) is $[MLT^{-1}]$. The dimensional formula for angular velocity (angle / time) is $[M^0L^0T^{-1}]$. Comparing the two, they have different dimensions for mass and length, but they share the exact same dimension for time, which is -1.

7. **Answer :** B) $[M^1L^0T^{-2}]$
The given unit is J/m^2. A Joule (J) is a unit of energy with dimensions $[ML^2T^{-2}]$. Dividing this by the dimension for area (m^2 or $[L^2]$) yields $\frac{[ML^2T^{-2}]}{[L^2]}$, which simplifies directly to $[MT^{-2}]$ or $[M^1L^0T^{-2}]$.

8. **Answer :** A) radian
Angular displacement measures the angle through which a point or line has been rotated in a specified sense about a specified axis. Its standard SI unit is the radian.

9. **Answer :** C) $[T^{-1}]$
Angular velocity is defined as the rate of change of angular displacement with respect to time. Since angular displacement (radians) is a dimensionless quantity, the dimensional formula for angular velocity is simply $\frac{1}{[T]}$, which is $[T^{-1}]$ or $[M^0L^0T^{-1}]$.

10. **Answer :** A) Work
Multiplying pressure and volume dimensionally yields the dimensions of work.

Pressure has dimensions $[ML^{-1}T^{-2}]$ and volume has dimensions $[L^3]$. Multiplying these gives $[ML^{-1}T^{-2}] \times [L^3] = [ML^2T^{-2}]$, which is the dimensional formula for Work or Energy.

11. **Answer :** C) Mass and time
Linear momentum ($p = mv$) has dimensions $[MLT^{-1}]$. Angular momentum ($L = mvr$) has dimensions $[ML^2T^{-1}]$. Comparing the two formulas, they have the same dimensional power for mass (1) and for time (-1), but differ in length.

12. **Answer :** D) Radian
A physical quantity is a property of a material or system that can be quantified by measurement. Power, momentum, and latent heat are all physical quantities. A radian is a unit of measurement for angles, not a physical quantity itself.

13. **Answer :** D) Volt
The erg, Joule, and electron volt are all recognized units of work and energy in various systems (CGS, SI, and atomic physics, respectively). The Volt is the SI unit for electric potential difference, not work.

14. **Answer :** D) Joule
Candela, Kelvin, and Mole are all fundamental base units in the SI system. The Joule is a derived unit, created from fundamental units to represent energy ($kg \cdot m^2/s^2$).

15. **Answer :** A) Kelvin
According to the International System of Units (SI), the fundamental base unit for thermodynamic temperature is the Kelvin.

16. **Answer :** C) Relative density
Relative density (or specific gravity) is the ratio of the density of a substance to the density of a reference material. Since it is a ratio of two identical physical quantities, their units cancel out, making it a dimensionless and unitless quantity.

17. **Answer :** B) Mass
According to Newton's Second Law of Motion, Force = Mass × Acceleration. Rearranging this equation gives Mass = Force / Acceleration. Therefore, dividing the unit of force by the unit of acceleration yields the unit of mass.

18. **Answer :** B) Power
Joules per second (J/s) measures the rate at which work is done or energy is transferred over time. This is the precise definition of Power. The SI derived unit for power is the Watt, which is exactly equal to one Joule per second.

19. **Answer :** A) Power
We must evaluate the dimensions of each option. Acceleration is $[LT^{-2}]$. Force constant is $[MT^{-2}]$. Stress is $[ML^{-1}T^{-2}]$. Power is Work/Time, which is $\frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$. In the formula for Power, the dimension (exponent) of length is 2.

20. **Answer :** D) Energy
An electron volt (eV) is defined as the amount of kinetic energy gained by a single electron accelerating from rest through an electric potential difference of one volt. It is a standard unit of Energy in particle physics.

21. **Answer :** B) Pressure = Energy per unit volume
We use dimensional analysis to check equality. Pressure has dimensions $[ML^{-1}T^{-2}]$. Energy per unit volume is Energy divided by Volume, which is $\frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}]$. Both sides match perfectly.

22. **Answer :** B) Momentum
The dyne is the CGS unit of force. Dyne-second represents Force × Time. According to the impulse-momentum theorem, the product of force and time equals Impulse, which translates directly to a change in Momentum.

23. **Answer :** A) $[ML^2T^{-2}]$
Energy is dimensionally equivalent to Work, which is calculated as Force × Displacement. Using fundamental dimensions, Force is $[MLT^{-2}]$ and Displacement is $[L]$. Multiplying these gives $[ML^2T^{-2}]$.

LEVEL-II

SINGLE CORRECT OPTION TYPE

1. **Answer :** B) 16
The dimensional formula for Power is $[ML^2T^{-3}]$. We can also express Power as Force × Velocity, which is Force × (Length / Time). If the unit of force is quadrupled (4), the unit of length is doubled (2), and the unit of time is halved (0.5), the new unit of power becomes proportional to $\frac{4\times2}{0.5}$. Calculating this gives $\frac{8}{0.5} = 16$. The unit of power changes by a factor of 16.

2. **Answer :** B) 2
We resolve the right side of the given equation into fundamental dimensions. Mass is $[M]$, radius is $[L]$, and time period is $[T]$. The right side becomes $\frac{[M]^2[L]^2}{[T]^4}$, which factors to $([MLT^{-2}])^2$. Since $[MLT^{-2}]$ is the dimensional formula for Force, the equation simplifies to $(\text{Force})^x = (\text{Force})^2$. Therefore, $x = 2$.

3. **Answer :** C) Energy
According to the principle of dimensional homogeneity, quantities added together must have the same dimensions. Thus, Pressure (P) must have the same dimensions as $\frac{a}{V^2}$, meaning $a = PV^2$. Similarly, Volume (V) must have the same dimensions as b, meaning $b = V$. The ratio $\frac{a}{b}$ is dimensionally equal to $\frac{PV^2}{V}$, which simplifies to PV. Pressure times Volume represents Work or Energy.

4. **Answer :** B) Momentum
The de Broglie wavelength equation relates wavelength to momentum via the formula $\lambda = \frac{h}{p}$, where p is momentum. Rearranging this equation yields $p = \frac{h}{\lambda}$. Therefore, the ratio $\frac{h}{\lambda}$ precisely carries the dimensions of Momentum.

5. **Answer :** A) $[F^0aT^2]$
We need to express length using acceleration (a) and time (T). The dimensional formula for acceleration is $[LT^{-2}]$. Rearranging this to solve for length gives Length = Acceleration × Time^2. Using the new fundamental units, this translates to a^1T^2. Since Force (F) is not required to define length in this setup, its power is 0. The final formula is $[F^0a^1T^2]$.

6. **Answer :** C) LT^{-2}, L and T
By the principle of dimensional homogeneity, all terms in an equation must have the same dimension. The term c is added to time t, so c must have the dimension of time, $[T]$. The term at must have the dimension of velocity V ($[LT^{-1}]$), so $a \times [T] = [LT^{-1}]$, making $a = [LT^{-2}]$. The entire term $\frac{b}{t+c}$ must also have the dimension of velocity. Since the denominator has dimension $[T]$, we get $\frac{b}{[T]} = [LT^{-1}]$, making $b = [L]$.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

7. **Answer :** A) Distance, B) Displacement, C) Wavelength
Distance, displacement, and wavelength are all fundamental measures of spatial length, meaning they all share the simple dimensional formula $[L]$. Speed is a rate of change of distance, possessing the dimensional formula $[LT^{-1}]$.

8. **Answer :** A) Dimensional formula for mass is $[V^{-1}FT]$, B) Dimensional formula for energy is $[VFT]$, C) Dimensional formula for pressure is $[FV^{-2}T^{-2}]$
To verify these, we express derived quantities using F, V, and T. Mass equals Force / Acceleration. Since acceleration is V/T, Mass $= \frac{F}{V/T} = FV^{-1}T$. Energy equals Force × Distance. Since distance is $V \times T$, Energy = FVT. Pressure equals Force / Area. Since area is Distance^2 or $(VT)^2$, Pressure $= \frac{F}{V^2T^2} = FV^{-2}T^{-2}$. All three of these relationships are dimensionally accurate.

9. **Answer :** A) Distance and displacement, B) Speed and velocity, C) Work and energy
Distance and displacement both have dimension $[L]$. Speed and velocity both have dimension $[LT^{-1}]$. Work and energy both have dimension $[ML^2T^{-2}]$. Planck's constant ($[ML^2T^{-1}]$) and torque ($[ML^2T^{-2}]$) do not share the same dimensions.

ASSERTION AND REASON TYPE

10. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Pressure is formally defined as force divided by area, which validates the Reason. Deriving the dimensional formula from this definition: Force is $[MLT^{-2}]$ and Area is $[L^2]$. Dividing these gives $\frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$, which validates the Assertion. The Reason directly leads to the Assertion

11. **Answer :** A) Both A and R are true, and R is the correct explanation of A
Angular velocity is calculated as the change in angular displacement (dimensionless) over time, yielding $[T^{-1}]$. Frequency is defined as cycles (dimensionless) per unit time, also yielding $[T^{-1}]$. Because both are fundamentally expressed as the reciprocal of time, they share the exact same dimensional formula.

COMPREHENSION TYPE (Passage-I)

12. **Answer :** B) $[ML^2T^{-3}]$
Power is defined as Work done per unit time. The dimensional formula for Work is $[ML^2T^{-2}]$. Dividing this by the dimension for time $[T]$ gives $\frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$.

13. **Answer :** D) $[MLT^{-1}]$
Impulse is mathematically defined as Force multiplied by Time. Taking the dimensions for Force ($[MLT^{-2}]$) and multiplying by Time ($[T]$) results in $[MLT^{-2}] \times [T] = [MLT^{-1}]$.

14. **Answer :** D) [I]
In dimensional analysis, electric current is treated as a foundational, fundamental base quantity. It is traditionally represented by its own specific dimensional symbol, which is $[I]$ or sometimes $[A]$.

COMPREHENSION TYPE (Passage-II)

15. **Answer :** D) $W^1F^{-1}V^{-1}$
We start with the relationship between Work, Force, and Distance: Work = Force × Distance. Distance can be rewritten as Velocity × Time. Substituting this into the first equation yields Work = Force × Velocity × Time. Rearranging this relation to

isolate Time gives Time = $\frac{\text{Work}}{\text{Force}\times\text{Velocity}}$. In dimensional notation using the given variables, this is $W^1F^{-1}V^{-1}$.

16. Answer : B) $FM^{-1}T^2$

From Newton's second law, we know that Force = Mass × Acceleration. Acceleration is the second derivative of position with respect to time, giving it dimensions of Length / Time^2. Substituting this into the force equation yields Force = Mass × $\frac{\text{Length}}{\text{Time}^2}$. Rearranging to solve for Length provides Length = $\frac{\text{Force}\times\text{Time}^2}{\text{Mass}}$, which maps to $FM^{-1}T^2$.

17. Answer : D) $F^1L^{-1}T^2$

We utilize the definition of force where Force = Mass × Acceleration. Since Acceleration is equal to Length divided by Time squared (L/T^2), we write Force = Mass × (L/T^2). Rearranging this equation to isolate Mass gives Mass = $\frac{\text{Force}\times\text{Time}^2}{\text{Length}}$, which dimensionally translates to $F^1L^{-1}T^2$.

MATRIX MATCHING TYPE

18. Answer : a) → R, b) → S, c) → Q, d) → P

The spring constant ($k = F/x$) has dimensions $\frac{[MLT^{-2}]}{[L]} = [MT^{-2}]$ or $[M^1L^0T^{-2}]$, which matches R. Pascal is the unit for Pressure (F/A), which has dimensions $[ML^{-1}T^{-2}]$, matching S. Hertz is the unit for frequency, which is 1/Time and has dimensions $[M^0L^0T^{-1}]$, matching Q. Joule is the unit for Energy or Work, having dimensions $[M^1L^2T^{-2}]$, which perfectly matches P.

19. Answer : a) → Q, b) → P, c) → S, d) → R

Work has the formula $[ML^2T^{-2}]$, so the exponent (dimension) of length is 2, matching Q. Energy has the formula $[ML^2T^{-2}]$, so the dimension of time is -2, matching P. Density is Mass/Volume with the formula $[ML^{-3}T^0]$, so the dimension of length is -3, matching S. Impulse is Force × Time with the formula $[MLT^{-1}]$, so the dimension of time is -1, matching R.

INTEGER TYPE

20. Answer : -2

The dimensional formula representing Work is $[M^1L^2T^{-2}]$. The power or exponent attached to the fundamental quantity of time (T) in this formula is strictly -2.

21. Answer : 1

The physical quantity of Force is represented by the dimensional formula $[M^1L^1T^{-2}]$. Looking specifically at the fundamental quantity of mass (M), its exponent is 1.

22. Answer : 2

The general dimensional formula for a physical quantity can be written as $[M^aL^bT^c]$. We know that the established dimensional formula for energy is $[M^1L^2T^{-2}]$. By directly comparing the exponents, the value corresponding to b (the dimension of length) is exactly 2.

3. Motion in a Straight Line

TOPIC-1

Fundamentals of One-Dimensional Motion (Kinematics)

LEVEL-I

1. Correct Answer: C

Explanation:

Take east as $+\hat{\imath}$and north as $+\hat{\jmath}$.

- First displacement:

  $\vec{d}_1 = 30\hat{\jmath}$
- Second displacement:

  $\vec{d}_2 = 20\hat{\imath}$
- Third displacement ($30\sqrt{2}$ m towards south-west):

  $\vec{d}_3 = 30\sqrt{2}\left(\frac{-\hat{\imath}-\hat{\jmath}}{\sqrt{2}}\right) = -30\hat{\imath} - 30\hat{\jmath}$

Hence,

$\vec{d} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3 = -10\hat{\imath}$

Therefore, the net displacement is

$\boxed{10 \text{ m towards west}}$

2. Correct Answer: B

Explanation:

For a round trip over equal distances with speeds $v_1$and v_2, the average speed is the harmonic mean:

$$\text{Average Speed} = \frac{2v_1v_2}{v_1 + v_2}$$

Substituting,

$$= \frac{2 \times 2.5 \times 4}{2.5 + 4} = \frac{20}{6.5} = \frac{40}{13} \text{ km/h}$$

$$\boxed{\text{Average Speed} = \frac{40}{13} \text{ km/h}}$$

3. Correct Answer: D

Explanation:

Let the total distance be D.

Time taken for the first one-third:

$$t_1 = \frac{D/3}{20} = \frac{D}{60}$$

Time taken for the remaining two-thirds:

$$t_2 = \frac{2D/3}{60} = \frac{D}{90}$$

Total time:

$$t = \frac{D}{60} + \frac{D}{90} = \frac{5D}{180} = \frac{D}{36}$$

Hence,

$$\text{Average Speed} = \frac{D}{D/36} = 36 \text{ km/h}$$

$\boxed{36 \text{ km/h}}$

4. Correct Answer: B

Explanation:

Total distance travelled:

$30 + 40 = 70$ m

Total time:

$t = \frac{30}{2} + \frac{40}{1.5} = 15 + \frac{80}{3} = \frac{125}{3}$ s

Average speed:

$= \frac{70}{125/3} = \frac{42}{25}$ m/s

Net displacement:

$= \sqrt{30^2 + 40^2} = 50$ m

Average velocity:

$= \frac{50}{125/3} = \frac{6}{5}$ m/s

$\boxed{\text{Average Speed} = \frac{42}{25} \text{ m/s, Average Velocity} = \frac{6}{5} \text{ m/s}}$

5. Correct Answer: D

Explanation:

Since the train and parrot move in opposite directions,

$v_{\text{relative}} = 10 + 5 = 15$ m/s

Time taken for the train to completely pass the parrot:

$t = \frac{150}{15} = 10$ s

$\boxed{10 \text{ s}}$

6. Correct Answer: B

Explanation:

Using

$v^2 - u^2 = 2as$

After travelling 3 cm,

$\left(\frac{u}{2}\right)^2 - u^2 = 2a(3)$

$a = -\frac{u^2}{8}$

For the remaining distance until it stops,

$0 - \left(\frac{u}{2}\right)^2 = 2as'$

Substituting a,

$-\frac{u^2}{4} = 2\left(-\frac{u^2}{8}\right)s'$

$s' = 1$ cm

$\boxed{1 \text{ cm}}$

7. Correct Answer: A

Explanation:

For Car A,

$x_A = 44t$

For Car B,

$x_B = \frac{1}{2}at^2 = \frac{1}{2}(2)t^2 = t^2$

Equating,

$t^2 = 44t$

$t = 44$ s

$\boxed{44 \text{ s}}$

8. Correct Answer: D

Explanation:

Using the first equation of motion,

$v = u + at$

$= 10 + 2(4) = 18$ m/s

$\boxed{18 \text{ m/s}}$

9. Correct Answer: A

Explanation:

Distance travelled during the n^{th}second:

$S_n = u + \frac{a}{2}(2n - 1)$

Given,

$u = 0, a = 8 \text{ m/s}^2, n = 5$

$S_5 = \frac{8}{2}(9) = 36$ m

$\boxed{36 \text{ m}}$

10. Correct Answer: D

Explanation:

For constant braking force,

$S \propto u^2$

The speed increases from 10 m/sto 30 m/s,

$\left(\frac{30}{10}\right)^2 = 9$

Hence,

$S = 20 \times 9 = 180$ m

$\boxed{180 \text{ m}}$

11. Correct Answer: A

Explanation:

For motion starting from rest,

$S_n \propto (2n - 1)$

Therefore,

$\frac{S_4}{S_3} = \frac{7}{5}$

$\boxed{\frac{7}{5}}$

12. Correct Answer: C

Explanation:

Distances covered in successive equal intervals are in the ratio

$1:3:5$

Hence,

$S_1 : S_2 : S_3 = 1:3:5$

or,

$\boxed{S_1 = \frac{1}{3}S_2 = \frac{1}{5}S_3}$

13. Correct Answer: B

Explanation:

For Body A (5th second),

$S_A = \frac{a_1}{2}(9) = \frac{9}{2}a_1$

Since Body B starts 2 seconds later, A's 5th second corresponds to B's 3rd second.

Thus,

$S_B = \frac{a_2}{2}(5) = \frac{5}{2}a_2$

Equating,

$\frac{9}{2}a_1 = \frac{5}{2}a_2$

$\boxed{\frac{a_1}{a_2} = \frac{5}{9}}$

14. Correct Answer: C

Explanation:

For the student to catch the bus,

$ut = 50 + \frac{1}{2}t^2$

or,

$t^2 - 2ut + 100 = 0$

For real solutions,

$b^2 - 4ac \geq 0$

$4u^2 - 400 \geq 0$

$u \geq 10$ m/s

Hence, the minimum speed is

$\boxed{10 \text{ m/s}}$

15. Correct Answer: C

Explanation:

Distance in first 5s:

$5u + 12.5a = 40$

$u + 2.5a = 8$

Total distance in 10s:

$10u + 50a = 105$

$u + 5a = 10.5$

Subtracting,

$2.5a = 2.5$

$a = 1$ m/s^2

Substituting,

$u = 5.5$ m/s

$\boxed{u = 5.5 \text{ m/s}}$

16. Correct Answer: C

Explanation:

Maximum speed reached is

$v = \sqrt{2fS}$

The constant-speed portion covers

$14S - 2S = 12S$

Time in the middle section:

$t = \frac{12S}{\sqrt{2fS}}$

Squaring,

$t^2 = \frac{72S}{f}$

Therefore,

$\boxed{S = \frac{ft^2}{72}}$

17. Correct Answer: C

Explanation:

Using

$v^2 - u^2 = 2aL$

Therefore,

$aL = \frac{v^2 - u^2}{2}$

When the midpoint passes the pole,

$v_{\text{mid}}^2 - u^2 = aL$

Substituting,

$v_{\text{mid}}^2 = u^2 + \frac{v^2 - u^2}{2} = \frac{u^2 + v^2}{2}$

Hence,

$\boxed{v_{\text{mid}} = \sqrt{\frac{u^2 + v^2}{2}}}$

18. Correct Answer: A

Explanation:

Time for the first half:

$t_1 = \frac{S}{v}$

For the second half,

$S = \frac{t_2}{2}(v_1 + v_2)$

$t_2 = \frac{2S}{v_1 + v_2}$

Therefore,

Average Speed $= \frac{2S}{t_1 + t_2} = \frac{2v(v_1 + v_2)}{2v + v_1 + v_2}$

$\boxed{\frac{2v(v_1 + v_2)}{2v + v_1 + v_2}}$

19. Correct Answer: C

Explanation:

Average speed is defined as

$\boxed{\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{s_1 + s_2 + \cdots + s_n}{t_1 + t_2 + \cdots + t_n}}$

20. Correct Answer: A

Explanation:

Let

- t_1= time of acceleration
- t_2= time of deceleration

where

$t = t_1 + t_2$

Maximum velocity is

$v_{\max} = \alpha t_1 = \beta t_2$

Therefore,

$t = \frac{v_{\max}}{\alpha} + \frac{v_{\max}}{\beta}$

$= v_{\max}\left(\frac{\alpha + \beta}{\alpha\beta}\right)$

Hence,

$\boxed{v_{\max} = \left(\frac{\alpha\beta}{\alpha + \beta}\right)t}$

LEVEL-II

Single Correct Option Type

1. Correct Option: D

Explanation:

Acceleration is given by:

$a = \frac{dv}{dt} = 2t + 5$

Integrating from $t = 0$(body starts from rest, so $v = 0$) to $t = 2$ s:

$v = \int_0^2 (2t + 5)\, dt$

$= [t^2 + 5t]_0^2$

$= (2)^2 + 5(2) = 4 + 10 = 14$ m/s

Hence, the velocity after 2s is

$\boxed{14 \text{ m/s}}$

2. Correct Option: D

Explanation:

Using relative motion with respect to Train A:

- Initial relative velocity:

  $u_{\text{rel}} = 0$
- Relative acceleration:

  $a_{\text{rel}} = 1\ \text{m/s}^2$
- Time:

  $t = 50$ s

  Relative displacement:

  $s_{\text{rel}} = \frac{1}{2}a_{\text{rel}}t^2$

  $= \frac{1}{2}(1)(50)^2 = 1250$ m

  This displacement consists of:
- Length of Train A = 400 m
- Length of Train B = 400 m

- Initial separation = x

Therefore,

$400 + 400 + x = 1250$

$x = 450\ \text{m}$

Hence,

$\boxed{x = 450\ \text{m}}$

3. Correct Option: C

Explanation:

Let the maximum velocity be $v_{\max}$.

Acceleration phase (distance = S)

$t_1 = \frac{2S}{v_{\max}}$

Uniform velocity phase (distance = $2S$)

$t_2 = \frac{2S}{v_{\max}}$

Retardation phase (distance = $5S$)

$t_3 = \frac{2(5S)}{v_{\max}} = \frac{10S}{v_{\max}}$

Total distance:

$S + 2S + 5S = 8S$

Total time:

$t = \frac{14S}{v_{\max}}$

Average velocity:

$v_{\text{avg}} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{8S}{\frac{14S}{v_{\max}}} = \frac{4}{7}v_{\max}$

Therefore,

$\boxed{\frac{v_{\text{avg}}}{v_{\max}} = \frac{4}{7}}$

4. Correct Option: B

Explanation:

For a body starting from rest under uniform acceleration,

$s \propto t^2$

Hence,

$t \propto \sqrt{s}$

If the body covers the total distance Sin 4s, then the time required to cover $\frac{S}{4}$ is

$t' = 4\sqrt{\frac{1}{4}} = 4\left(\frac{1}{2}\right) = 2\ \text{s}$

Thus,

$\boxed{2\ \text{s}}$

Assertion and Reason Type

9. Correct Option: A

Explanation:

The area under a velocity–time graph is

$\int v\,dt$

which represents displacement.

The Reason is also true because velocity is the rate of change of displacement.

Therefore,

Velocity × Time = Displacement

Hence, both the Assertion and Reason are true, and the Reason correctly explains the Assertion.

10. Correct Option: A

Explanation:

Uniform velocity means the velocity remains constant throughout the motion.

Acceleration is

$a = \frac{dv}{dt}$

Since the velocity does not change,

$\frac{dv}{dt} = 0$

Therefore,

$a = 0$

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

11. Correct Option: D

Explanation:

The slope of a position–time graph is

$\frac{dx}{dt}$

which represents velocity, not acceleration.

Therefore, the Assertion is false.

The Reason correctly states that the slope of a position–time graph gives velocity.

Hence,

- Assertion is false.
- Reason is true.

12. Correct Option: A

Explanation:

When a body returns to its starting point (for example, after completing one circular path),

- Net displacement = 0
- Distance travelled > 0

This is because displacement depends only on the initial and final positions, whereas distance depends on the actual path travelled.

Therefore, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Comprehension Type

13. Correct Option: C

Explanation:

The minimum distance between the particles occurs when their relative velocity becomes zero.

Given,

$v_A = 10\ \text{m/s}$

$v_B = 2t$

Relative velocity,

$v_{\text{rel}} = 10 - 2t$

For minimum separation,

$v_{\text{rel}} = 0$

$10 - 2t = 0$

$t = 5\ \text{s}$

Hence,

$\boxed{t = 5\ \text{s}}$

14. Correct Option: B

Explanation:

At $t = 5\text{s}$,

Distance travelled by particle A:

$s_A = 10 \times 5 = 50\ \text{m}$

Distance travelled by particle B:

$s_B = \frac{1}{2}(2)(5)^2 = 25\ \text{m}$

Since B was initially 40m ahead,

Position of B $= 40 + 25 = 65\ \text{m}$

Separation:

$65 - 50 = 15\ \text{m}$

Therefore,

$\boxed{15\ \text{m}}$

15. Correct Option: C

Explanation:

Up to $t = 5\text{s}$,

Particle A covers

$50\ \text{m}$

Particle B covers

$25\ \text{m}$

Relative gain by A:

$50 - 25 = 25\ \text{m}$

Hence, for a collision to just occur, the initial separation must be

$\boxed{25\ \text{m}}$

16. Correct Option: B

Explanation:

Let

- Acceleration time = t_1
- Retardation time = t_2

Then,

$t_1 + t_2 = t$

Maximum velocity,

$v_{max} = \alpha t_1 = \beta t_2$

Therefore,

$t_1 = \frac{v_{max}}{\alpha}, t_2 = \frac{v_{max}}{\beta}$

Hence,

$t = v_{max}\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

$= v_{max}\left(\frac{\alpha + \beta}{\alpha\beta}\right)$

Therefore,

$$\boxed{v_{max} = \left(\frac{\alpha\beta}{\alpha + \beta}\right)t}$$

17. Correct Option: B

Explanation:

The distance travelled equals the area under the triangular velocity–time graph.

Distance $= \frac{1}{2} \times$ Base $\times$ Height

$= \frac{1}{2} \times t \times v_{max}$

Substituting

$v_{max} = \frac{\alpha\beta}{\alpha + \beta}t$

gives

$$\boxed{\text{Distance} = \frac{\alpha\beta}{2(\alpha + \beta)}t^2}$$

18. Correct Option: A

Explanation:

The literal average acceleration over the complete journey is

$\frac{\Delta v}{\Delta t} = 0$

since the initial and final velocities are equal.

However, among the given options, only Option A has the correct dimensions of acceleration (m/s^2). Hence, Option A is the intended answer.

Matrix Matching Type

19. Answer:

a) P, Q

b) R

c) S

d) S

20. Answer:

a) Q, R

b) R

c) Q, R

d) Q, R

Integer Type

21. Answer: 50

Explanation:

Using the second equation of motion,

$s = ut + \frac{1}{2}at^2$

Since the body starts from rest,

$u = 0$

Therefore,

$s = \frac{1}{2}(4)(5)^2 = 2 \times 25 = 50\,\text{m}$

$\boxed{50}$

22. Answer: 22

Explanation:

Using the first equation of motion,

$v = u + at$

$= 10 + (2)(6)$

$= 10 + 12 = 22\,\text{m/s}$

$\boxed{22}$

23. Answer: 10

Explanation:

After correcting the typographical error ("min" → "m in"), the particle travels:

- 30 m in 3 s

Hence,

$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{30}{3} = 10\,\text{m/s}$

$\boxed{10}$

TOPIC-2: MOTION UNDER GRAVITY

LEVEL-I

1. Correct Option: C

Explanation:

For a freely falling body dropped from rest,

$t = \sqrt{\frac{2h}{g}}$

Since $g$is constant,

$t \propto \sqrt{h}$

Therefore, the ratio of times is

$\boxed{\sqrt{a}:\sqrt{b}}$

2. Correct Option: A

Explanation:

In a perfect vacuum, there is no air resistance. Hence, all objects experience the same acceleration due to gravity (g), irrespective of their mass, shape, or size. Therefore, all bodies fall at the same rate.

3. Correct Option: A

Explanation:

Using the equation of motion,

$h = \frac{1}{2}gt^2$

Given:

- $g = 10\ \text{m/s}^2$
- $t = 4\ \text{s}$

$h = \frac{1}{2} \times 10 \times 4^2 = 5 \times 16 = 80\ \text{m}$

4. Correct Option: B

Explanation:

For a body falling from rest,

$v^2 = 2gS$

Given that the numerical value of velocity equals the displacement,

$v = S$

Substituting,

$v^2 = 2gv$

$v = 2g$

5. Correct Option: A

Explanation:

For the first stone,

$44.1 = \frac{1}{2}(9.8)t_1^2$

$t_1 = 3\ \text{s}$

The second stone is thrown 1 s later, so its time of travel is

$t_2 = 3 - 1 = 2\ \text{s}$

Using

$44.1 = u(2) + \frac{1}{2}(9.8)(2)^2$

$44.1 = 2u + 19.6$

$u = 12.25\ \text{m/s}$

6. Correct Option: C

Explanation:

Let the maximum height reached be h.

The time taken to fall from the highest point to the ground is equal to the time taken to rise.

During the last $\frac{t}{2}$of the descent, the body falls

$\frac{1}{2}g\left(\frac{t}{2}\right)^2 = \frac{1}{4}\left(\frac{1}{2}gt^2\right) = \frac{h}{4}$

Hence, the height of the body above the ground at that instant is

$h - \frac{h}{4} = \boxed{\frac{3h}{4}}$

7. Correct Option: B

Explanation:

Using

$S = ut + \frac{1}{2}gt^2$

Given:

- $S = 30\ \text{m}$
- $t = 1\ \text{s}$
- $g = 9.8\ \text{m/s}^2$

$30 = u + \frac{1}{2}(9.8)$

$30 = u + 4.9$

$u = 25.1$ m/s

8. Correct Option: D

Explanation:

Maximum height,

$H = \frac{u^2}{2g}$

At half the maximum height,

$y = \frac{H}{2} = \frac{u^2}{4g}$

Using

$v^2 = u^2 - 2gy$

$v^2 = u^2 - \frac{u^2}{2} = \frac{u^2}{2}$

$\boxed{v = \frac{u}{\sqrt{2}}}$

9. Correct Option: B

Explanation:

Total distance is equal to the total area under the velocity-time graph.

- Triangle (0–1 s):
  $\frac{1}{2} \times 1 \times 20 = 10$ m
- Rectangle (1–2 s):
  $1 \times 20 = 20$ m
- Trapezium (2–3 s):
  $\frac{1}{2}(20 + 10) \times 1 = 15$ m
- Rectangle (3–4 s):
  $1 \times 10 = 10$ m

Total distance

$10 + 20 + 15 + 10 = 55$ m

10. Correct Option: C

Explanation:

Distance covered during the last second,

$S_n = \frac{g}{2}(2n - 1)$

Total height,

$H = \frac{1}{2}gn^2$

Given,

$S_n = 0.36H$

$\frac{g}{2}(2n - 1) = 0.36\left(\frac{1}{2}gn^2\right)$

$0.36n^2 - 2n + 1 = 0$

$n = 5$ s

Therefore,

$H = \frac{1}{2} \times 9.8 \times 5^2 = 122.5$ m

11. Correct Option: C

Explanation:

Using

$S = ut - \frac{1}{2}gt^2$

Given:

- $u = 14.7$ m/s
- $t = 5$ s

$S = 14.7(5) - \frac{1}{2}(9.8)(25)$

$S = 73.5 - 122.5 = -49$ m

The negative sign indicates that the body is 49 m below the point of projection.

Hence, the height of the tower is

$\boxed{49 \text{ m}}$

12. Correct Option: C

Explanation:

Distance covered in the n^{th}second,

$x = \frac{g}{2}(2n - 1)$

Distance covered in the $(n + 1)^{\text{th}}$second,

$y = \frac{g}{2}(2n + 1)$

Therefore,

$y - x = \frac{g}{2}[(2n + 1) - (2n - 1)] = g$

13. Correct Option: B

Explanation:

For the first distance h,

$v^2 = 2gh$

After falling through $2h$,

$v'^2 = 2g(2h) = 4gh = 2v^2$

Hence,

$\boxed{v' = \sqrt{2}\, v}$

14. Correct Option: B

Explanation:

Average velocity,

$v_{\text{avg}} = \frac{v}{2}$

Given,

$v_{\text{avg}} = 7$ m/s

Therefore,

$v = 14$ m/s

Using

$v^2 = 2gh$

$14^2 = 2(10)h$

$196 = 20h$

$h = 9.8$ m

15. Correct Option: A

Explanation:

The slope of a displacement-time graph represents velocity.

The graph starts with a steep positive slope that gradually decreases to zero.
Hence, the particle slows down continuously until it comes to rest.

16. Correct Option: D

Explanation:

- During flight, the ball has constant acceleration ($-g$), so the velocity decreases linearly with time.
- The velocity changes from positive → zero (highest point) → negative.
- In Graph D, the sloping straight line correctly represents this motion.
- The horizontal line after the negative velocity indicates that the motion has ended (the ball has hit the ground) and the graph is no longer describing the ball's flight. It is not indicating continued motion with constant negative velocity.

Therefore, the expected answer is D

17. Correct Option: A

Explanation:

Areas under the velocity-time graph:

- 0–2 s:
  $4 \times 2 = 8$ m
- 2–4 s:
  $-2 \times 2 = -4$ m
- 4–6 s:
  $2 \times 2 = 4$ m

Displacement,
$8 - 4 + 4 = 8$ m
Distance,
$8 + 4 + 4 = 16$ m

18. Correct Option: D

Explanation:

Acceleration is the slope of the velocity-time graph.
The steepest segment is between 30 sand 40 s.

$$a = \frac{\Delta v}{\Delta t} = \frac{80 - 20}{40 - 30} = \frac{60}{10} = 6 \text{ cm/s}^2$$

19. Correct Option: A

Explanation:

Uniform motion means the body moves with constant velocity.
Since acceleration is zero, the velocity-time graph is a horizontal straight line parallel to the time axis.

20. Correct Answer: D

The area under an acceleration–time (a–t) graph is obtained by multiplying acceleration by time.

$$\text{Area} = \left(\frac{\text{m}}{\text{s}^2}\right) \times (\text{s}) = \frac{\text{m}}{\text{s}}$$

The resulting unit, m/s, is the unit of velocity. Therefore, the area under an acceleration–time graph represents the change in velocity (Δv) of the object during that time interval.

Conclusion:

The area under an acceleration–time graph gives the change in velocity of the object over the given time period.

LEVEL-II

Single Correct Option Type

1. Correct Option: A

The time taken to fall freely from a height $h$is:

$$t = \sqrt{\frac{2h}{g}}$$

Hence,

$t \propto \sqrt{h}$

Therefore,

$$\frac{t_1}{t_2} = \sqrt{\frac{h}{2h}} = \frac{1}{\sqrt{2}}$$

Ratio = $1:\sqrt{2}$.

2. Correct Option: A

The maximum height reached by a vertically projected body is:

$$h = \frac{V_0^2}{2g}$$

Thus,

$V_0 \propto \sqrt{h}$

To make the maximum height three times,

$V' = \sqrt{3}\, V_0$

3. Correct Option: B

The distance covered during the last $t$seconds of ascent is equal to the distance covered during the first $t$seconds of free fall from rest from the highest point.

$$s = \frac{1}{2} g t^2$$

4. Correct Option: A

Displacement equals the area under the velocity–time graph.

Positive triangular area:

$\frac{1}{2} \times 5 \times 5 = 12.5$ m

Negative triangular area:

$\frac{1}{2} \times 5 \times (-5) = -12.5$ m

Net displacement:

$12.5 - 12.5 = 0$

Therefore,

Average velocity = 0 m s^{-1}

5. Correct Option: A

A packet dropped from an aircraft retains the aircraft's horizontal velocity:

$v_x = u$

Vertical velocity on reaching the ground:

$v_y = \sqrt{2gh}$

Hence, resultant velocity:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{u^2 + 2gh}$$

6. Correct Option: A

According to Galileo's law of odd numbers, the distances covered in successive equal intervals of time by a freely falling body are in the ratio:

$1:3:5:7:\cdots$

7. Correct Option: C

The balloon descends with a speed of $12\ \text{m s}^{-1}$. Hence, the stone's initial downward velocity is:

$u = 12\ \text{m s}^{-1}$

Using

$s = ut + \frac{1}{2}gt^2$

with $t = 10$ s,

$s = 12(10) + \frac{1}{2}(9.8)(10)^2$

$= 120 + 490 = 610$ m

8. Correct Option: B

Taking downward as positive,

Initial velocity:

$v_i = -u$

Final velocity:

$v_f = 3u$

Using

$v_f^2 - v_i^2 = 2gh$

$(3u)^2 - (-u)^2 = 2gh$

$9u^2 - u^2 = 2gh$

$8u^2 = 2gh$

$$\boxed{h = \frac{4u^2}{g}}$$

9. Correct Option: B

For a body falling freely from rest,

$v^2 = 2gh$

If the final velocity becomes $2v$,

$(2v)^2 = 2gH$

$4v^2 = 2gH$

Since

$v^2 = 2gh$,

$4(2gh) = 2gH$

$$\boxed{H = 4h}$$

10. Correct Option: B

Relative acceleration:

$a_{\text{rel}} = g - g = 0$

Relative velocity:

$u_{\text{rel}} = 50 - 0 = 50\ \text{m s}^{-1}$

Time taken to meet:

$$t = \frac{\text{Separation}}{\text{Relative velocity}} = \frac{100}{50} = 2\ \text{s}$$

Assertion and Reason Type

16. Correct Option: A

Both Assertion and Reason are true, and the Reason correctly explains the Assertion.

When a body moves upward, upward is taken as the positive direction. Since gravity acts downward, acceleration due to gravity is negative with respect to the chosen positive direction.

17. Correct Option: A

Both Assertion and Reason are true, and the Reason correctly explains the Assertion.

A vertically projected body has zero velocity at its highest point because gravity continuously reduces its upward velocity until it becomes zero.

18. Correct Option: D

The Assertion is false, but the Reason is true.

In vacuum, all bodies fall with the same acceleration because there is no air resistance. Also,

$g = \frac{GM}{R^2}$

which is independent of the mass of the falling body.

19. Correct Option: A

Both Assertion and Reason are true, and the Reason correctly explains the Assertion.

Earth exerts a gravitational force on every object toward its centre. Hence, the direction toward Earth's centre is called the vertical downward direction.

Comprehension Type (Passage–I)

20. Correct Option: B

As the body falls, its speed continuously increases due to gravity.

Therefore, it takes more time to cover the first 50 mthan the second 50 m.

Hence,

$\boxed{t_1 > t_2}$

21. Correct Option: A

Time to cover the first 50 m:

$t_1 = \sqrt{\frac{2(50)}{g}} = \sqrt{\frac{100}{g}}$

Total time to fall 100 m:

$t = \sqrt{\frac{2(100)}{g}} = \sqrt{2}\, t_1$

Therefore,

$t_2 = (\sqrt{2} - 1)t_1$

Hence,

$\frac{t_1}{t_2} = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 \approx 2.414$

Closest ratio:

$\boxed{5:2}$

22. Correct Option: B

Total time to hit the ground:

$t = \sqrt{2}\, t_1$

Therefore,

$\boxed{t : t_1 = \sqrt{2} : 1}$

Hence, the required ratio is:

$\boxed{\sqrt{2} : 1}$

23. Correct Answer: A

Let the maximum velocity attained be $v_{\max}$.

Since the acceleration is $5\ \text{m/s}^2$,

$v_{\max} = 5t$

As the magnitude of deceleration is equal to the acceleration, the deceleration also lasts for $t$seconds.

Hence, the total time of motion is

$T = 20 + t$

The total displacement is the area under the velocity–time graph (trapezium):

$S = \frac{1}{2} \times$ (sum of parallel sides) $\times$ (height)

$= \frac{1}{2} \times [(20 - t) + (20 + t)] \times 5t$

$= 100t$

Given that the average velocity is $20\ \text{m/s}$,

$v_{\text{avg}} = \frac{S}{T}$

$20 = \frac{100t}{20 + t}$

$400 + 20t = 100t$

$80t = 400$

$\boxed{t = 5\ \text{s}}$

Answer: A

24. Correct Answer: B

Using the value obtained in Q23,

$t = 5\ \text{s}$

The maximum velocity is

$v_{\max} = 5t$

$= 5 \times 5$

$= \boxed{25\ \text{m/s}}$

Answer: B

25. Correct Answer: A

The particle moves with uniform velocity from $t = 5$ sto 20 s.

Hence,

Time $= 20 - 5 = 15$ s

Distance travelled during uniform motion is

Distance $= v_{\max} \times$ time

$= 25 \times 15$

$= \boxed{375\ \text{m}}$

Answer: A

Passage – III (Q26 to Q28)

26. Correct Answer: C

For vertical motion,

$h = ut - \frac{1}{2}gt^2$

Rearranging,

$\frac{1}{2}gt^2 - ut + h = 0$

Let the two roots be $t_1$and t_2.

Using the product of roots,

$t_1 t_2 = \frac{h}{\frac{1}{2}g}$

Therefore,

$\boxed{h = \frac{1}{2}gt_1 t_2}$

27. Correct Answer: A

Using the same quadratic equation,

$\frac{1}{2}gt^2 - ut + h = 0$

The sum of the roots is

$t_1 + t_2 = \frac{-(-u)}{\frac{1}{2}g}$

$= \frac{2u}{g}$

Hence,

$\boxed{u = \frac{1}{2}g(t_1 + t_2)}$

28. Correct Answer: C

Maximum height reached is

$H = \frac{u^2}{2g}$

Substituting

$u = \frac{1}{2}g(t_1 + t_2)$

$H = \frac{\left[\frac{1}{2}g(t_1 + t_2)\right]^2}{2g}$

$= \frac{\frac{1}{4}g^2(t_1 + t_2)^2}{2g}$

$= \boxed{\frac{1}{8}g(t_1 + t_2)^2}$

Answer: C

Matrix Matching Type (Q29 & Q30)

29. (a) → S (1 s)

The time of flight is given by:

$T = \frac{2v_0}{g}$

Substituting v_0 = 4.9 m/sand $g = 9.8\,\mathrm{m/s^2}$:

$T = \frac{2 \times 4.9}{9.8} = 1$ s

(b) → P (4.9 m/s)

When the body returns to the point of projection, its speed is equal to its initial speed.

$v = v_0 = 4.9$ m/s

(c) → No Match (0 m)

Since the body is projected vertically upward, there is no horizontal displacement.

Range = 0 m

Hence, there is no matching option.

(d) → P (4.9 m/s)

The change in velocity in 0.5 sis:

$\Delta v = g \times t = 9.8 \times 0.5 = 4.9$ m/s

Answer:

- (a) → S
- (b) → P
- (c) → No Match
- (d) → P

30. Correct Matching:

Answer: a → Q, b → S, c → P, d → R

- a → Q (Maximum Height):

  The maximum height reached by a body projected vertically upward is

  $H = \frac{u^2}{2g}$.

  (The option incorrectly shows u^2 instead of $\frac{u^2}{2g}$.)

- b → S (Time of Flight):

  The total time of flight is

  $T = \frac{2u}{g}$.

  (The option incorrectly shows $\frac{2h}{g}$ instead of $\frac{2u}{g}$.)

- c → P (Velocity at Ground):

  When the body returns to the point of projection, its speed is equal to the initial speed:

  $v = u = \sqrt{2gh}$.

- d → R (Velocity at Highest Point):

  At the highest point of its motion, the body's instantaneous velocity is

  $\boldsymbol{v = 0}$.

Integer Type (Q31 to Q33)

31. Answer: 4

For a freely falling body,

$t = \sqrt{\frac{2h}{g}}$

Substituting,

$t = \sqrt{\frac{2 \times 80}{10}}$

$= \sqrt{16}$

$= \boxed{4\ \text{s}}$

32. Answer: 4

The total time of flight for a body projected vertically upward is

$T = \frac{2u}{g}$

Substituting,

$T = \frac{2 \times 20}{10}$

$= \boxed{4\ \text{s}}$

33. Answer: 5

The time taken to reach the maximum height is

$t = \frac{u}{g}$

Substituting,

$t = \frac{50}{10}$

$= \boxed{5\ \text{s}}$

PREVIOUS YEAR QUESTIONS

Question 1

Explanation:

Average velocity is defined as:

$\text{Average Velocity} = \frac{\text{Displacement}}{\text{Time}}$

A particle moves from A to B along a semicircular path of radius $R = 1.0\ \text{m}$. The displacement is the straight-line distance between A and B, which is the diameter:

Displacement $= 2R = 2 \times 1.0 = 2.0\ \text{m}$

Given that the time taken is 1.0 s,

$\text{Average Velocity} = \frac{2.0}{1.0} = 2.0\ \text{m/s}$

Question 2

Explanation:

The time taken to reach the highest point is

$t_1 = \frac{u}{g}$

If the total time taken to hit the ground is $n$times the upward time,

$t_2 = nt_1 = \frac{nu}{g}$

Using the second equation of motion for the downward displacement ($-H$),

$-H = ut_2 - \frac{1}{2}gt_2^2$

Substituting $t_2 = \frac{nu}{g}$,

$-H = u\left(\frac{nu}{g}\right) - \frac{1}{2}g\left(\frac{nu}{g}\right)^2$

Multiplying both sides by $2g$,

$2gH = nu^2(n-2)$

Hence,

$\boxed{2gH = nu^2(n-2)}$

Question 3

Explanation:

Given,

$v = \frac{dx}{dt} = b\sqrt{x}$

Separating the variables,

$\frac{dx}{\sqrt{x}} = b\,dt$

Integrating,

$\int_0^x \frac{dx}{\sqrt{x}} = \int_0^t b\,dt$

$2\sqrt{x} = bt$

or,

$\sqrt{x} = \frac{bt}{2}$

Substituting into the velocity equation,

$v = b\sqrt{x} = b\left(\frac{bt}{2}\right) = \frac{b^2 t}{2}$

Therefore, at $t = \tau$,

$\boxed{v = \frac{b^2\tau}{2}}$

Question 4

Explanation:

During upward motion, both gravity and air resistance act downward.

Hence,

$\frac{dv}{dt} = -(g + \gamma v^2)$

Separating variables,

$\int_{v_0}^{0} \frac{dv}{g + \gamma v^2} = -\int_0^T dt$

Evaluating,

$T = \frac{1}{\sqrt{\gamma g}} \tan^{-1}\left(\sqrt{\frac{\gamma}{g}}\, v_0\right)$

Thus,

$\boxed{T = \frac{1}{\sqrt{\gamma g}} \tan^{-1}\left(\sqrt{\frac{\gamma}{g}}\, v_0\right)}$

Question 5

Explanation:

At equilibrium,

- Horizontal forces:

  $qE = T\sin\theta$

- Vertical forces:

  $mg = T\cos\theta$

  Therefore,

  $\tan\theta = \frac{qE}{mg}$

Substituting the values,

$\tan\theta = \frac{(5\times10^{-6})(2000)}{(2\times10^{-3})(10)} = \frac{0.01}{0.02} = 0.5$

Hence,

$\boxed{\theta = \tan^{-1}(0.5)}$

Question 6

Explanation:

Using

$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$

Given

$\vec{r}_0 = 2\hat{\imath} + 4\hat{\jmath}, \vec{v}_0 = 5\hat{\imath} + 4\hat{\jmath}, \vec{a} = 4\hat{\imath} + 4\hat{\jmath}, t = 2\text{ s}$

Substituting,

$\vec{r}(2) = (2\hat{\imath} + 4\hat{\jmath}) + (5\hat{\imath} + 4\hat{\jmath})(2) + \frac{1}{2}(4\hat{\imath} + 4\hat{\jmath})(2)^2$

$\vec{r}(2) = 20\hat{\imath} + 20\hat{\jmath}$

The distance from the origin is

$|\vec{r}| = \sqrt{20^2 + 20^2} = 20\sqrt{2}\text{ m}$

Hence,

$\boxed{20\sqrt{2}\text{ m}}$

Question 7

Explanation:

For car A,

$v_A = \sqrt{2a_1 S}$

For car B,

$v_B = \sqrt{2a_2 S}$

Therefore,

$v = v_A - v_B = \sqrt{2S}(\sqrt{a_1} - \sqrt{a_2})$

The corresponding times are

$t_A = \sqrt{\frac{2S}{a_1}}, t_B = \sqrt{\frac{2S}{a_2}}$

Thus,

$t = t_B - t_A = \sqrt{2S}\left(\frac{\sqrt{a_1} - \sqrt{a_2}}{\sqrt{a_1 a_2}}\right)$

Dividing the two expressions,

$\boxed{v = \sqrt{a_1 a_2}\, t}$

Question 8

Explanation:

Initial velocity,

$\vec{v}_i = 5\hat{\imath}$

Final velocity,

$\vec{v}_f = 5\hat{\jmath}$

Hence,

$\Delta\vec{v} = \vec{v}_f - \vec{v}_i = -5\hat{\imath} + 5\hat{\jmath}$

which is directed towards the North-West.

Magnitude of average acceleration,

$a_{\text{avg}} = \frac{|\,\Delta\vec{v}\,|}{\Delta t} = \frac{\sqrt{(-5)^2 + 5^2}}{10} = \frac{5\sqrt{2}}{10} = \frac{1}{\sqrt{2}}\ \text{m/s}^2$

Therefore,

$\boxed{a_{\text{avg}} = \frac{1}{\sqrt{2}}\ \text{m/s}^2}$

Question 9

Explanation:

For a body starting from rest with constant acceleration,

$s_n = \frac{a}{2}(2n - 1)$

Similarly,

$s_{n+1} = \frac{a}{2}(2n + 1)$

Therefore,

$\frac{s_n}{s_{n+1}} = \frac{2n - 1}{2n + 1}$

Hence,

$\boxed{\frac{2n - 1}{2n + 1}}$

Question 10

Explanation:

Given,

$\vec{r}(t) = \cos\omega t\,\hat{\imath} + \sin\omega t\,\hat{\jmath}$

Differentiating,

$\vec{v}(t) = -\omega\sin\omega t\,\hat{\imath} + \omega\cos\omega t\,\hat{\jmath}$

The dot product,

$\vec{r} \cdot \vec{v} = 0$

Therefore, the velocity is always perpendicular to the position vector.

Differentiating once more,

$\vec{a}(t) = -\omega^2\cos\omega t\,\hat{\imath} - \omega^2\sin\omega t\,\hat{\jmath} = -\omega^2\vec{r}$

Hence, the acceleration is always directed towards the origin (centripetal acceleration).

$\boxed{\vec{a} = -\omega^2\vec{r}}$

4. Newton's Laws of Motion

Force

LEVEL-I

1. Answer: B) Unbalanced force

Explanation:

According to Newton's Second Law, $F_{\text{net}} = ma$. A body changes its state of motion only when the net force acting on it is not zero. Balanced forces cancel each other and cannot change the motion of an object. Therefore, only unbalanced forces produce acceleration.

2. Answer: A) Balanced forces

Explanation:

When a toothpaste tube is squeezed from opposite sides, equal and opposite forces act on it. These forces balance each other, so the tube does not move. However, the balanced forces deform the tube and change its shape.

3. Answer: A) Balanced force act on the balloon

Explanation:

When a balloon is pressed between two hands, equal and opposite forces act on it. Since the net force is zero, the balloon does not move. The balanced forces only change its shape.

4. Answer: D) All the above

Explanation:

An unbalanced force produces acceleration, which may:

- Increase or decrease the speed,
- Change the direction of motion, or
- Change the shape of an object.

Hence, all the given effects are possible.

5. Answer: B) Forward

Explanation:

When a moving car stops suddenly, the lower part of the passenger's body stops with the car, but the upper body continues moving due to inertia of motion. Therefore, the passenger jerks forward.

6. Answer: A) Backward

Explanation:

When a stationary vehicle starts suddenly, the lower body moves with the vehicle while the upper body tends to remain at rest due to inertia of rest. As a result, the passenger jerks backward.

7. Answer: C) Inertia of direction

Explanation:

Mud sticking to a rotating wheel moves in a circular path. When it detaches, the centripetal force disappears and the mud moves tangentially in a straight line due to inertia of direction.

8. Answer: C) Mass of the object

Explanation:

Inertia is the tendency of an object to resist changes in its state of motion. The amount of inertia depends only on the mass of the object. Greater mass means greater inertia.

9. Answer: A) Newton's First Law of Motion

Explanation:

When a tree branch is shaken, the branch moves suddenly while the fruits tend to remain at rest because of inertia. This illustrates Newton's First Law of Motion.

10. Answer: B) Move forward

Explanation:

When a moving tanker stops suddenly, the water inside continues moving forward due to inertia of motion, causing it to surge toward the front of the tanker.

11. Answer: D) Mass

Explanation:

Mass is the measure of inertia. An object with greater mass offers greater resistance to changes in its state of motion.

12. Answer: C) Inertia

Explanation:

Newton's First Law is also called the Law of Inertia because it explains the tendency of objects to resist changes in their state of rest or uniform motion.

13. Answer: A) Inertia of direction

Explanation:

When a bus takes a sharp turn, passengers tend to continue moving in their original straight-line path due to inertia of direction, making them appear to be thrown outward.

14. Answer: A) (i) is true & (ii) is false

Explanation:

A passenger jumping from a moving bus should run in the direction of the bus to gradually reduce forward momentum. Jumping without running may cause the person to fall forward due to inertia.

15. Answer: A) Inertia of rest

Explanation:

When a stationary vehicle starts suddenly, passengers tend to remain at rest due to inertia of rest, causing them to lean backward.

16. Answer: B) Force

Explanation:

Newton's Second Law states:

$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$

Thus, this law defines the quantitative relationship between force, mass, and acceleration.

17. Answer: C) Net force

Explanation:

According to Newton's Second Law,

$\vec{F}_{\text{net}} = m\vec{a}$

Since mass is always positive, the acceleration is always in the same direction as the net force.

18. Answer: A) $2mv$

Explanation:

Initial momentum:

$p_i = mv$

Final momentum after rebounding:

$p_f = -mv$

Change in momentum:

$\Delta p = p_f - p_i = (-mv) - mv = -2mv$

Magnitude of change:

$|\Delta p| = 2mv$

19. Answer: B) 10 N

Explanation:

Using

$s = ut + \frac{1}{2}at^2$

Given:

- $u = 0$
- $s = 10$ m
- $t = 2$ s

$10 = \frac{1}{2}a(2)^2$

$10 = 2a$

$a = 5\ \text{m/s}^2$

Using Newton's Second Law:

$F = ma$

$F = 2 \times 5 = 10$ N

20. Answer: $\boxed{\vec{s} = 78125\hat{\imath} + 39062.5\hat{\jmath}\ \text{m}}$

Explanation:

Given:

- $m = 10$ g $= 0.01$ kg
- $\vec{F} = 10\hat{\imath} + 5\hat{\jmath}$

Acceleration:

$\vec{a} = \frac{\vec{F}}{m} = \frac{10\hat{\imath} + 5\hat{\jmath}}{0.01} = 1000\hat{\imath} + 500\hat{\jmath}\ \text{m/s}^2$

Position after $t = 12.5$ s:

$\vec{s} = \frac{1}{2}\vec{a}t^2$

$\vec{s} = \frac{1}{2}(1000\hat{\imath} + 500\hat{\jmath})(12.5)^2$

$\boxed{\vec{s} = 78125\hat{\imath} + 39062.5\hat{\jmath}\ \text{m}}$

21. Answer: B) 6250 N

Explanation:

Given:

- $u = 250$ m/s
- $v = 0$
- $s = 5$ cm $= 0.05$ m
- $m = 10$ g $= 0.01$ kg

Using

$v^2 - u^2 = 2as$

$0 - 250^2 = 2a(0.05)$

$-62500 = 0.1a$

$a = -625000\ \text{m/s}^2$

Force:

$F = m \mid a \mid$

$F = 0.01 \times 625000$

$\boxed{F = 6250\ \text{N}}$

22. Answer: D) 180°

Explanation:

According to Newton's Third Law, action and reaction forces are equal in magnitude and opposite in direction. Since they act along the same line but in opposite directions, the angle between them is:

$\boxed{180°}$

LEVEL-II

1. Answer: C

Explanation:

When a train stops suddenly, the lower part of a passenger's body, which is in contact with the seat, comes to rest along with the train. However, the upper part of the body tends to continue moving due to inertia of motion, causing the passenger to lurch forward.

2. Answer: C

Explanation:

When a hanging carpet is beaten with a stick, the carpet is set into sudden motion. The dust particles tend to remain at rest due to inertia of rest, causing them to separate from the carpet and fall off.

3. Answer: D

Explanation:

Given:

- Mass of each ball, $m = 1\,\text{g} = 10^{-3}\,\text{kg}$
- Speed, $v = 100\,\text{m/s}$
- Number of balls striking per second = 10,000
- Area $= 1\,\text{cm}^2 = 10^{-4}\,\text{m}^2$

For a perfectly elastic collision,

$\Delta p = 2mv$

$= 2 \times 10^{-3} \times 100 = 0.2\,\text{kg m/s}$

Force,

$F = (10{,}000) \times 0.2 = 2000\,\text{N}$

Pressure,

$P = \frac{F}{A} = \frac{2000}{10^{-4}} = 2 \times 10^7\,\text{N/m}^2$

Therefore,

$\boxed{P = 2 \times 10^7\,\text{N/m}^2}$

4. Answer: A

Explanation:

Using Newton's Second Law,

$F = \frac{m(v-u)}{t}$

Given:

- $m = 100\,\text{kg}$
- $u = 5\,\text{m/s}$
- $v = 0$
- $t = 0.1\,\text{s}$

$F = \frac{100(0-5)}{0.1} = -5000\,\text{N}$

The negative sign indicates the force acts opposite to the direction of motion.

$\boxed{\text{Magnitude of force} = 5000\,\text{N}}$

5. Answer: B

Explanation:

For a perfectly elastic collision, the change in momentum of each ball is

$\Delta p = 2mu$

If $n$balls strike the surface every second, then

$F = n\Delta p = n(2mu)$

Hence,

$\boxed{F = 2mnu}$

6. Answer: D

Explanation:

Velocity of a body falling freely from height H:

$v = \sqrt{2gH}$

For the first body:

$v_1 = \sqrt{2g(2h)} = 2\sqrt{gh}$

For the second body:

$v_2 = \sqrt{2gh}$

Momentum ratio,

$\frac{p_1}{p_2} = \frac{m_1 v_1}{m_2 v_2} = \frac{m_1(2\sqrt{gh})}{m_2(\sqrt{2gh})} = \frac{\sqrt{2}\,m_1}{m_2}$

Therefore,

$\boxed{\frac{p_1}{p_2} = \frac{\sqrt{2}\,m_1}{m_2}}$

7. Answer: C

Explanation:

Using Newton's Second Law,

$m_1 = \frac{F}{a_1}, m_2 = \frac{F}{a_2}$

Combined mass,

$M = m_1 + m_2 = F\left(\frac{1}{a_1} + \frac{1}{a_2}\right) = F\left(\frac{a_1 + a_2}{a_1 a_2}\right)$

Acceleration of the combined system,

$a = \frac{F}{M} = \frac{a_1 a_2}{a_1 + a_2}$

Hence,

$\boxed{a = \frac{a_1 a_2}{a_1 + a_2}}$

8. Answer: A

Explanation:

By the Law of Conservation of Linear Momentum,

$m_{\text{bullet}} v_{\text{bullet}} + M_{\text{gun}} V_{\text{gun}} = 0$

Given,

- Bullet mass = $0.02\,\text{kg}$
- Bullet speed = $200\,\text{m/s}$

- Gun mass = 1 kg
  $(0.02)(200) + (1)V = 0$
  $4 + V = 0$
  $V = -4\ \text{m/s}$
  The negative sign indicates recoil in the opposite direction.
  $\boxed{\text{Recoil speed} = 4\ \text{m/s}}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

9. Correct Options: A, B, C

Explanation:

- A. Correct: According to Newton's Second Law, force is the rate of change of momentum.
  $F = \frac{\Delta p}{\Delta t}$
  If a body is brought to rest in 1 s, then
  $\Delta t = 1$ s
  Hence,
  $F = \Delta p$
  Therefore, the numerical value of force equals the change in momentum.
- B. Correct: Momentum is defined as the product of mass and velocity.
  $\vec{p} = m\vec{v}$
  - SI unit: kg·m/s
  - CGS unit: g·cm/s
- C. Correct: Linear momentum is historically defined as the quantity of motion possessed by a moving body.
- D. Incorrect: Momentum depends on velocity, which is a vector quantity. Therefore, momentum is also a vector, not a scalar.

10. Correct Options: A, B, C, D

Explanation:

All the given options are valid units of force.

- A. Newton (N): Absolute SI unit of force.
- B. Dyne: Absolute CGS unit of force.
  1 N = 10^5 dyne
- C. Kilogram-force (kg-f): Gravitational SI unit of force.
  1 kg-f ≈ 9.8 N
- D. Gram-force (gm-f): Gravitational CGS unit of force.
  1 gm-f ≈ 980 dyne

Hence, all four options are correct.

11. Correct Options: A, B, D

Explanation:

- A. Correct: When a bullet is fired forward, the gun experiences an equal and opposite force backward, causing recoil. This illustrates Newton's Third Law of Motion.
- B. Correct: A swimmer pushes water backward, and the water pushes the swimmer forward with an equal and opposite force.
- C. Incorrect: Action and reaction forces do not act on the same body. They always act simultaneously on two different interacting bodies.
- D. Correct: Every force in nature exists as part of an action–reaction pair. Therefore, forces always occur in pairs.

12. Correct Options: A, D

Explanation:

A 5 kg block is in equilibrium. Therefore, the net force along both the horizontal and vertical directions is zero.

Horizontal Equilibrium

$T_1 \cos 30° = T_2 \cos 60°$

Substituting the trigonometric values,

$T_1 \left(\frac{\sqrt{3}}{2}\right) = T_2 \left(\frac{1}{2}\right)$

$T_2 = \sqrt{3}\, T_1$

Vertical Equilibrium

$T_1 \sin 30° + T_2 \sin 60° = mg$

Taking $g = 10\ \text{m/s}^2$,

$mg = 5 \times 10 = 50$ N

Substituting the values,

$T_1 \left(\frac{1}{2}\right) + T_2 \left(\frac{\sqrt{3}}{2}\right) = 50$

Replacing T_2 with $\sqrt{3}\, T_1$,

$\frac{T_1}{2} + \frac{(\sqrt{3}\, T_1)\sqrt{3}}{2} = 50$

$\frac{T_1 + 3T_1}{2} = 50$

$2T_1 = 50$

$T_1 = 25$ N

Now,

$T_2 = \sqrt{3} \times 25 = 25\sqrt{3}$ N

Therefore,

- A. $T_1 = 25$ N✓
- D. $T_2 = 25\sqrt{3}$ N✓

Hence, the correct options are A and D.

Assertion and Reason

13. Answer: A

Explanation:

- Assertion: Balanced forces produce zero net force. (True)
- Reason: According to Newton's First Law, an object changes its state of rest or uniform motion only when acted upon by a non-zero net external force. (True)

The reason correctly explains the assertion.

14. Answer: A

Explanation:

- Assertion: A passenger falls forward when a moving bus stops suddenly. (True)
- Reason: The upper part of the body continues moving due to inertia of motion. (True)

The reason correctly explains the assertion.

15. Answer: D

Explanation:

- Assertion: Magnetic force is a contact force. (False)
- Reason: Magnets exert force without physical contact. (True)

Since the assertion is false but the reason is true, the correct answer is D.

16. Answer: A

Explanation:

- Assertion: Momentum depends on both mass and velocity. (True)
- Reason: Momentum is defined as

  $p = mv$

  where $m$is mass and $v$is velocity.

  Since momentum is the product of mass and velocity, the reason correctly explains the assertion.

Passage - I

17. Correct Option: B

Explanation:

Using Newton's Second Law,

$F = ma \Rightarrow m = \frac{F}{a}$

$m_1 = \frac{5}{10} = 0.5 \text{ kg}$

Hence, the mass of the first body is 0.50 kg.

18. Correct Option: C

Explanation:

Using Newton's Second Law,

$m_2 = \frac{F}{a} = \frac{5}{20} = 0.25 \text{ kg}$

Hence, the mass of the second body is 0.25 kg.

19. Correct Option: D

Explanation:

Combined mass,

$M = m_1 + m_2 = 0.5 + 0.25 = 0.75 \text{ kg}$

Applying Newton's Second Law,

$a = \frac{F}{M} = \frac{5}{0.75} = 6.67 \text{ m/s}^2$

Therefore, the acceleration of the combined system is 6.67 m/s².

Passage - II

20. Correct Option: A

Explanation:

Balancing the vertical forces,

$T_a \cos 60° = mg$

Since,

$\cos 60° = \frac{1}{2}$

$T_a \left(\frac{1}{2}\right) = mg$

$T_a = 2mg$

Hence, the tension in string a is $2mg$.

21. Correct Option: B

Explanation:

Balancing the horizontal forces,

$T_b = T_a \sin 60°$

Substituting $T_a = 2mg$,

$T_b = 2mg\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}\,mg$

Therefore, the tension in string b is $\sqrt{3}\,mg$.

22. Correct Option: B

Explanation:

After string a is burnt, the ball swings like a pendulum of length ℓ.

The vertical fall of the ball is

$h = \ell$

Using conservation of mechanical energy,

$mgh = \frac{1}{2}mv^2$

$mg\ell = \frac{1}{2}mv^2$

$v = \sqrt{2g\ell}$

Hence, the speed of the ball just before collision is

$\boxed{\sqrt{2g\ell}}$

Matrix Matching Type

23. Correct Matching

- **a → S**
- **b → R**
- **c → Q**
- **d → P**

Explanation

- Inertia → Measurement of mass: Mass is the quantitative measure of inertia.
- Linear momentum → Quantity of motion: Momentum represents the quantity of motion possessed by a body.
- Newton's Second Law → Rate of change of momentum: $F = \frac{dp}{dt}$.
- Newton's Third Law → Action = − Reaction: Every action has an equal and opposite reaction.

24. Correct Matching

- **a → R**
- **b → R**
- **c → P**
- **d → Q**

Explanation

Given,

$p = mv = 24$ kg.m/s

Twice mass, same speed:

$2mv = 48$ kg.m/s

- Same mass, twice speed:

$m(2v) = 48$ kg.m/s

- Half mass, twice speed:

$\frac{m}{2} \times 2v = 24$ kg\cdotpm/s

- Same mass, half speed:

$m\left(\frac{v}{2}\right) = 12$ kg\cdotpm/s

Integer Type Questions

25. Answer: 2

Explanation:

Acceleration,

$a = \frac{F}{m} = \frac{100}{5} = 20 \text{ cm/s}^2$

Velocity after 10 s,

$v = at = 20 \times 10 = 200 \text{ cm/s}$

Since,

$200 = 2 \times 100$

the required integer is

$\boxed{2}$

26. Answer: 20

Explanation:

Impulse,

$J = F \times t$

$J = 10 \times 2 = 20 \text{ N}\backslash\text{cdotps}$

Hence, the required integer is

$\boxed{20}$

27. Answer: 20

Explanation:

Using Hooke's Law,

$F = kx$

$F = 200 \times 0.10 = 20 \text{ N}$

Hence, the required integer is

$\boxed{20}$

TOPIC-2: Forces On a Rigid Body or Turning Forces

LEVEL-I

1. Answer: A

Explanation:

The moment of a force (or torque) is the turning effect produced by a force about a fixed point or axis. It is equal to the product of the magnitude of the force and the perpendicular distance of its line of action from the axis of rotation.

$\tau = F \times d_{\perp}$

where:

- F= Force
- $d_{\perp}$= Perpendicular distance from the pivot

2. Answer: C

Explanation:

Since,

Moment of Force = Force × Distance

SI unit:

$\boxed{\text{Newton} \times \text{metre} = \text{N m}}$

3. Answer: A

Explanation:

The SI unit of force is Newton (N), while the CGS unit is dyne.

$\boxed{1 \text{ N} = 10^5 \text{ dynes}}$

4. Answer: B

Explanation:

Given:

- Force $F = 50$ N
- Moment $M = 10$ N m

Using,

$M = F \times d$

$d = \frac{M}{F} = \frac{10}{50} = 0.2$ m

$\boxed{d = 0.2 \text{ m}}$

5. Answer: A

Explanation:

Given:

- Force $F = 80$ N
- Distance $d = 1.75$ m

$M = F \times d$

$M = 80 \times 1.75 = 140$ N m

$\boxed{140 \text{ N m}}$

6. Answer: A

Explanation:

According to the Principle of Moments, when a body is in rotational equilibrium,

$\boxed{\text{Clockwise moments} = \text{Anticlockwise moments}}$

or equivalently,

$\boxed{\text{Algebraic sum of all moments} = 0}$

7. Answer: B

Explanation:

Given:

- Moment $M = 1575$ dyne-cm
- Distance $d = 45$ cm

$F = \frac{M}{d}$

$F = \frac{1575}{45} = 35$ dynes

$\boxed{35 \text{ dynes}}$

8. Answer: A

Explanation:

A couple consists of two equal, opposite, and parallel forces acting along different lines of action. Its turning effect is known as the moment of a couple.

$\boxed{\text{Moment of a couple}}$

9. Answer: $\boxed{d = 0.206 \text{ m}}$

Explanation:

Given:

- Force $F = 125$ N
- Moment $M = 25.75$ N m

$d = \frac{M}{F}$

$d = \frac{25.75}{125} = 0.206$ m

$\boxed{d = 0.206 \text{ m}}$

10. Answer: B

Explanation:

Two equal and opposite parallel forces acting along different lines of action form a

$\boxed{\text{Couple}}$

A couple produces rotation without causing translation.

11. Answer: B

Explanation:

The centre of gravity of the uniform scale is at 25 cm.

Taking moments about the 20 cm mark,

Anticlockwise moment:

$100 \times (20 - 10) = 100 \times 10 = 1000$ gf-cm

Clockwise moments:

$50 \times (30 - 20) + W \times (25 - 20)$

$1000 = 50 \times 10 + 5W$

$1000 = 500 + 5W$

$5W = 500$

$W = 100$ gf

$\boxed{100 \text{ gf}}$

12. Answer: A

Explanation:

Given:

- Force $F = 525$ N
- Moment $M = 420$ N m

$d = \frac{420}{525} = 0.8$ m

$\boxed{0.8 \text{ m}}$

13. Answer: B

Explanation:

From the diagram,

- $F_1 = 3$ N
- Distance from pivot = 3 m

Since the force acts downward on the left side of the pivot, it produces an anticlockwise moment.

$M = F \times d$

$M = 3 \times 3 = 9$ N m

$\boxed{9 \text{ N m (Anticlockwise)}}$

14. Answer: A

Explanation:

From the diagram,

- $F_2 = 3$ N
- Distance from pivot = 12 m

The force acts downward on the left side of the pivot, producing an anticlockwise moment.

$M = 3 \times 12 = 36$ N m

$\boxed{36 \text{ N m (Anticlockwise)}}$

15. Answer: B

Explanation:

Given:

- Force $F = 3$ N
- Perpendicular distance $d = 22$ m

$M = F \times d$

$M = 3 \times 22 = 66$ N m

66 N m

16. Answer: A

Explanation:

From the diagram,

- $F_1 = 12$ N
- Distance from pivot = 4 m

The force acts downward on the left side of the pivot, producing an anticlockwise moment.

$M = F \times d$

$M = 12 \times 4 = 48$ N m

48 N m (Anticlockwise)

Question 17

Explanation:

The clockwise moment is produced by the downward forces F_3 and F_2.

Clockwise Moment $= (\mathbf{22 \times 3}) + (\mathbf{5 \times 6})$

$= \mathbf{66 + 30}$

$= \mathbf{96\ N \cdot m}$

Answer: $\mathbf{96\ N \cdot m}$

18. Answer: $\mathbf{220\ N \cdot m}$

Explanation:

The force $F_2 = 11$ Nacts at a perpendicular distance of 20 m from the pivot.

Using the moment formula,

Moment $= F \times d$

$= 11 \times 20 = 220\ \text{N} \cdot \text{m}$

19. Answer: $\mathbf{36\ N \cdot m}$

Explanation:

The anticlockwise moment is produced by the force F_1.

Given,

$F_1 = 3$ N, $d = 12$ m

Using the moment formula,

Anticlockwise Moment $= 3 \times 12 = 36\ \text{N} \cdot \text{m}$

LEVEL-II

1. Answer: *A*) $\mathbf{3.08\ m}$

Explanation:

The see-saw is balanced about its center (pivot). Therefore, the clockwise moment must be equal to the anticlockwise moment.

Moment produced by the two boys:

Total Moment $= (30\ \text{kgf} \times 1.5\ \text{m}) + (40\ \text{kgf} \times 3.5\ \text{m})$

$= 45 + 140$

$= 185\ \text{kgf} \cdot \text{m}$

Let the woman sit at a distance $d$from the pivot.
Using the principle of moments,
$60 \text{ kgf} \times d = 185 \text{ kgf} \cdot \text{m}$
$d = \frac{185}{60} \approx 3.08 \text{ m}$

2. Answer:D) $\boxed{\mathbf{225\ g}}$

Explanation:
A uniform metre scale has its centre of gravity at the 50 cm mark. The scale is pivoted at the 55 cm mark, so the centre of gravity is 5 cm to the left of the pivot. To balance the scale, a 25 g mass is placed at the 100 cm mark, which is 45 cm from the pivot.
Applying the principle of moments,
Mass of scale $\times 5 = 25 \times 45$
Mass of scale $= \frac{25 \times 45}{5} = \frac{1125}{5} = 225 \text{ g}$

3. Answer:A) $\boxed{\mathbf{2\ N \cdot m}}$

Explanation:
A force of 2 N acts upward at one end of the rod, while another 2 N force acts downward at the opposite end. These forces form a couple, producing rotation in the same direction.
The distance of each force from the central pivot is:
0.5 m
Hence,
Resultant Moment $= (2 \times 0.5) + (2 \times 0.5)$
$= 1 + 1$
$= 2 \text{ N} \cdot \text{m}$

4. Answer:B) $\boxed{\mathbf{77.5\ cm}}$

Explanation:
The metre scale weighs 50 gf, and its weight acts through the 50 cm mark.
Since the pivot is at the 40 cm mark, the weight of the scale produces a clockwise moment:
$50 \times 10 = 500 \text{ gf} \cdot \text{cm}$
A 100 gf weight placed at the 5 cm mark produces an anticlockwise moment:
$100 \times 35 = 3500 \text{ gf} \cdot \text{cm}$
Let the 80 gf weight be placed at a distance $x$to the right of the pivot.
Using the principle of moments,
$3500 = 500 + 80x$
$3000 = 80x$
$x = 37.5 \text{ cm}$
Therefore, its position on the scale is
$40 + 37.5 = 77.5 \text{ cm}$

5. Answer: $\boxed{\mathbf{0\ N \cdot m}}$

Explanation:
The 20 N force acts exactly through the pivot. Therefore, its perpendicular distance from the pivot is zero.
Moment of force,
Moment $= F \times d$
$= 20 \times 0 = 0 \text{ N} \cdot \text{m}$
Hence, the force produces no turning effect.

Question 6.

Answer: $\boxed{\mathbf{66\ N \cdot m}}$

Explanation:

The moment due to force $F_3$is calculated using:

Moment = $F \times d$

Given,

$F_3 = 22$ N, $d = 3$ m

Moment = $22 \times 3 = 66$ N · m

Q5 (Correct Option: C)

Explanation:

The moment of a force is given by:

τ = Force × Perpendicular Distance

The 20 Nforce acts through the pivot, so its perpendicular distance from the pivot is zero.

$\tau = 20 \times 0 = 0$ N · m

Hence, the force produces no turning effect (zero moment).

Q6 (Correct Option: B)

Explanation:

The force $F_3 = 22$ Nacts at a perpendicular distance of 3 mto the right of the pivot.

$\tau = F \times d = 22 \times 3 = 66$ N · m

Since the force acts downward on the right side, it produces a clockwise moment.

Moment = 66 N · m(Clockwise).

Q7 (Correct Option: A)

Explanation:

The 6 Nforce acts at a perpendicular distance of 70 cm.

$\tau = F \times d = 6 \times 70 = 420$ N · cm

Therefore, the moment of the force is:

$\boxed{420\ \text{N} \cdot \text{cm}}$

Q8 (Correct Option: B)

Explanation:

Clockwise moments:

$(22 \times 3) + (5 \times 6) = 66 + 30 = 96$ N · m

Anticlockwise moment:

$3 \times 12 = 36$ N · m

Resultant moment:

$96 - 36 = 60$ N · m

Since the clockwise moment is greater, the net moment is:

$\boxed{60\ \text{N} \cdot \text{m (Clockwise)}}$

Q12 (Correct Option: B)

Explanation:

The Assertion is true because a rigid body is ideally defined as a body that does not undergo deformation when a force is applied.

The Reason is also true because most solid objects behave approximately as rigid bodies under ordinary conditions.

However, the Reason only gives a practical example and does not explain why a rigid body is defined as non-deformable.

Therefore, both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.

Q13 (Correct Option: A)

Explanation:

The Principle of Transmissibility of Forces states that a force acting on a rigid body may be shifted to any other point along its line of action without changing the body's external motion or equilibrium.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Q14 (Correct Option: D)

Explanation:

The Assertion is false because applying a force near the hinges produces a smaller turning effect due to the shorter perpendicular distance from the pivot.

The Reason is true because the turning effect (moment of force) is given by:

$\tau = F \times d$

Thus, increasing the perpendicular distance from the pivot increases the moment of force.

Therefore, the Assertion is false, but the Reason is true.

Q15 (Correct Option: A)

Explanation:

A body remains in rotational equilibrium when the total clockwise moment equals the total anticlockwise moment.

Mathematically,

$\sum \tau_{\text{clockwise}} = \sum \tau_{\text{anticlockwise}}$

or equivalently,

$\sum \tau = 0$

Hence, the algebraic sum of all moments acting on the body is zero.

Q16 (Correct Option: B)

Explanation:

Given:

- Force = 50 dynes
- Perpendicular distance = 75 cm

Using

$\tau = F \times d$

$\tau = 50 \times 75 = 3750 \text{ dyne} \cdot \text{cm}$

Therefore,

$\boxed{3750 \text{ dyne} \cdot \text{cm}}$

Q17 Correct Option: $\boxed{1.4 \text{ N} \cdot \text{m}}$

Explanation:

Given:

- Force = 80 N
- Perpendicular distance = 1.75 cm = 0.0175 m

Using

$\tau = F \times d$

$\tau = 80 \times 0.0175 = 1.4 \text{ N} \cdot \text{m}$

The correct answer is:

$\boxed{1.4 \text{ N} \cdot \text{m}}$

Q18 (Correct Option: C)

Explanation:

Given:

- Force = 40 dynes
- Distance = 60 cm

Using

$\tau = F \times d$

$\tau = 40 \times 60 = 2400 \text{ dyne} \cdot \text{cm}$

Hence,

$\boxed{2400 \text{ dyne} \cdot \text{cm}}$

Q19 (Correct Option: A)

Explanation:

For equilibrium, clockwise moment equals anticlockwise moment.

$W \times 20 = 100 \times 30$

$20W = 3000$

$W = \frac{3000}{20} = 150 \text{ N}$

Therefore,

$\boxed{W = 150 \text{ N}}$

Q20 (Correct Option: A)

Explanation:

Moment of force is given by

$\tau = F \times d$

Given:

- Force = 100 N
- Distance = 30 cm

$\tau = 100 \times 30 = 3000 \text{ N} \cdot \text{cm}$

Hence,

$\boxed{3000 \text{ N} \cdot \text{cm}}$

21 (Correct Option: B)

Explanation:

Initially,

Clockwise Moment = $100 \times 30 = 3000 \text{ N} \cdot \text{cm}$

If the distance is doubled,

New Clockwise Moment = $100 \times 60 = 6000 \text{ N} \cdot \text{cm}$

The anticlockwise moment remains unchanged at $3000 \text{ N} \cdot \text{cm}$.

Since

$6000 > 3000$,

the clockwise moment becomes greater, causing the ruler to rotate clockwise.

Therefore, the correct answer is:

$\boxed{\text{The ruler will rotate clockwise.}}$

Question 22 — Match the Following

Correct Matching:

- a → R ($1 \text{ N} = 10^5$ dynes)

  Explanation: This is the standard conversion factor between the SI and CGS units of force.
- b → P (Moment of force = $F \times$ Perpendicular distance)

  Explanation: The moment of a force (torque) is defined as the product of the applied

force and the perpendicular distance of its line of action from the pivot or axis of rotation.

- c → S (CGS unit of moment of force = dyne-cm)
  Explanation: In the CGS system, force is measured in dynes and distance in centimeters. Hence, the CGS unit of moment of force is dyne-cm.
- d → Q (Arm of a couple = Perpendicular distance)
  Explanation: The arm of a couple is the perpendicular distance between the lines of action of the two equal and opposite parallel forces.

Question 23 — Match the Following

Correct Matching:

- a → 3 (Point of Action)
  Explanation: The point of action is the exact point on a body where the force is applied.
- b → 2 (Line of Action)
  Explanation: The line of action is an imaginary straight line extending in the direction along which the force acts.
- c → 1 (Principle of Transmissibility)
  Explanation: According to the principle of transmissibility, the external effect of a force on a rigid body remains unchanged if the force is moved anywhere along its line of action.
- d → 4 (Rigid Body)
  Explanation: A rigid body is an ideal body in which the distance between any two points remains constant, even when external forces act on it.

Question 24. Answer: **66.67 gf**

A uniform metre scale has its centre of gravity at the 50 cm mark. It is pivoted at the 20 cm mark.

The weight of the scale (W)acts downward at the 50 cm mark, producing a clockwise moment.

Clockwise moment = $W \times (50 - 20)$

$= W \times 30$ cm

A 100 gf weight is suspended at the 0 cm mark, producing an anticlockwise moment.

Anticlockwise moment = $100 \times (20 - 0)$

$= 100 \times 20$

By the Principle of Moments,

Clockwise moment = Anticlockwise moment

$W \times 30 = 100 \times 20$

$W = \frac{2000}{30} = 66.67$ gf

Question 25. Answer: **77.5 cm mark**

A metre scale of weight 50 gf is pivoted at the 40 cm mark.

The weight of the scale acts at its centre of gravity (50 cm mark), producing a clockwise moment.

Clockwise moment = $50 \times (50 - 40)$

$= 50 \times 10 = 500$ gf $\cdot$ cm

A 100 gf weight is suspended at the 5 cm mark, producing an anticlockwise moment.

Anticlockwise moment = $100 \times (40 - 5)$

$= 100 \times 35 = 3500 \text{ gf} \cdot \text{cm}$

Hence, the excess anticlockwise moment is

$3500 - 500 = 3000 \text{ gf} \cdot \text{cm}$

Let the 80 gf weight be placed at a distance $x$to the right of the pivot.

$80x = 3000$

$x = \frac{3000}{80} = 37.5 \text{ cm}$

Therefore, the required position on the metre scale is

$40 + 37.5 = 77.5 \text{ cm}$

26. Answer: 3 m

The moment of a couple is given by

Moment of Couple = $F \times d$

where

- F= magnitude of one of the forces,
- d= perpendicular distance (arm of the couple) between the two forces.

Given,

Moment = $12 \text{ N} \cdot \text{m}$

$F = 4 \text{ N}$

Substituting,

$12 = 4 \times d$

$d = \frac{12}{4} = 3 \text{ m}$

5. Friction

1. Answer: C) Frictional force

Explanation: Frictional force is the tangential resistance that acts between contacting surfaces to oppose relative motion or the tendency of motion between them.

2. Answer: B) Static friction

Explanation: Static friction acts between two surfaces that are at rest relative to each other, perfectly balancing any applied force up to a certain limit to prevent sliding. (Note: "Stationary friction" in option A is linguistically synonymous but not the standard scientific term.)

3. Answer: A) Friction

Explanation: As the ball rolls, rolling friction acts in the direction opposite to its motion, continuously taking away kinetic energy until the ball comes to a complete stop.

4. Answer: C) Smoothness

Explanation: Friction depends on the roughness or smoothness of the contacting surfaces because microscopic irregularities interlock with each other to create the resistive force.

5. Answer: B) Rolling friction

Explanation: During normal driving without skidding, the surface of the tyre rolls over the road rather than sliding across it, which involves rolling friction.

6. Answer: C) Newton

Explanation: Frictional force is fundamentally a force, and the standard International System (SI) unit for all types of force is the Newton (N).

7. Answer: D) When sliding

Explanation: Sliding friction is significantly larger than rolling friction because sliding requires continuously breaking and reforming microscopic bonds between the two surfaces over a flat contact area.

8. Answer: D) No Units

Explanation: The coefficient of friction (μ) is calculated as the ratio of frictional force to the normal reaction force:

$$\mu = \frac{f}{N}$$

Since it is a ratio of two identical units (Newtons/Newtons), the units cancel out, making it dimensionless.

9.

Answer: D) Both (B) & (C)

Explanation: If the applied force exceeds the maximum static limit, the box will begin to move (Statement B). If the applied force is below this threshold, the static friction counteracts it entirely, and the box remains stationary (Statement C).

10. Answer: B) Sliding friction

Explanation: Without rollers or wheels, the flat bottom surface of the suitcase directly scrapes and slides across the floor, producing sliding (kinetic) friction.

11. Answer: A) Limiting friction

Explanation: Limiting friction is specifically defined as the maximum value of static friction acting when a body is on the verge of motion.

12. Answer: A) 15 N

Explanation: The minimum force required equals the limiting friction.

$f = \mu N = \mu mg$

$f = 0.3 \times 5 \times 10 = 15\ \text{N}$

13. Answer: B) 0.40

Explanation: Using

$\mu = \frac{F}{N}$

$\mu = \frac{200}{500} = 0.40$

14. Answer: D) Both B and C

Explanation: Since the applied force (10 N) is less than the limiting friction (15 N), the body does not move (Statement B). However, it has a tendency to move, which is opposed by static friction (Statement C).

15. Answer: B) Dynamic friction

Explanation: Dynamic friction (also called kinetic friction) acts between two surfaces that are moving relative to each other.

16. Answer: C) Remains unaffected due to a change in velocity

Explanation: Limiting friction is the maximum value of static friction that acts just before an object starts moving. It is given by:

$F_{\text{limiting}} = \mu_s N$

where:

- μ_s= coefficient of static friction
- N= normal reaction

The force of limiting friction does not depend on the velocity of the object because it acts before motion begins. It depends only on the nature of the surfaces in contact and the normal force.

17. Answer: B) Unchanged

Explanation: The coefficient of friction (μ) is a property of the contacting surfaces and remains unchanged when the normal reaction changes.

18. Answer: C) $\mu_s = \tan\theta$

Explanation: The angle of friction (θ) is related to the coefficient of static friction by

$\mu_s = \tan\theta$

19. Answer: A) $\mu_s N$

Explanation: At the point of impending motion, static friction reaches its maximum value:

$f_s = \mu_s N$

20. Answer: C) Rubber provides more friction than leather

Explanation: Rubber has a higher coefficient of friction than leather, giving athletes better grip and reducing the chances of slipping.

21. Correct Answer: C

Explanation:

When an object is pulled at an upward angle, the upward component of the applied force reduces the normal reaction acting on the surface.

$N = mg - F\sin\theta$

Since friction is directly proportional to the normal reaction,

$f = \mu N$

a smaller normal force results in less friction. Therefore, pulling an object is easier than pushing it.

22. Correct Answer: A

Explanation:

Friction is a self-adjusting and reactive force. It acts only when an external force tends to produce relative motion between two surfaces. If no external force is applied, there is no tendency of motion, so the frictional force is zero.

23. Correct Answer: C

Explanation:

The limiting friction is given by

$f_L = \mu N$

Given:

- $\mu = 0.4$
- $N = 200$ N

$f_L = 0.4 \times 200 = 80$ N

Hence, the limiting friction is 80 N.

24. Correct Answer: B

Explanation:

First, calculate the normal reaction:

$N = mg$

$N = 6 \times 10 = 60$ N

Now, calculate the limiting friction:

$f_L = \mu N$

$f_L = 0.1 \times 60 = 6$ N

Hence, the limiting friction is 6 N.

25. Correct Answer: C

Explanation:

The net horizontal force acting on the block is

$F_{\text{net}} = F_{\text{applied}} - f$

$F_{\text{net}} = 100 \text{ N} - 40 \text{ N} = 60 \text{ N}$

Hence, the net force acting on the block is 60 N.

26. Correct Answer: A

Explanation:

The coefficient of friction and the angle of friction are related by

$\tan\theta = \mu$

Given:

$\mu = 0.5$

$\theta = \tan^{-1}(0.5) \approx 26.56°$

The nearest value is 26°.

27. Correct Answer: C

Explanation:

The maximum static friction is also known as the limiting friction.

It is given by

$f_{\max} = \mu_s N$

Given:

- $\mu_s = 0.6$
- $N = 50$ N

$f_{\max} = 0.6 \times 50 = 30$ N

Hence, the maximum static friction is 30 N.

28. Correct Answer: C

Explanation:

The kinetic friction is given by

$f_k = \mu_k N$

Rearranging,

$N = \frac{f_k}{\mu_k}$

Given:

- $f_k = 15$ N
- $\mu_k = 0.3$

$N = \frac{15}{0.3} = 50$ N

Hence, the normal reaction is 50 N.

LEVEL-II

1. Correct Option: B

Explanation:

Constant velocity means the net acceleration is zero. Therefore, the engine force must exactly balance the kinetic frictional force.

Calculation:

$F = \mu mg = 0.2 \times 120 \times 10 = 240$ N

2. Correct Option: C

Explanation:

When an object moves with uniform velocity, the applied force is equal to the kinetic frictional force.

Calculation:

$F = \mu mg$

$400 = \mu \times 100 \times 10$

$\mu = \frac{400}{1000} = 0.4$

3. Correct Option: C

Explanation:

The minimum force required to start sliding a body is equal to the limiting static friction.

Calculation:

$F = \mu N$

$N = \frac{F}{\mu} = \frac{150}{0.35} \approx 428.6$ N

4. Correct Option: C

Explanation:

When a block is pressed against a vertical wall, the normal force is equal to the applied force.

$N = F$

To prevent the block from falling, the upward frictional force must balance its weight.

$f = mg$

Since

$f \leq \mu N$

$mg \leq \mu F$

$F \geq \frac{mg}{\mu}$

Hence, the minimum applied force is

$F = \frac{mg}{\mu}$

5. Correct Option: A

Explanation:

According to Newton's second law,

$F_{\text{net}} = ma$

Since kinetic friction opposes the applied force,

$F_{\text{net}} = F - F_k$

Therefore,

$a = \frac{F - F_k}{m}$

6. Correct Option: D

Explanation:

For the block to slide downward with uniform velocity, the upward kinetic friction must equal its weight.

$f_k = 20$ N

Using

$f_k = \mu_k F$

$20 = 0.4F$

$F = 50$ N

7. Correct Option: A

Explanation:

The mass of the block is

$m = \frac{W}{g} = \frac{20}{10} = 2$ kg

Using Newton's second law,

$W - f_k = ma$

$20 - f_k = 2 \times 2$

$f_k = 16$ N

Using

$f_k = \mu_k F$

$16 = 0.4F$

$F = 40$ N

8. Correct Option: A

Explanation:

The coefficient of static friction is related to both the angle of friction and the angle of repose.

$\tan\theta = \mu$

$\tan\alpha = \mu$

Therefore,

$\theta = \alpha$

9. Correct Option: B

Explanation:

The maximum static friction is

$f_{s,\max} = \mu mg$

Calculation:

$f_{s,\max} = 0.5 \times 10 \times 10 = 50$ N

Since the applied force is 100 N,

$100 > 50$

the block starts moving.

Net force,

$F_{\text{net}} = 100 - 50 = 50$ N

Acceleration,

$a = \frac{F_{\text{net}}}{m} = \frac{50}{10} = 5 \text{ m/s}^2$

10. Correct Option: D

Explanation:

A body on an inclined plane is acted upon by:

- Weight acting vertically downward.
- Normal reaction acting perpendicular to the plane.
- Frictional force acting parallel to the plane, opposing motion or the tendency of motion.

11. Correct Option: C

Explanation:

When a body is pulled at an angle θabove the horizontal, the upward component of the pulling force reduces the normal reaction.

$N + F\sin\theta = Mg$

Hence,

$N = Mg - F\sin\theta$

Since friction depends on the normal reaction, pulling at an upward angle reduces friction.

12. Correct Option: A

Explanation:

When a body is pushed at an angle θbelow the horizontal, the downward component of the applied force increases the normal reaction.

$N = Mg + F\sin\theta$

This increases the frictional force.

13. Correct Option: A

Explanation:

The force just sufficient to move the block is equal to the limiting friction.

$F = \mu W$

Calculation:

$W = \frac{F}{\mu}$

$W = \frac{165}{0.30} = 550$ N

14. Correct Option: C

Explanation:

The coefficient of kinetic friction is defined as the ratio of kinetic friction to the normal reaction.

$\mu_k = \frac{f_k}{N}$

It is a dimensionless quantity.

15. Correct Option: A

Explanation:

Streamlining reduces air resistance or drag, which is a type of fluid friction. This allows objects to move more efficiently through air or water.

16. Correct Option: C

Explanation:

According to the laws of friction, friction depends on:

- The nature of the surfaces in contact.
- The normal reaction between the surfaces.

It does not depend on the apparent area of contact.

17. Correct Option: B

Explanation:

Rainwater acts as a lubricant on railway tracks, reducing the coefficient of friction. Spreading sand increases the roughness of the contact surfaces, thereby increasing friction and preventing the train wheels from slipping.

18. Correct Option: D

Explanation:

Rubber tyres provide better grip because rubber has a much higher coefficient of friction with road surfaces than iron. This increased friction improves traction during acceleration, braking, and turning, making vehicles safer to operate.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

19. Correct Options: A, B

Explanation:

When a block is pressed against a vertical wall with a horizontal force F, the wall exerts an equal normal reaction on the block. Hence,

$N = F$

- A is correct: If the wall is smooth ($\mu = 0$), no friction acts between the block and the wall. Since gravity is unopposed, the block slides downward under its weight (mg).
- B is correct: If the wall is rough, static friction acts upward to oppose the tendency of the block to slide downward. The maximum static friction is

  $f_{\max} = \mu N = \mu F$

  If

  $\mu F \geq mg,$

  the frictional force balances the weight and the block remains at rest.
- C is incorrect: The normal reaction is not equal to the weight. It depends only on the applied horizontal force, i.e.,

  $N = F.$
- D is incorrect: Even if the wall is smooth, the normal reaction does not become zero. The wall still exerts a normal force equal to the applied horizontal force.

20. Correct Options: A, C

Explanation:

To just start moving a block on a horizontal surface using a horizontal force, the applied force must equal the limiting friction.

The limiting friction is

$F = \mu N.$

Since the surface is horizontal,

$N = mg.$

Therefore,

$F = \mu mg.$

Using the relation between the coefficient of friction and the angle of friction,
$\mu = \tan\phi$,
we obtain
$F = mg\tan\phi$.
Hence, both A and C are correct.
Note: If the force could be applied at an optimum angle instead of horizontally, the minimum force required would be
$F = mg\sin\phi$.
However, since the specifically states that the force is horizontal, this expression is not applicable. Therefore, B is incorrect.

21. Correct Options: A, C, D

Explanation:

- A is correct: Static friction acts when there is a tendency of relative motion between two surfaces, but no actual sliding occurs.
- C is correct: Static friction is a self-adjusting force. It varies from zero up to its maximum value to balance the applied force and prevent motion.
- D is correct: The maximum value of static friction, just before motion begins, is called limiting friction.
- B is incorrect: Static friction is not constant. Its magnitude changes according to the applied force until it reaches its limiting value.

22. Correct Options: A, B, D

Explanation:

- A. Lubrication – Correct: Lubricants such as oil or grease form a thin layer between the surfaces, reducing direct contact and thereby decreasing friction.
- B. Polishing – Correct: Polishing removes microscopic irregularities on the surfaces, reducing interlocking between them and hence decreasing friction.
- D. Ball bearings – Correct: Ball bearings replace sliding friction with rolling friction, which is much smaller and allows smoother motion.
- C. Increasing roughness – Incorrect: Rougher surfaces have greater microscopic interlocking, which increases friction instead of reducing it.

23. Answer A)

Explanation:

When an object is in motion, the microscopic irregularities of the two surfaces do not get enough time to interlock as strongly as they do when the object is at rest. Hence, the kinetic friction is always less than the limiting static friction.
Therefore, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

24. Answer A)

Explanation:

The frictional force on a horizontal surface is given by:
$f = \mu N$
where:

- μ= coefficient of friction
- N= normal reaction

For a given pair of surfaces and a fixed load, friction depends only on the coefficient of friction and the normal reaction. It does not depend on the apparent area of contact.
Therefore, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

25. Answer C)

Explanation:

A horizontal force of 10 N presses the block against the vertical wall.

For horizontal equilibrium,

$N = F$

$N = 10$ N

Hence, the normal reaction exerted by the wall is:

$\boxed{10 \text{ N}}$

26. $8.5\ \text{m/s}^2$

Explanation:

The weight of the block is

$W = mg = 2 \times 10 = 20$ N

The maximum static friction is

$f_{s,\max} = \mu_s N = 0.4 \times 10 = 4$ N

Since

$20 \text{ N} > 4 \text{ N}$,

the block cannot remain at rest and starts sliding downward.

The kinetic friction acting upward is

$f_k = \mu_k N = 0.3 \times 10 = 3$ N

Hence, the net downward force is

$20 - 3 = 17$ N

Therefore,

$a = \frac{17}{2} = 8.5\ \text{m/s}^2$

27. Answer: A)

Explanation:

To prevent the block from sliding, the maximum static friction must at least balance its weight.

$\mu_s N_{\text{total}} = 20$

$0.4 N_{\text{total}} = 20$

$N_{\text{total}} = 50$ N

Since 10 N is already applied,

Additional force = $50 - 10 = 40$ N

Hence, the required additional horizontal force is

$\boxed{40 \text{ N}}$

28. Answer: A)

Explanation:

The box moves with constant velocity, so the net horizontal force is zero.

The horizontal component of the applied force is

$F_x = 30\cos 30°$

$= 30 \times \frac{\sqrt{3}}{2} = 15\sqrt{3}$ N

Therefore, the kinetic friction must be equal to this value.

$\boxed{f_k = 15\sqrt{3} \text{ N}}$

29. Answer: D)

Explanation:

The upward component of the pulling force is

$F_y = 30\sin 30° = 15$ N

Hence,

$N = mg - F_y$

$N = (5 \times 10) - 15$

$N = 35$ N

Since

$f_k = \mu_k N$

$\mu_k = \frac{f_k}{N} = \frac{15\sqrt{3}}{35} = \frac{3\sqrt{3}}{7}$

Therefore,

$\boxed{\mu_k = \frac{3\sqrt{3}}{7}}$

30. Explanation:

- a → T: Pulling a body at an angle produces an upward component, reducing the normal reaction.
  $N = mg - F\sin\theta$
- b → S: Pushing a body at an angle produces a downward component, increasing the normal reaction.
  $N = mg + F\sin\theta$
- c → P: The coefficient of friction is related to the angle of friction by
  $\tan\phi = \mu$
- d → R: The coefficient of kinetic friction is defined as
  $\mu_k = \frac{f_k}{N}$

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Explanation:

The applied force is

$F = 20\sqrt{2}$ N

Its components are

$F_x = 20\sqrt{2}\cos 45° = 20$ N

$F_y = 20\sqrt{2}\sin 45° = 20$ N

The normal reaction is

$N = mg - F_y$

$= (4 \times 10) - 20$

$N = 20$ N

So,

a → Q

The limiting friction is

$f_{s,\max} = 0.8 \times 20 = 16$ N

Since

$20 > 16$,

the block starts moving.

d → R

The kinetic friction is

$f_k = 0.6 \times 20 = 12$ N

b → P

The acceleration is

$a = \frac{F_x - f_k}{m}$

$= \frac{20 - 12}{4}$

$= 2\ \text{m/s}^2$

c → S

32. $\boxed{10\ \text{N}}$

Explanation:

The limiting friction is

$f_s = \mu N$

$= 0.5 \times 20$

$= 10\ \text{N}$

Hence,

$\boxed{10\ \text{N}}$

33. $\boxed{30\ \text{N}}$

Explanation:

The net force acting on the block is

$F_{\text{net}} = F_{\text{applied}} - f$

$= 40 - 10$

$= 30\ \text{N}$

Hence,

$\boxed{30\ \text{N}}$

34. $\boxed{\mu = 1}$

Explanation:

The coefficient of static friction is equal to the tangent of the angle of repose.

$\mu = \tan\theta$

Given,

$\theta = 45°$

Therefore,

$\mu = \tan 45° = 1$

Hence,

$\boxed{\mu = 1}$

PREVIOUS YEAR QUESTIONS

1. Correct Answer : B. Surface Y produces more friction

Explanation: In a distance–time graph, slope = speed $\left(v = \frac{d}{t}\right)$. Surface X has a steeper slope ⇒ higher speed ⇒ less friction acting. Surface Y has a smaller slope ⇒ lower speed ⇒ more friction acting on the bicycle.

2. Correct Answer : C. Towel, newspaper, dry marble floor, wet marble floor

Explanation: Friction depends on the roughness of the surface. Rougher surfaces produce more friction. Towel (very rough) > Newspaper (moderately rough) > Dry marble floor (smooth) > Wet marble floor (very smooth due to water acting as a lubricant, reducing friction).

3. Correct Answer : B) P

Explanation: In a pedalled wheel moving forward, the tyre pushes the ground backward. By Newton's Third Law, the ground exerts a forward static frictional force on the wheel at the point of contact. Hence, friction acts in the direction of motion.

4. Correct Answer : (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)

Explanation:
Limiting friction is the maximum value of static friction ⇒ (a)–(iii).
Static friction is a self-adjusting force ⇒ (b)–(i).
Dynamic (sliding) friction is less than limiting friction ⇒ (c)–(iv).
Rolling friction is less than sliding friction ⇒ (d)–(ii).

5. Correct Answer : A. Limiting friction is greater than dynamic friction.

Explanation: At the start, the horse must overcome limiting friction ($f_{lim} = \mu_s N$). Once the cart starts moving, dynamic friction ($f_k = \mu_k N$)acts, and since $\mu_s > \mu_k$, more force is needed initially.

6. Correct Answer : B. Drag Speed

Shape Nature

Explanation: The frictional force exerted by fluids on moving objects is called drag. Drag depends on the speed of the object relative to the fluid, the shape of the object, and the nature (viscosity) of the fluid.

7. Correct Answer : A. P

Explanation: Friction acts at the **contact surface** between the box and the floor. The rough floor offers greater friction compared to the polished floor. Point **P** lies on the rough surface where friction is maximum. Points Q (polished floor) and R (side of box) do not have maximum frictional contact with the floor.

8. Correct Answer : D. Irrespective of whether the objects are solid, liquid or gas.

Explanation: Friction occurs whenever two surfaces or layers in contact move or tend to move relative to each other. This applies to solids (dry friction) and also to fluids (liquids and gases) where friction is called drag or fluid friction.

6. Sound

TOPIC-1: Wave Motion and Sound

LEVEL-I

1. Correct Answer: C

Explanation: The echo travels to the wall and back, so the total distance covered is:

$2 \times 85\ \text{m} = 170\ \text{m}$

Using the formula,

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$t = \frac{170}{340} = 0.5\ \text{s}$$

2. Correct Answer: A

Explanation: Sound travels fastest in solids because their molecules are closely packed and strongly bonded. Aluminium is a solid metal, so the speed of sound is highest in it (approximately $6420\ \text{m/s}$).

3. Correct Answer: A

Explanation: Ultrasonic waves have very high frequencies and carry significant energy. They produce rapid vibrations that thoroughly mix substances, resulting in a perfectly homogeneous solution.

4. Correct Answer: D

Explanation: A slinky can produce:

- Longitudinal waves when compressed and released along its length.
- Transverse waves when moved up and down perpendicular to its length.

Hence, it can produce both types of waves.

5. Correct Answer: C

Explanation: Compressing a spring with a sudden jerk creates regions of compression that travel along the length of the spring. Since the particles vibrate parallel to the direction of wave propagation, the wave is longitudinal.

6. Correct Answer: D

Explanation: Crests (highest points) and troughs (lowest points) are characteristic features found only in transverse waves.

7. Correct Answer: A

Explanation: The normal range of human hearing extends from:

$20\ \text{Hz}$ to $20{,}000\ \text{Hz}$

8. Correct Answer: B

Explanation: A compression is the region in a longitudinal wave where particles are crowded closer together, resulting in higher density and pressure.

9. Correct Answer: D

Explanation: Loudness depends directly on the amplitude of the sound wave. Increasing the amplitude makes a feeble sound louder.

10. Correct Answer: A

Explanation: Taking the speed of sound as:

$v = 332\ \text{m/s}$

Distance to the first cliff:

$$d_1 = \frac{332 \times 2}{2} = 332\ \text{m}$$

Distance to the second cliff:

$d_2 = \frac{332 \times 4.5}{2} = 747\ \text{m}$

Total distance between the two cliffs:

$332 + 747 = 1079\ \text{m}$

11. Correct Answer: B

Explanation: Stationary (standing) waves are formed when two waves of the same frequency and amplitude travel in opposite directions. They do not transfer energy from one place to another because the energy remains confined between the nodes.

12. Correct Answer: B

Explanation: The total echo time is:

$4\ \text{s}$

One-way travel time:

$\frac{4}{2} = 2\ \text{s}$

Depth:

$\text{Depth} = 1500 \times 2 = 3000\ \text{m}$

13. Correct Answer: B

Explanation: Using the echo formula:

$\text{Depth} = \frac{vt}{2}$

$= \frac{1500 \times 0.4}{2} = 300\ \text{m}$

14. Correct Answer: B

Explanation: The wave equation is:

$v = f\lambda$

Algebraically, wave velocity is expressed as the product of frequency and wavelength, so it is directly proportional to both quantities in this equation.

15. Correct Answer: C

Explanation: Striking a piano key harder increases the amplitude of vibration, making the sound louder. The frequency (pitch) remains unchanged because it depends on the string's length, tension, and mass.

16. Correct Answer: D

Explanation: Pitch is the sensation perceived by the human ear, whereas frequency is the actual measurable number of vibrations per second. Therefore, they are related but not identical.

17. Correct Answer: A

Explanation: When a wave enters another medium:

- Frequency remains constant because it depends on the source.
- Speed and wavelength change according to the properties of the new medium.

18. Correct Answer: B

Explanation: Sound intensity level (loudness level) is measured in decibels (dB).

19. Correct Answer: A

Explanation: A larger vibrating surface displaces more air molecules, transferring more energy to the surrounding air and producing a louder sound.

20. Correct Answer: B

Explanation: Loudness is directly proportional to the square of the amplitude:

$\text{Loudness} \propto (\text{Amplitude})^2$

21. Correct Answer: B

Explanation: The distance from the 5th crest to the 15th crest represents:

$15 - 5 = 10$

wavelengths.

Given:

$10\lambda = 20\,\text{cm}$

Therefore,

$\lambda = \frac{20}{10} = 2\,\text{cm}$

22. Correct Answer: A

Explanation: Timbre (quality) is the characteristic of sound that enables us to distinguish between voices or musical instruments producing the same pitch and loudness. It depends on the waveform produced by different overtones and harmonics.

LEVEL-II

1. Answer: D

Explanation:

Wavelength, $\lambda = 100\,\text{m}$ and wave velocity, $v = 20\,\text{m/s}$.

Using the formula,

$f = \frac{v}{\lambda}$

$f = \frac{20}{100} = 0.2\,\text{Hz}$

Hence, the correct answer is D.

2. Answer: C

Explanation:

Loudness depends on:

- Amplitude of vibration
- Surface area of the vibrating body
- Distance from the source

It does not depend on the frequency of vibration. Frequency determines the pitch (shrillness) of sound.

Hence, the correct answer is C.

3. Answer: C

Explanation:

Tightening or loosening a string changes its tension, which changes the frequency at which it vibrates. This alters the pitch of the sound, allowing the instrument to be tuned.

Hence, the correct answer is C.

4. Answer: C

Explanation:

The frequency of a wave is determined only by its source. When a wave travels from one medium (air) to another (water), its speed and wavelength change, but its frequency remains unchanged.

Hence, the correct answer is C.

5. Answer: A

Explanation:

Frequency,

$f = \frac{v}{\lambda} = \frac{1500}{3} = 500\ \text{Hz}$

Assuming the wavelength is 68 cm = 0.68 m,

$v = f\lambda$

$v = 500 \times 0.68 = 340\ \text{m/s}$

Hence, the correct answer is A.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

6. Correct Options: A, C

- **Explanation:**
  - A is correct because waves transport energy and momentum through a medium or space without causing a net transport of matter.
  - B is incorrect because a pulse is a *short, sudden* disturbance, not a sustained long-duration one.
  - C is correct because mechanical waves require a material medium (like sound), while electromagnetic waves travel via oscillating electric and magnetic fields through a vacuum.
  - D is incorrect because particles only oscillate around their fixed equilibrium positions; they do not move permanently from one place to another.

7. Correct Options: A, B, C

- **Explanation:**
  - A is correct by definition; in transverse waves, the medium particles oscillate perpendicular to the direction of wave propagation.
  - B is correct because the displacements form high peaks (crests) and low valleys (troughs).
  - C is correct because transverse mechanical waves require shear elasticity to propagate. Solids possess shear elasticity, and liquid surfaces have surface tension behaving similarly, allowing transverse waves to travel through them.
  - D is incorrect because compressions and rarefactions are characteristics of longitudinal waves, not transverse waves.

8. Correct Options: A, C, D

- **Explanation:**
  - A is correct by definition; in longitudinal waves, particles oscillate parallel to the direction of wave travel.
  - B is incorrect because compressions are regions where particles are crowded together, resulting in *high* density and pressure.
  - C is correct because rarefactions are regions where particles are spread apart, resulting in *low* density and pressure.
  - D is correct because sound waves propagating through fluids (like air or water) are classic examples of longitudinal waves.

9. Correct Options: A, C, D

- **Explanation:**
  - A is correct because sound waves require a material medium (solid, liquid, or gas) to propagate, making them mechanical waves.
  - B is incorrect because sound cannot travel through a vacuum due to the absence of matter/particles to transmit the vibrations.

- C is correct because sound originates from the mechanical vibration of objects (like vocal cords, strings, or tuning forks).
- D is correct because the speed of sound depends on the elasticity and density of the medium ($v = \sqrt{E/\rho}$). Solids have much higher elastic moduli than liquids and gases, which makes sound travel fastest through them.

Assertion and Reason Type

10. Answer: A

Explanation:

Both the Assertion and the Reason are true.

Sound is a mechanical wave and requires a material medium for propagation. Since the Moon has no atmosphere (vacuum), sound cannot travel there. Thus, the Reason correctly explains the Assertion.

11. Answer: A

Explanation:

Both the Assertion and the Reason are true.

Pitch is the perception of the frequency of a sound. A higher frequency produces a higher pitch (shriller sound). Therefore, the Reason correctly explains the Assertion.

12. Answer: A

Explanation:

Both the Assertion and the Reason are true.

Ultrasonic waves are reflected by internal cracks or defects in metals. This principle of reflection of sound waves is used for detecting flaws inside metal objects.

13. Answer: C

Explanation:

The Assertion is false because transverse waves consist of crests and troughs, not compressions and rarefactions.

The Reason is true because, in a transverse wave, particles vibrate perpendicular to the direction of wave propagation.

Therefore, the correct answer is C.

Matrix Matching Type

14

Column I	Column II
A. Wave	S. A moving disturbance
B. Mechanical wave	P. Requires a material medium
C. Transverse wave	R. Particles vibrate perpendicular to the direction of propagation
D. Oscillatory motion	Q. Back-and-forth motion about the mean position

Answer:

- **A → S**
- **B → P**
- **C → R**
- **D → Q**

15

Column I	Column II
A. Audible sound	Q. Frequency between 20 Hz and 20,000 Hz
B. Ultrasonic sound	R. Used by bats for navigation
C. Infrasonic sound	P. Frequency less than 20 Hz

Column I	Column II
D. SONAR	S. Based on the echo-ranging technique

Answer:

- **A → Q**
- **B → R**
- **C → P**
- **D → S**

16. Correct Answer: B and C

Explanation:

- B is correct because ultrasonic waves used in SONAR are mechanical longitudinal waves.
- C is correct because the speed of ultrasonic waves depends on the properties of the medium, so it changes when the medium changes.
- A is incorrect because mechanical waves cannot travel through a vacuum; they require a material medium.
- D is incorrect because longitudinal waves consist of compressions and rarefactions, whereas crests and troughs are characteristics of transverse waves.

17. Correct Answer: B) 3000 m

Explanation:

The given time (4 s) is the round-trip time.

Time taken to reach the obstacle $= \frac{4}{2} = 2$ s

Using,

Distance = Speed × Time

$= 1500 \times 2 = 3000$ m

Distance of the obstacle = 3000 m.

18. Correct Answer: C) They undergo reflection even from very small obstacles.

Explanation:

Ultrasonic waves have very high frequencies and therefore very short wavelengths. Due to their short wavelengths, they are reflected even by tiny cracks and small underwater objects instead of bending around them. This property makes them ideal for SONAR and non-destructive testing.

19. Correct Answer: B and D

Explanation:

- B is correct because transverse waves consist of crests and troughs.
- D is correct because, in transverse waves, the particles of the medium oscillate perpendicular to the direction of wave propagation.
- A is incorrect because particle motion parallel to the direction of propagation is a characteristic of longitudinal waves.
- C is incorrect because waves transfer energy, not matter. The particles only oscillate about their equilibrium positions.

20. Correct Answer: B) 60 m/s

Explanation:

Wave speed is given by

$v = f\lambda$

Substituting the values,

$v = 5 \times 12 = 60$ m/s

Speed of the wave = 60 m/s.

21

Correct Answer: C) Have equal displacement magnitudes but move in opposite directions.

Explanation:

Two particles are 180° out of phase when they are in exactly opposite stages of vibration. At any instant, they have equal magnitudes of displacement from the equilibrium position but are moving in opposite directions.

22. Correct Answer: 75

Explanation:

For an echo to be heard distinctly, the reflected sound must return after at least 0.1 s.

Round-trip distance:

$2d = vt$

$2d = 1500 \times 0.1 = 150$ m

$d = \frac{150}{2} = 75$ m

Answer = 75 m

23. Correct Answer: 40

Explanation:

Using the wave equation,

$v = f\lambda$

$v = 5 \times 8 = 40$ m/s

Answer = 40 m/s

24. Correct Answer: 340

Explanation:

Speed is calculated using

$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

$= \frac{680}{2} = 340$ m/s

Answer = 340 m/s

TOPIC-2:

Propagation, Reflection, and Applications of Sound

1. Correct Answer: A) Reverberation of sound

Explanation:

Reverberation is the persistence of sound due to repeated reflections from the walls and roof of a large room or auditorium. Sound-absorbing materials, such as compressed fiberboards and rough plaster, absorb sound waves instead of reflecting them, thereby reducing unwanted reverberation.

2. Correct Answer: C) 17.2 m

Explanation:

The human ear retains the sensation of sound for about 0.1 s (persistence of hearing). For an echo to be heard distinctly, the reflected sound must reach the listener after this interval.

Using the formula:

$d = \frac{\text{Speed} \times \text{Time}}{2}$

$d = \frac{344 \times 0.1}{2} = 17.2$ m

3. Correct Answer: C) Multiple reflections

Explanation:

A stethoscope works on the principle of multiple reflections. The sound produced by the patient's heartbeat undergoes successive reflections along the inner walls of the stethoscope tube before reaching the doctor's ears.

4. Correct Answer: C) Frequency, speed, and wavelength

Explanation:

When a sound wave is reflected back into the same medium:

- Speed remains unchanged because it depends only on the medium.
- Frequency remains unchanged because it depends on the source.
- Since

$\lambda = \frac{v}{f}$

and both $v$and $f$remain constant, the wavelength also remains unchanged.

5. Correct Answer: B) Longitudinal waves

Explanation:

Longitudinal waves propagate through alternating compressions (high pressure and density) and rarefactions (low pressure and density). Therefore, a graph showing changes in density with distance represents a longitudinal wave.

6. Correct Answer: B) 4×10^6 N/m^2

Explanation:

The speed of a longitudinal wave is given by:

$v = \sqrt{\frac{E}{\rho}}$

Rearranging,

$E = v^2\rho$

Substituting the values:

$E = (400)^2 \times 25$

$E = 160000 \times 25$

$E = 4 \times 10^6$ N/m^2

7. Correct Answer: B) 2 : 1

Explanation:

The velocity of sound in a gas is directly proportional to the square root of its absolute temperature.

$v \propto \sqrt{T}$

$\frac{v_1}{v_2} = \sqrt{\frac{400}{100}} = \sqrt{4} = 2$

Hence,

$v_1 : v_2 = 2 : 1$

8. Correct Answer: B) Half

Explanation:

At constant pressure,

$v \propto \frac{1}{\sqrt{\rho}}$

If the density becomes four times,

$v_{new} = \frac{v}{\sqrt{4}} = \frac{v}{2}$

Therefore, the velocity becomes half of its original value.

9. Correct Answer: B) 2 : 1

Explanation:

At constant temperature,

$v \propto \frac{1}{\sqrt{M}}$

where $M$is the molecular weight.

$\frac{v_1}{v_2} = \sqrt{\frac{16}{4}} = \sqrt{4} = 2$

Therefore,

- $v_1 : v_2 = 2 : 1$

LEVEL-II

Single Correct Option Type s

1. Correct Answer: A) Statement 1 is true but Statement 2 is false.

Explanation:

- Statement 1: True. Hard, smooth, and polished surfaces are good reflectors of sound and are necessary for producing an echo.
- Statement 2: False. Soft and porous materials absorb sound instead of reflecting it efficiently.

2. Correct Answer: D) Density and elasticity of gas

Explanation:

The speed of sound in a gas depends on its elasticity and density, according to the Newton–Laplace equation:

$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{E}{\rho}}$

Thus, the determining factors are the density and elasticity of the gas.

3. Correct Answer: A) The horizontal straight-line graph

Explanation:

At constant temperature,

$v = \sqrt{\frac{\gamma P}{\rho}}$

Since pressure and density increase proportionally, the ratio P/ρremains constant. Therefore, the speed of sound does not change with pressure and is represented by a horizontal straight line.

4. Correct Answer: C) Atmospheric pressure

Explanation:

Changes in atmospheric pressure produce proportional changes in air density. Since these changes cancel each other in the speed equation, atmospheric pressure has no effect on the speed of sound.

5. Correct Answer: B) 34.4 m

Explanation:

The sound travels to the reflecting surface and back in 0.2 s.

Distance to the wall:

$d = \frac{vt}{2}$

$d = \frac{344 \times 0.2}{2} = 34.4 \text{ m}$

6. Correct Answer: C) 340 m/s

Explanation:

Using the wave equation:

$v = f\lambda$

$v = 850 \times 0.4 = 340 \text{ m/s}$

7. Correct Answer: A) Double

Explanation:

Since

$v \propto \sqrt{T}$

$\frac{v_{\text{new}}}{v_{\text{initial}}} = \sqrt{\frac{1200}{300}} = \sqrt{4} = 2$

Hence, the speed of sound becomes double.

8. Correct Answer: B) 17.2 m

Explanation:

The minimum time interval required to hear a distinct echo is 0.1 s.

$d = \frac{vt}{2}$

$d = \frac{344 \times 0.1}{2} = 17.2 \text{ m}$

9. Correct Answer: A) 100 m/s

Explanation:

The speed of sound is inversely proportional to the square root of molecular weight.

$v \propto \frac{1}{\sqrt{M}}$

If the molecular weight becomes 9 times the original value,

$v_{\text{new}} = \frac{300}{\sqrt{9}} = \frac{300}{3} = 100 \text{ m/s}$

10. Correct Answer: B) 34.4 m

Explanation:

The asks for the total distance travelled by the sound (forward and backward).

Distance = vt

$= 344 \times 0.1 = 34.4 \text{ m}$

Therefore, the total distance travelled by the sound is 34.4 m.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Correct Answer: A

Explanation:

When a piano key is struck harder, more mechanical energy is transferred, increasing the amplitude of the vibration, which makes the sound louder. However, the frequency (pitch) depends entirely on the intrinsic properties of the string (its length, tension, and mass), which do not change. Thus, the pitch remains the same.

8. Correct Answer: D

Explanation:

The characteristic that enables us to distinguish between two sounds having the same loudness and pitch played by different instruments is called quality (or

timbre). It depends on the unique waveform and overtones produced by the instrument.

9. Correct Answer: A

Explanation:

Using the wave speed formula:

$v = f \times \lambda$

Given:

- Frequency, $f = 10$ kHz $= 10{,}000$ Hz
- Wavelength, $\lambda = 0.033$ m

$v = 10{,}000 \times 0.033 = 330$ m/s

10. Correct Answer: A

Explanation:

Loudness is a subjective sensation that is directly related to the objective amplitude of the sound wave. Specifically, the intensity of a wave is proportional to the square of its amplitude:

$I \propto A^2$

11. Correct Answers: A, C, D

Explanation:

The velocity of sound in a medium is determined by the formula:

$$v = \sqrt{\frac{E}{\rho}}$$

where:

- E= elasticity of the medium
- ρ= density of the medium
- Option A: Correct, assuming elasticity remains constant:

  $v \propto \frac{1}{\sqrt{\rho}} \Rightarrow \frac{1}{\sqrt{9}} = \frac{1}{3}$

  Hence, the velocity becomes one-third.
- Option C: Correct because the velocity of sound depends directly on the elasticity of the medium.
- Option D: Correct because if both elasticity and density increase by a factor of 9,

  $\frac{9E}{9\rho} = \frac{E}{\rho}$

  Therefore, the velocity remains unchanged.

12. Correct Answers: A, B, C

Explanation:

We verify each combination using the relation:

$v = f \times \lambda$

For the speed of sound to be 340 m/s:

- A: 170 Hz × 2 m = 340 m/s✓
- B: 850 Hz × 0.4 m = 340 m/s✓
- C: 34 Hz × 10 m = 340 m/s✓
- D: 680 Hz × 0.25 m = 170 m/s✗

Hence, the correct options are A, B, and C.

13. Correct Answers: A, B, D

Explanation:

For an echo to be heard distinctly:

- Option A: Correct because the persistence of hearing in humans is approximately 0.1 s.
- Option B: Correct because:
  Distance travelled by sound = $344 \times 0.1 = 34.4$ m
  Since the sound travels to the wall and back,
  Minimum distance to the wall = $\frac{34.4}{2} = 17.2$ m
- Option D: Correct because an echo is the repetition of sound caused by its reflection from a distant surface.

14. Correct Answers: A, B, D

Explanation:

The velocity of sound in gases varies directly with the square root of the absolute temperature:

$v \propto \sqrt{T}$

- Option A: Correct because increasing the absolute temperature increases the velocity of sound.
- Option B: Correct because the velocity is directly proportional to $\sqrt{T}$.
- Option D: Correct because if the absolute temperature becomes $4T$,
  $v \propto \sqrt{4T} = 2\sqrt{T}$
  Hence, the velocity doubles.

15. Answer: A

Explanation:

The speed of sound in a medium is given by:

$$v = \sqrt{\frac{E}{\rho}}$$

where:

- E= Elasticity (Elastic Modulus) of the medium
- ρ= Density of the medium

Although solids are much denser than gases, their elasticity is enormously greater. Therefore, the ratio $\frac{E}{\rho}$is highest for solids and lowest for gases. As a result, sound travels:

- Fastest in solids
- Slower in liquids
- Slowest in gases

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

16. Answer: A

Explanation:

The human ear retains the sensation of sound for approximately 0.1 s, a phenomenon known as persistence of hearing.

If the reflected sound returns in less than 0.1 s, it merges with the original sound and cannot be heard separately.

Therefore, an echo is heard only when the reflected sound reaches the listener after at least 0.1 s.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

17. Answer: B

Explanation:

- Assertion: A transverse wave is one in which the particles of the medium vibrate perpendicular to the direction of wave propagation. This statement is true.
- Reason: Due to this perpendicular vibration, alternating crests and troughs are formed. This statement is also true.

However, the formation of crests and troughs is a result of perpendicular vibration rather than the reason why the particles vibrate perpendicularly.

Therefore, both statements are true, but the Reason is not the correct explanation of the Assertion.

18. Answer: A

Explanation:

Ultrasound waves have very high frequencies and short wavelengths, allowing them to travel through solid materials with minimal spreading.

When these waves encounter an internal crack or defect, part of the wave is reflected due to the difference in acoustic properties at the boundary. The reflected waves are detected and used to locate internal flaws.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Matrix Matching Type

19. Answer:

- **A → R**
- **B → P**
- **C → Q**
- **D → S**

---

20. Answer:

- **A → Q**
- **B → P**
- **C → R**
- **D → S**

Explanation:

The speed of sound in gases is given by

$$v = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

where:

- T= Absolute temperature
- ρ= Density
- M= Molecular mass
- E= Elasticity

Using proportional relationships:

- A → Q: $v \propto \sqrt{T}$

  If temperature becomes 4 times, speed becomes 2 times.
- B → P: $v \propto \frac{1}{\sqrt{\rho}}$

  If density becomes 4 times, speed becomes half.

- C → R: $v \propto \frac{1}{\sqrt{M}}$
  If molecular mass becomes 9 times, speed becomes one-third.
- D → S: $v \propto \sqrt{E}$
  If elasticity becomes 16 times, speed becomes 4 times.

Passage-I

21. Answer: D) 5440 m

Explanation:

Distance travelled by the jet:

Distance = Speed × Time

= 680 × 8 = 5440 m

22. Answer: B) 2720 m

Explanation:

Distance travelled by the sound wave:

Distance = 340 × 8 = 2720 m

23. Answer: B) 1360 m

Explanation:

For a stationary source, the total echo time is 8 s.

Time taken by the sound to reach the mountain:

$\frac{8}{2}$ = 4 s

Therefore,

Distance = 340 × 4 = 1360 m

Passage-II

24. Answer: B) 1700 m

Explanation:

The flash of lightning is seen almost instantly because the speed of light is extremely high.

The 5-second delay is therefore due entirely to the sound travelling through air.

Distance = 340 × 5 = 1700 m

25. Answer: B) 340 m

Explanation:

Using only the given echo information:

Echo time = 2 s (round trip)

One-way travel time:

$\frac{2}{2}$ = 1 s

Therefore,

Distance = 340 × 1 = 340 m

26. Answer: B) 680 m

Explanation:

The echo travels from the boy to the hill and back.

Total distance travelled:

Distance = 340 × 2 = 680 m

Hence, the total path covered by the reflected sound wave is 680 m**.**

27

Answer: 4500

Explanation:

For an echo, sound travels to the obstacle and returns back.

Given:

- Speed of sound in water,
  $v = 1500$ m/s
- Time,
  $t = 6$ s

Using

$2d = vt$

$2d = 1500 \times 6 = 9000$ m

$d = \frac{9000}{2} = 4500$ m

28

Answer: 4

Explanation:

The distance between two consecutive crests is one wavelength.

From the 3rd crest to the 13th crest, there are

$13 - 3 = 10$

wavelengths.

Given:

$10\lambda = 40$ cm

$\lambda = \frac{40}{10} = 4$ cm

29

Answer: 20

Explanation:

Frequency is the number of vibrations completed in one second.

Given:

- Total vibrations = 2400
- Time = 2 minutes

Convert time into seconds:

$2 \times 60 = 120$ s

Using

$f = \frac{\text{Number of vibrations}}{\text{Time}}$

$f = \frac{2400}{120} = 20$ Hz

PREVIOUS YEAR QUESTIONS

1. Answer: A. 1.7 cm

Explanation:

According to the wave equation,

$v = f\lambda$

where:

- v= wave speed
- f= frequency
- λ= wavelength

For waves travelling through the same uniform medium, the wave speed remains constant. Therefore, frequency and wavelength are inversely proportional.

From 2, the frequency is doubled, so the wavelength must become half of its original value.

Among the given options, only 1.7 cm becomes 0.85 cm when halved:

$\frac{1.7 \text{ cm}}{2} = 0.85 \text{ cm}$

Hence, the correct answer is A. 1.7 cm.

2. Answer: C. 0.85 cm

Explanation:

Since the frequency is doubled while the wave travels in the same medium, the wave speed remains constant.

Using

$v = f\lambda$

If

$f \rightarrow 2f,$

then

$\lambda \rightarrow \frac{\lambda}{2}$

Given the original wavelength is 1.7 cm,

New wavelength $= \frac{1.7}{2} = 0.85$ cm

Therefore, the correct answer is C. 0.85 cm.

3. Answer: D. 8000 Hz

Explanation:

From the displacement–distance graph, one complete wave occupies a distance of

$\lambda = 0.04$ m

Given:

$v = 320$ m/s

Using

$f = \frac{v}{\lambda}$

$f = \frac{320}{0.04} = 8000$ Hz

Hence, the correct answer is D. 8000 Hz.

4. Answer: D. T, T, T, F

Explanation:

Using

$\lambda = \frac{v}{f}$

where

$v = 340$ m/s

(i) True

The highest audible frequency is

20 000 Hz

$\lambda_{\text{min}} = \frac{340}{20\,000} = 0.017$ m

(ii) True

The lowest audible frequency is

20 Hz

$\lambda_{\text{max}} = \frac{340}{20} = 17$ m

(iii) True

Pitch depends only on frequency. As frequency increases from 20 Hz to 20 kHz, the pitch also increases.

(iv) False
Pitch does not decrease with increasing frequency; it increases.
Therefore, the correct sequence is:
T, T, T, F

5. Answer: C. III

Explanation:
Noise is produced by irregular, non-periodic vibrations. Its waveform has random variations in amplitude and frequency.
Among the given graphs, Graph III shows an irregular and chaotic waveform, which is characteristic of noise.
Therefore, the correct answer is C. III.

6. Answer: A. II

Explanation:
Pitch depends directly on the frequency of a sound wave.

- Higher frequency → Higher pitch
- Lower frequency → Lower pitch

Among the four waveforms drawn over the same horizontal distance, Graph II has the fewest complete cycles, indicating the lowest frequency.
Hence, it has the lowest pitch.
Therefore, the correct answer is A. II.

7. Correct Option: B) $50\ \text{m s}^{-1}$

Explanation:
The speed of sound is calculated using the formula:
$v = \frac{d}{t}$
where:

- d= distance travelled by the sound
- t= time taken by the sound

Given:

- Distance from the volcano to the UFO = 5 km = 5000 m
- Time between seeing the flash and hearing the sound = 100 s

Since light travels much faster than sound ($\approx 3 \times 10^8\ \text{m s}^{-1}$), the flash is seen almost instantaneously. Hence, the entire 100 srepresents the time taken by the sound to travel the distance.

Calculation:
$v = \frac{5000}{100} = 50\ \text{m s}^{-1}$

8. Correct Option: D) Energy

Explanation:
A wave is a disturbance that transfers energy from one place to another without transporting matter.
As a wave propagates:

- The particles of the medium only vibrate about their mean positions.
- They transfer the disturbance to neighbouring particles.
- The particles themselves do not travel along with the wave.

Therefore, a wave transfers energy, not matter.

9. Correct Answer: A, D

Explanation:

A pleasant (musical) sound is produced by regular, periodic vibrations. Its waveform repeats itself at equal intervals of time.

- A) Regular, repeating sinusoidal wave → Pleasant (musical) sound
- D) Although it is periodic, the standard representation of a pure musical tone is the smooth, regular waveform shown in A.

7. Light

TOPIC-1: Refraction of Light and Optical Phenomena

LEVEL-I

1. Correct Option: C) 48.6°

Explanation:

According to Snell's Law,

$$\frac{\sin i}{\sin r} = \frac{V_A}{V_B}$$

Given,

- $\frac{V_A}{V_B} = \frac{3}{2}$
- $r = 30°$

$$\frac{\sin i}{\sin 30°} = \frac{3}{2}$$

Since,

$$\sin 30° = \frac{1}{2}$$

$$\sin i = \frac{3}{2} \times \frac{1}{2} = \frac{3}{4} = 0.75$$

Therefore,

$i = \sin^{-1}(0.75) \approx 48.6°$

Hence, the correct answer is C) 48.6°.

2. Correct Option: B) definitely decrease

Explanation:

The lateral shift produced by a parallel glass slab is

$$d = t\sin i\left(1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}\right)$$

where

- t= thickness of the slab
- i= angle of incidence
- μ= refractive index of the slab

If the refractive index μdecreases while $t$and $i$remain constant,

- $\sqrt{\mu^2 - \sin^2 i}$decreases,
- therefore $\frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}}$increases,
- so the quantity inside the brackets decreases.

Hence, the lateral shift definitely decreases.

Therefore, the correct answer is B) definitely decrease.

3. Correct Option: C) 800 nm

Explanation:

The relative refractive index of medium 2 with respect to medium 1 is

$$^1\mu_2 = \frac{\lambda_1}{\lambda_2}$$

Given,

$$^1\mu_2 = \frac{3}{4}, \lambda_1 = 600 \text{ nm}$$

Substituting,

$$\frac{3}{4} = \frac{600}{\lambda_2}$$

$\lambda_2 = \frac{600 \times 4}{3} = 800$ nm

Hence, the correct answer is C) 800 nm.

4. Correct Option: A) both media have equal refractive index

Explanation:

Snell's Law states

$\mu_1 \sin i = \mu_2 \sin r$

If

$\sin i = \sin r$

for every angle of incidence, then

$i = r$

Substituting into Snell's Law,

$\mu_1 = \mu_2$

Thus, both media have the same refractive index, so the light ray passes without bending.

Hence, the correct answer is A) both media have equal refractive index.

5. Correct Option: D) 2×10^8 m/s

Explanation:

Using Snell's Law in terms of speeds,

$\frac{\sin i}{\sin r} = \frac{V_X}{V_Y}$

Therefore,

$V_Y = V_X \left(\frac{\sin r}{\sin i}\right)$

Given,

- $i = 45°$
- $r = 35°$
- $V_X = 2.5 \times 10^8$ m/s

Using

$\sin 35° \approx 0.5736, \sin 45° \approx 0.7071$

$V_Y = (2.5 \times 10^8) \times \frac{0.5736}{0.7071}$

$V_Y \approx 2.03 \times 10^8$ m/s

This rounds to

2×10^8 m/s

Hence, the correct answer is D) 2×10^8 m/s.

6. Correct Option: B) 1.8

Explanation:

For successive relative refractive indices,

${}^C\mu_A = {}^B\mu_A \times {}^C\mu_B$

Given,

${}^B\mu_A = 1.2, \quad {}^C\mu_B = 1.5$

Therefore,

${}^C\mu_A = 1.2 \times 1.5 = 1.8$

Hence, the correct answer is B) 1.8.

7. Correct Option: C) $i = 0°$

Explanation:

The lateral displacement produced by a parallel slab is

$d = t\dfrac{\sin(i-r)}{\cos r}$

For zero lateral displacement,

$d = 0$

which requires

$\sin(i-r) = 0$

Thus,

$i = r$

For two different media, this occurs only during normal incidence, where

$i = r = 0°$

Hence, the correct answer is C) $i = 0°$.

8. Correct Option: B) $\lambda_a > \lambda_w > \lambda_g$

Explanation:

The refractive indices satisfy

$\mu_{\text{air}} < \mu_{\text{water}} < \mu_{\text{glass}}$

Since

$\lambda \propto \dfrac{1}{\mu}$,

the wavelength is inversely proportional to the refractive index.

Therefore,

$\lambda_a > \lambda_w > \lambda_g$

Hence, the correct answer is B) $\lambda_a > \lambda_w > \lambda_g$.

9. Correct Option: C) 1.5

Explanation:

According to Snell's Law,

$\dfrac{\sin i}{\sin r} = \mu_{21}$

where $\mu_{21}$is the refractive index of the second medium with respect to the first.

Since the refractive index of a pair of media (for a given wavelength) is constant, changing the angle of incidence changes the angle of refraction in such a way that this ratio remains unchanged.

Hence,

$\dfrac{\sin i}{\sin r} = 1.5$

Therefore, the correct answer is C) 1.5.

10. Correct Option: B) refracted ray grazes the surface

Explanation:

When light travels from a denser medium to a rarer medium and the angle of incidence equals the critical angle, the angle of refraction becomes

$90°$

Thus, the refracted ray travels along the interface, i.e., it grazes the surface.

Hence, the correct answer is B) refracted ray grazes the surface.

11. Correct Option: B) 9 cm

Explanation:

The apparent depth is

$d' = \dfrac{d}{\mu}$

Given,

- Real depth of first object = 24cm

- Real depth of second object = 36cm
- $\mu = \frac{4}{3}$

Apparent depth of first object,

$d_1' = \frac{24}{4/3} = 18$ cm

Apparent depth of second object,

$d_2' = \frac{36}{4/3} = 27$ cm

Therefore,

Apparent separation $= 27 - 18 = 9$ cm

Hence, the correct answer is B) 9 cm.

12. Correct Option: D) Angles involved must be small

Explanation:

The formula for refraction at a spherical surface,

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$,

is derived using the paraxial approximation.

This approximation assumes that all rays make very small angles with the principal axis, so that

$\sin\theta \approx \tan\theta \approx \theta$

(in radians).

Hence, the correct answer is D) Angles involved must be small.

13. Correct Option: 25.66°

Explanation:

For multiple parallel interfaces,

$\mu_1 \sin\theta_1 = \mu_4 \sin\theta_4$

Given,

$\mu_1 = 1, \theta_1 = 60^\circ, \mu_4 = 2$

Substituting,

$1 \times \sin 60^\circ = 2\sin\theta_4$

$\frac{\sqrt{3}}{2} = 2\sin\theta_4$

$\sin\theta_4 = \frac{\sqrt{3}}{4} \approx 0.433$

Therefore,

$\theta_4 = \sin^{-1}(0.433) \approx 25.66^\circ$

14. Correct Option: B) $\frac{\mu_2}{\mu_1}$

Explanation:

The general formula for refraction at a spherical surface is

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

For a plane surface,

$R \to \infty$

Hence,

$\frac{\mu_2}{v} = \frac{\mu_1}{u}$

Rearranging,

$\frac{v}{u} = \frac{\mu_2}{\mu_1}$

Hence, the correct answer is B) $\frac{\mu_2}{\mu_1}$.

15. Correct Option: B) On image side only

Explanation:

The second principal focus is obtained by considering an object at infinity.

Using the spherical refraction formula,

$$\frac{\mu_2}{f_2} = \frac{\mu_2 - \mu_1}{R}$$

Therefore,

$$f_2 = \frac{\mu_2 R}{\mu_2 - \mu_1}$$

Since

- $R > 0$,
- $\mu_2 > \mu_1$,

the focal length $f_2$is positive.

A positive focal length indicates that the second principal focus lies in the second medium, i.e., on the image side.

Hence, the correct answer is B) On image side only.

16. Correct Option: B) $\frac{v}{u}$

Explanation:

The lateral magnification for refraction at a spherical surface is

$$m = \frac{\mu_1 v}{\mu_2 u}$$

If both media are identical,

$$\mu_1 = \mu_2$$

Therefore,

$$m = \frac{\mu_1 v}{\mu_1 u} = \frac{v}{u}$$

Hence, the correct answer is B) $\frac{v}{u}$.

17. Correct Option: C) Apparent depth predicted by small-angle

Explanation:

The formula

$$d' = \frac{d}{\mu}$$

is valid only under the small-angle (paraxial) approximation, where the observer looks nearly perpendicular to the surface.

When the observer moves to a highly oblique position, the actual apparent depth changes and the simple formula no longer gives the correct value.

Note: The is conceptually flawed because the formula itself predicts the same apparent depth regardless of the viewing angle. The intended idea is that the actual observed apparent depth changes, while the value predicted by the small-angle approximation remains unchanged.

Hence, the intended correct answer is C) Apparent depth predicted by small-angle approximation.

LEVEL-II

SINGLE CORRECT ANSWER TYPE

1

Answer: C) Opposite

Explanation:

Absolute refractive indices (μ_1 and μ_2) are always positive values greater than or equal to 1. Since

$\frac{f_1}{f_2} = -\frac{\mu_1}{\mu_2}$,

the ratio is negative. Therefore, $f_1$and $f_2$must have opposite signs.

2

Answer: B) Parallel to incident ray

Explanation:

According to Snell's law across multiple parallel refracting surfaces,

$\mu_1 \sin i_1 = \mu_n \sin i_n$.

Since the first and last media have the same refractive index ($\mu_n = \mu_1$),

$i_n = i_1$.

Hence, the emergent ray is parallel to the incident ray.

3

Answer: C) Negative according to sign convention

Explanation:

A virtual image formed by a refracting surface lies on the same side as the object. According to the Cartesian sign convention, distances measured opposite to the direction of incident light are negative. Therefore, the image distance $v$is negative.

4

Answer: B) +20 cm

Explanation:

For a thin lens placed in the same medium on both sides,

$f_2 = -f_1$.

Given

$f_1 = -20$ cm,

therefore,

$f_2 = +20$ cm.

5

Answer: C) Converging thin lens

Explanation:

A lens with

- $R_1 > 0$
- $R_2 < 0$

is thicker at the center and thinner at the edges, making it a biconvex lens. In air, a biconvex lens behaves as a converging lens.

6

Answer: C) $\frac{\mu_2 - \mu_1}{R}$

Explanation:

The refracting power of a spherical surface separating two media is

$P = \frac{\mu_2 - \mu_1}{R}$,

where

- μ_1= refractive index of the first medium,
- μ_2= refractive index of the second medium,
- R= radius of curvature.

7

Answer: C) Passes undeviated

Explanation:

The optical center of a thin lens is the point through which an incident light ray passes without any deviation or lateral displacement.

8

Answer: D) Plano-concave lens

Explanation:

A lens that is thinner at the center than at its edges is a concave (diverging) lens. A plano-concave lens satisfies this condition.

9

Answer: $\boldsymbol{m_L = \frac{\mu_2}{\mu_1} m^2}$.

Explanation:

For refraction at a spherical surface,

$m_L = \frac{dv}{du} = \frac{\mu_1}{\mu_2}\left(\frac{v}{u}\right)^2$.

Since the transverse magnification is

$m = \frac{\mu_1 v}{\mu_2 u}$,

we obtain

$m_L = \frac{\mu_2}{\mu_1} m^2$.

10

Answer: B) Appear to diverge from the second focus

Explanation:

A concave lens diverges parallel incident rays. After refraction, the rays spread out and their backward extensions meet at the second principal focus.

11

Answer: C) It joins the centres of curvature of the two spherical surfaces

Explanation:

The principal axis is the straight line passing through the centres of curvature of both spherical surfaces of the lens.

12

Answer: B) $\boldsymbol{\sqrt{2}}$

Explanation:

When the surrounding medium is the same on both sides of the lens,

$m_L = m^2$.

Given

$\frac{dv}{du} = 2$,

we have

$m^2 = 2$

and hence,

$m = \sqrt{2}$.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

13. Correct Options: A, B, D

Explanation

Option A is correct:

The absolute refractive index of a medium is defined as

$\mu = \frac{c}{v}$

where $c$is the speed of light in vacuum and $v$is the speed of light in the medium. Since light travels fastest in vacuum ($v \leq c$), the absolute refractive index of any ordinary medium is always

$\mu \geq 1$.

Therefore, it can never be less than 1.

Option B is correct:

The relative refractive index of medium 2 with respect to medium 1 is

$^1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{V_1}{V_2}$.

If light travels from a denser medium to a rarer medium (for example, glass to water), then

$\mu_2 < \mu_1$,

making

$^1\mu_2 < 1$.

Hence, the relative refractive index can be less than 1.

Option C is incorrect:

During refraction, the frequency of light remains unchanged because it is determined solely by the source. Only the speed and wavelength change when light enters a different medium.

Option D is correct:

When a ray strikes the interface normally,

$i = 0°$.

Using Snell's law,

$\mu_1 \sin i = \mu_2 \sin r$,

$\mu_1 \sin 0° = \mu_2 \sin r$,

$\sin r = 0 \Rightarrow r = 0°$.

Thus, the ray passes straight through the interface without any deviation.

14. Correct Options: A, B, C

Explanation

Option A is correct:

An optically denser medium is one in which light travels more slowly. Since

$V_1 > V_2$,

light moves slower in medium 2. Therefore, medium 2 is optically denser than medium 1.

Option B is correct:

The relative refractive index of medium 2 with respect to medium 1 is

$^1\mu_2 = \frac{V_1}{V_2}$.

Since

$V_1 > V_2$,

it follows that

$\frac{V_1}{V_2} > 1.$

Hence,

${}^1\mu_2 > 1.$

Option C is correct:

When light travels from an optically rarer medium to an optically denser medium, its speed decreases and it bends towards the normal.

Option D is incorrect:

The relation between speed, frequency, and wavelength is

$V = f\lambda.$

Since the frequency remains constant during refraction,

$V \propto \lambda.$

As

$V_2 < V_1,$

the wavelength also decreases in medium 2. Therefore, the statement is incorrect.

15. Correct Options: A, B, C, D

Explanation

Option A is correct:

When a light ray enters a rectangular glass slab obliquely, the emergent ray is parallel to the incident ray but is shifted sideways. This sideways displacement is called lateral displacement.

Option B is correct:

A rectangular glass slab has parallel refracting surfaces. The bending produced at the first surface is exactly compensated by the bending at the second surface, causing the emergent ray to remain parallel to the incident ray.

Option C is correct:

Since the opposite faces of the slab are parallel and the surrounding medium on both sides is the same,

$\angle i = \angle e,$

where $i$is the angle of incidence and $e$is the angle of emergence.

Option D is correct:

The light ray undergoes refraction twice:

- once while entering the slab (air → glass),
- once while leaving the slab (glass → air).

Hence, it experiences two refractions.

16. Correct Options: A, B, D

Explanation

Option A is correct:

When an object is placed inside an optically denser medium and viewed from a rarer medium, it appears closer to the surface than its actual position due to refraction.

Option B is correct:

The apparent depth is given by

$$\text{Apparent depth} = \frac{\text{Real depth}}{\mu}.$$

Since

$\mu > 1,$

the apparent depth is always smaller than the real depth.

Option C is incorrect:
Suppose two objects are at real depths $d_1$and d_2.
The real separation is
$\Delta d_{\text{real}} = d_2 - d_1$.
Their apparent separation becomes
$\Delta d_{\text{apparent}} = \frac{d_2}{\mu} - \frac{d_1}{\mu} = \frac{d_2 - d_1}{\mu} = \frac{\Delta d_{\text{real}}}{\mu}$.
Since
$\mu > 1$,
the apparent separation decreases, not increases.
Option D is correct:
The apparent upward shift is
Shift = Real depth $\left(1 - \frac{1}{\mu}\right)$.
Therefore, the amount of shift depends directly on the refractive index of the medium.

17. Correct Options: A, B, D

Explanation

Option A is correct:
A convex lens is thicker at the centre and thinner at the edges. Its outwardly curved surfaces enable it to converge parallel rays of light.
Option B is correct:
According to the Cartesian sign convention, the second principal focus (F_2) of a convex lens lies on the positive side of the optical axis. Therefore,
$f_2 > 0$.
Option D is correct:
Convex lenses are manufactured in different shapes depending on their application, such as:

- Biconvex lens (both surfaces convex)
- Plano-convex lens (one plane surface and one convex surface)
- Positive meniscus lens (one convex and one concave surface, thicker at the centre)

All these are types of convex lenses.

Assertion–Reason s

18. Answer: A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

Explanation:

The relative refractive index is
$\mu_{21} = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}$.
It can be less than 1 when light travels from an optically denser medium to a rarer medium. Since it is defined as the ratio of the speeds of light in the two media, the Reason correctly explains the Assertion.

19. Answer: A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

Explanation:

The optical density of a medium is directly related to its refractive index (μ). Since
$v = \frac{c}{\mu}$,

the speed of light decreases as the refractive index increases. Therefore, light travels slower in an optically denser medium, making the Reason the correct explanation.

20. Answer: A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

Explanation:

A plane refracting surface can be considered a spherical surface with $R = \infty$.

Substituting this into the spherical refraction formula yields the apparent depth relation

$$d_{\text{apparent}} = \frac{d_{\text{real}}}{\mu},$$

showing that apparent depth is inversely proportional to the refractive index. Hence, the Reason correctly explains the Assertion.

21. Answer: A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

Explanation:

The derivation of the spherical refraction formula is based on paraxial rays, which travel close to the principal axis and make very small angles with it. Under these conditions, the small-angle approximation

$\sin\theta \approx \theta$

is valid, allowing the mathematical derivation. Therefore, the Reason correctly explains the Assertion.

22. Correct Matches:

- **(A) → (Q)**
- **(B) → (P)**
- **(C) → (R)**
- **(D) → (S)**

Explanation

(A) → (Q): Absolute Refractive Index

The formula

$$\mu = \frac{c}{V_m}$$

represents the ratio of the speed of light in vacuum (c)to the speed of light in a medium (V_m). Hence, it defines the Absolute Refractive Index.

(B) → (P): Relative Refractive Index

The formula

$$^1\mu_2 = \frac{V_1}{V_2}$$

gives the ratio of the speed of light in medium 1 to that in medium 2. Therefore, it represents the Relative Refractive Index of medium 2 with respect to medium 1.

(C) → (R): Frequency Remains Constant

The wave equation is

$V = \nu\lambda$

where $V$is the speed, $\nu$is the frequency, and $\lambda$is the wavelength.

When light passes from one medium to another:

- Frequency (ν)remains constant.
- Speed changes.
- Wavelength changes proportionally.

Thus,

$\lambda \propto V$.

(D) → (S): Emergent Ray from a Glass Slab

A ray passing through a parallel-sided glass slab undergoes refraction at both surfaces. Since the two surfaces are parallel, the deviations cancel each other. Hence, the emergent ray is parallel to the incident ray, experiencing only a lateral displacement.

23. Correct Matches:

- **(A) → (Q)**
- **(B) → (P)**
- **(C) → (R)**
- **(D) → (S)**

Explanation

The refraction formula for a spherical surface is

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

(A) → (Q): First Principal Focal Length

For the first principal focus,

$v = \infty$

Substituting into the refraction formula,

$\frac{\mu_2}{\infty} - \frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R}$

$-\frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R}$

Therefore,

$\boxed{f_1 = -\frac{\mu_1 R}{\mu_2 - \mu_1}}$

(B) → (P): Second Principal Focal Length

For the second principal focus,

$u = -\infty$

Substituting,

$\frac{\mu_2}{f_2} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{R}$

$\frac{\mu_2}{f_2} = \frac{\mu_2 - \mu_1}{R}$

Hence,

$\boxed{f_2 = \frac{\mu_2 R}{\mu_2 - \mu_1}}$

(C) → (R): Lateral Magnification

The lateral magnification of a spherical refracting surface is

$\boxed{m = \frac{\mu_1 v}{\mu_2 u}}$

(D) → (S): Plane Refracting Surface

For a plane surface,

$R \to \infty$

Hence,

$\frac{\mu_2 - \mu_1}{R} = 0$

The refraction formula becomes

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = 0$

or

$\boxed{\frac{v}{u} = \frac{\mu_2}{\mu_1}}$

24. Answer: B) $\frac{d}{\mu}$

Explanation

When viewed normally from air, the apparent depth (or apparent distance) is given by

$h' = \frac{h}{\mu}$

where

- h= real depth,
- h'= apparent depth,
- μ= refractive index.

If the real separation between two consecutive beads is d, then their apparent separation becomes

$\boxed{\frac{d}{\mu}}$

25. Answer: C) $\mu \sin\theta$

Explanation

According to Snell's law,

$\mu \sin\theta$ = constant

across successive parallel boundaries.

Hence, the invariant quantity is

$\boxed{\mu \sin\theta}$

26. Answer: D) Decrease further

Explanation

The apparent depth is

$h' = \frac{h}{\mu}$

Since the apparent depth is inversely proportional to the refractive index, replacing the liquid with one having a larger refractive index makes the apparent depth decrease further.

27. Answer: B) Parallel to the principal axis

Explanation

When an object is placed at the first principal focus of a convex lens, the emerging rays become

$\boxed{\text{Parallel to the principal axis}}$

28. Answer: C) Infinity

Explanation

Parallel rays after refraction meet at

$\boxed{\text{Infinity}}$

Hence, the image is formed at infinity.

29. Answer: C) An object at the first focus forms image at infinity

Explanation

A convex lens produces parallel emergent rays when the object is placed at its first principal focus. Consequently, the image is formed at infinity.

Hence, the correct answer is

An object at the first focus forms image at infinity

30. Answer: 1.5

Explanation

The absolute refractive index is

$\mu = \frac{c}{v}$

Given,

$c = 3 \times 10^8$ m/s, $v = 2 \times 10^8$ m/s

Therefore,

$\mu = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$

1.5

31. Answer: 1.52

Explanation

Using Snell's law,

$\frac{\sin i}{\sin r} = \frac{\sin 40^\circ}{\sin 25^\circ}$

Substituting,

$= \frac{0.6428}{0.4226} \approx 1.52$

Hence,

1.52

32. Answer: 1.6875×10^8 m/s

Explanation

The relative refractive index is

$^w\mu_g = \frac{V_{\text{water}}}{V_{\text{glass}}}$

Given,

$^w\mu_g = \frac{4}{3}, V_{\text{water}} = 2.25 \times 10^8$ m/s

Therefore,

$\frac{4}{3} = \frac{2.25 \times 10^8}{V_{\text{glass}}}$

$V_{\text{glass}} = 2.25 \times 10^8 \times \frac{3}{4}$

$V_{\text{glass}} = 1.6875 \times 10^8$ m/s

TOPIC-2:

Image Formation by Lenses and Optical Instruments

LEVEL-I

1. Correct Option: B) 20 cm

Explanation:

Using the lens formula,

$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$

Given:

- Image distance, $v=+24$ cm
- Object distance, $u=-120$ cm

Substituting,

$\frac{1}{f}=\frac{1}{24}-\left(-\frac{1}{120}\right)=\frac{1}{24}+\frac{1}{120}=\frac{5+1}{120}=\frac{6}{120}=\frac{1}{20}$

$\boxed{f=20 \text{ cm}}$

2. Correct Option: C) Real and inverted at 30 cm

Explanation:

The focal length of the convex lens is

$f=15$ cm

Hence,

$2f=2\times 15=30$ cm

When an object is placed at $2f$, a convex lens forms:

- a real image,
- inverted,
- of the same size,
- at $2f$ on the opposite side of the lens.

Therefore, the image is formed at

$\boxed{30 \text{ cm}}$

3. Correct Option: C) 6

Explanation:

For a simple microscope with the final image formed at the near point ($D=25$ cm),

$m=1+\frac{D}{f}$

Given,

$D=25$ cm, $f=5$ cm

$m=1+\frac{25}{5}=1+5=6$

$\boxed{m=6}$

4. Correct Option: B) 12 cm

Explanation:

For two thin lenses in contact,

$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}$

Given,

$f_1=20$ cm, $f_2=30$ cm

$\frac{1}{f}=\frac{1}{20}+\frac{1}{30}=\frac{3+2}{60}=\frac{5}{60}=\frac{1}{12}$

Hence,

$\boxed{f=12 \text{ cm}}$

5. Correct Option: B) -200 cm

Explanation:

A person who cannot see clearly beyond

200 cm

has myopia (short-sightedness).
To correct myopia, a concave lens is used whose focal length is equal to the far point (with a negative sign).
Therefore,
$\boxed{f = -200 \text{ cm}}$

6. Correct Option: B) 16 cm

Explanation:
Using the lens displacement method,
$f = \frac{D^2 - d^2}{4D}$
Given,
$D = 100 \text{ cm}, d = 60 \text{ cm}$
$f = \frac{100^2 - 60^2}{4 \times 100} = \frac{10000 - 3600}{400} = \frac{6400}{400} = 16 \text{ cm}$
$\boxed{f = 16 \text{ cm}}$

7. Correct Option: B) Concave lens

Explanation:
An air bubble has a refractive index
$n_{\text{air}} \approx 1.0$
while the surrounding water has
$n_{\text{water}} \approx 1.33$.
Since the bubble has a lower refractive index than the surrounding medium, light rays diverge after passing through it.
Hence, an air bubble in water behaves like a
$\boxed{\text{Concave (diverging) lens.}}$

8. Correct Option: B) 50 cm

Explanation:
Power and focal length are related by
$P = \frac{100}{f}$
where $f$is in centimetres.
Given,
$P = +2 \text{ D}$
$f = \frac{100}{2} = 50 \text{ cm}$
$\boxed{f = 50 \text{ cm}}$

9. Correct Option: C) 9

Explanation:
Magnification is given by
$m = \frac{v}{u}$
Given,
$v = 18 \text{ cm}, u = 2 \text{ cm}$
$m = \frac{18}{2} = 9$
$\boxed{m = 9}$

10. Correct Option: B) Hypermetropia

Explanation:

A person who can see distant objects clearly but has difficulty seeing nearby objects suffers from

Hypermetropia (long-sightedness).

This defect is corrected using a convex lens.

11. Correct Option: C) 30 cm

Explanation:

Using the lens formula,

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Given,

$f = +20 \text{ cm}, u = -60 \text{ cm}$

$\frac{1}{20} = \frac{1}{v} + \frac{1}{60}$

$\frac{1}{v} = \frac{1}{20} - \frac{1}{60} = \frac{3-1}{60} = \frac{2}{60} = \frac{1}{30}$

Therefore,

$v = 30 \text{ cm}$

12. Correct Option: B) 25 cm

Explanation:

The focal length of a lens is related to its power by

$f = \frac{100}{P}$

Given,

$P = +4 \text{ D}$

$f = \frac{100}{4} = 25 \text{ cm}$

Thus,

$f = 25 \text{ cm}$

13. Correct Option: C) 14°

Explanation:

Angular dispersion is given by

$\theta = \delta_v - \delta_r$

Given,

- Angular dispersion = 4°
- Deviation of violet light = 18°

Therefore,

$4^\circ = 18^\circ - \delta_r$

$\delta_r = 18^\circ - 4^\circ = 14^\circ$

Hence,

$\delta_r = 14^\circ$

14. Correct Option: A) $+\frac{1}{2}$

Explanation:

A concave lens always forms a virtual and erect image.

Given,

$v = -12 \text{ cm}, u = -24 \text{ cm}$

Magnification,

$m = \frac{v}{u} = \frac{-12}{-24} = \frac{1}{2}$

$\boxed{m = +\frac{1}{2}}$

15. Correct Option: B) 50

Explanation:

For a compound microscope in normal adjustment,

$m \approx \frac{L \times D}{f_o \times f_e}$

Given,

- Tube length, $L = 20$ cm
- Least distance of distinct vision, $D = 25$ cm
- Objective focal length, $f_o = 2$ cm
- Eyepiece focal length, $f_e = 5$ cm

Substituting,

$m = \frac{20 \times 25}{2 \times 5} = \frac{500}{10} = 50$

Therefore,

$\boxed{m = 50}$

Note: If the microscope is assumed to produce the final image at the near point instead of normal adjustment, the magnifying power may be 60. Based on the standard normal-adjustment formula, the correct answer is 50.

16. Correct Option: B) 20 cm

Explanation:

Magnification produced by a lens is

$m = \frac{v}{u}$

Since the image is real and formed on a screen, it is inverted.

Hence,

$m = -2$

Given,

$v = 40$ cm

$-2 = \frac{40}{u}$

$u = -20$ cm

Therefore, the magnitude of the object distance is

$\boxed{20 \text{ cm}}$

LEVEL-II

SINGLE CORRECT ANSWER TYPE

1. Correct Option: B

Explanation:

The relationship between the critical angle $C$and the refractive index $\mu$is:

$\sin C = \frac{1}{\mu}$

Substituting $\mu = 1.5$:

$\sin C = \frac{1}{1.5} = 0.667$

$C = \sin^{-1}(0.667) \approx 42°$

Hence, the correct option is B.

2. Correct Option: B

Explanation:

For a simple microscope (magnifying glass) with the final image at infinity,

$M = \frac{D}{f}$

where $D = 25$ cm.

Given,

$3 = \frac{25}{f}$

$f = \frac{25}{3} \approx 8.3$ cm

Hence, the correct option is B.

3. Correct Option: A

Explanation:

When a light ray is incident normally on a surface,

$i = 0°$

By Snell's law,

$r = 0°$

Thus, the ray passes straight through the surface without any deviation.

Hence, the angle of deviation,

$\delta = 0°$

Therefore, the correct option is A.

4. Correct Option: C

Explanation:

According to the ray diagram of a convex lens, a light ray passing through the principal focus before refraction emerges parallel to the principal axis after refraction. Hence, the image is formed at infinity.

Therefore, the correct option is C.

5. Correct Option: B

Explanation:

For a myopic eye,

$f = -50$ cm $= -0.5$ m

Power of the correcting lens is

$P = \frac{1}{f}$

$P = \frac{1}{-0.5} = -2$ D

Hence, the correct option is B.

6. Correct Option: C

Explanation:

According to Cauchy's relation, the refractive index of a medium is greater for shorter wavelengths.

$\mu_{\text{violet}} > \mu_{\text{red}}$

Since deviation increases with refractive index, violet light deviates more than red light.

Hence, the correct option is C.

7. Correct Option: B

Explanation:

For two thin lenses in contact,

$P_{eq} = P_1 + P_2$

Given,

$P_{eq} = +3 + 2 = +5$ D

Hence, the correct option is B.

8. Correct Option: C

Explanation:

Twinkling of stars occurs due to atmospheric refraction. Layers of air having different temperatures and densities continuously change the apparent position and brightness of stars.

Hence, the correct option is C.

9. Correct Option: B

Explanation:

For a thin prism,

$\delta = (\mu - 1)A$

Given,

$4° = (\mu - 1) \times 8°$

$\mu - 1 = \frac{4}{8} = 0.5$

$\mu = 1.5$

Hence, the correct option is B.

10. Correct Option: B

Explanation:

For correction of hypermetropia,

$u = -25$ cm, $v = -50$ cm

Using the lens formula,

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{f} = \frac{1}{-50} - \frac{1}{-25}$

$\frac{1}{f} = -\frac{1}{50} + \frac{1}{25} = \frac{1}{50}$

$f = +50$ cm

Hence, the correct option is B.

11. Correct Option: C

Explanation:

A ray passing through the optical center of a thin lens undergoes equal and opposite refractions at the two surfaces and emerges practically undeviated.

Hence, the correct option is C.

12. Correct Option: B

Explanation:

Given,

$f = -40$ cm $= -0.4$ m

Power,

$P = \frac{1}{f}$

$P = \frac{1}{-0.4} = -2.5$ D

Hence, the correct option is B.

13. Correct Option: C

Explanation:

For minimum deviation in a prism,

$i = e$

The light ray travels symmetrically inside the prism.

Hence, the correct option is C.

14. Correct Option: B

Explanation:

The magnifying power of a compound microscope is

$M \approx \frac{L}{f_o} \times \frac{D}{f_e}$

where f_o and f_e are the focal lengths of the objective and eyepiece.

A smaller objective focal length gives greater magnification.

Hence, the correct option is B.

15. Correct Option: B

Explanation:

For an object at infinity, the image forms at the focal plane.

Given,

$f = 10$ cm $= 0.1$ m

Power,

$P = \frac{1}{0.1} = 10$ D

Hence, the correct option is B.

16. Correct Option: C

Explanation:

Total internal reflection occurs only when

$i > C$

Since the angle of incidence is

$i = 45°$,

the critical angle must be slightly less than $45°$. For ordinary crown glass,

$C \approx 42°$.

Hence, the correct option is C.

17. Correct Option: C

Explanation:

According to the Lens Maker's Formula,

$\frac{1}{f} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{medium}}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

If

$\mu_{\text{lens}} = \mu_{\text{medium}}$,

then

$\left(\frac{\mu_{\text{lens}}}{\mu_{\text{medium}}} - 1\right) = 0.$

Therefore,

$\frac{1}{f} = 0$

which implies

$f = \infty.$

Hence, the correct option is C.

18. Correct Option: C

Explanation:

Clouds contain water droplets and ice crystals that are much larger than the wavelength of visible light. They scatter all colours almost equally (Mie scattering), making clouds appear white.

Hence, the correct option is C.

19. Correct Option: C

Explanation:

In the lens displacement method, a real image can be obtained only if

$D \geq 4f$,

where $D$is the distance between the object and the screen.

If

$D < 4f$,

no real image can be formed on the screen.

Hence, the correct option is C.

20. Correct Option: B

Explanation:

Dispersion occurs because different colours of light have different refractive indices in a medium.

If every colour had the same refractive index, all colours would deviate by the same amount and no dispersion would occur.

Hence, the correct option is B.

21. Correct Option: B

Explanation:

Bifocal spectacles are used to correct presbyopia along with defects in distant vision.

- Upper portion: Concave lens (for distant vision)
- Lower portion: Convex lens (for near vision/reading)

Hence, the correct option is B.

22. Correct Option: A

Explanation:

In a compound microscope:

- The objective lens forms a real, inverted, magnified intermediate image.
- The eyepiece acts as a simple microscope and magnifies this image.

Therefore, the final image remains inverted.

Hence, the correct option is A.

23. Correct Option: B

Explanation:

The least distance of distinct vision (near point) for a normal human eye is 25 cm.

Hence, the correct option is B.

24. Correct Option: B

Explanation:

A concave lens always forms a virtual, erect, and diminished image.

A convex lens forms a virtual, erect, and magnified image only when the object is placed between the optical center and the principal focus.

Hence, the correct option is B.

25. Correct Option: B

Explanation:

At the critical angle,

$i = C$.

The refracted ray makes an angle of

90°

with the normal and travels along the interface.

Hence, the correct option is B.

26. Correct Option: D

Explanation:

According to Rayleigh's scattering law,

$I \propto \frac{1}{\lambda^4}$

where I is the intensity of scattered light and λ is its wavelength.

Since 400 nm is the shortest wavelength among the given options, it undergoes the maximum scattering.

Hence, the correct option is D.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

27. Correct Options: A, B, C

Explanation:

The focal length of the convex lens is $f = +20$ cmand the object distance is $u = -10$ cm.

Since the object is placed between the optical center and the principal focus ($|u| < |f|$), a convex lens forms a virtual, erect, and magnified image on the same side of the lens as the object. Therefore, Statements A and B are correct.

Using the magnification formula,

$m = \frac{f}{f+u}$

$m = \frac{20}{20+(-10)} = \frac{20}{10} = +2$

Since the magnification is positive and greater than 1, the image is erect and magnified. Hence, Statement C is also correct.

A virtual image cannot be obtained on a screen, so Statement D is incorrect.

28. Correct Options: A, B, C

Explanation:

At the position of minimum deviation ($\delta = \delta_m$), the path of light inside the prism becomes symmetrical.

Therefore,

- The angle of incidence equals the angle of emergence ($i = e$). Hence, Statement A is correct.
- The angles of refraction at the two faces are equal ($r_1 = r_2$). Hence, Statement C is correct.
- In a prism of standard geometry (such as an equilateral prism), the refracted ray inside the prism travels parallel to the base. Hence, Statement B is correct. Although the deviation is minimum, it is not zero. Therefore, Statement D is incorrect.

29. Correct Options: A, B, C

Explanation:

Total Internal Reflection (TIR) occurs only when the following two conditions are satisfied:

1. Light travels from an optically denser medium to an optically rarer medium (for example, glass to air). Therefore, Statement B is correct.
2. The angle of incidence in the denser medium is greater than the critical angle ($i > \theta_c$). Therefore, Statement A is correct.

When the angle of incidence is equal to the critical angle,

$i = \theta_c$

the angle of refraction becomes

$r = 90°$,

which means the refracted ray just grazes the interface. Hence, Statement C is correct.

Therefore, the correct options are A, B, and C.

30. Correct Options: A, B, C

Explanation:

The magnifying power of a compound microscope (when the final image is formed at the near point) is approximately given by

$M \approx \frac{L}{f_o}\left(1+\frac{D}{f_e}\right)$,

where

- L= tube length,
- f_o= focal length of the objective,
- f_e= focal length of the eyepiece,
- D= least distance of distinct vision.

From the formula:

- Magnifying power is inversely proportional to the focal length of the objective. Therefore, using an objective with a smaller focal length increases magnification. Hence, Statement A is correct.
- Magnifying power is directly proportional to the tube length. Therefore, increasing the tube length increases magnification. Hence, Statement B is correct.
- Magnifying power is also inversely proportional to the focal length of the eyepiece. Therefore, using an eyepiece with a smaller focal length increases magnification. Hence, Statement C is correct.

Therefore, the correct options are A, B, and C.

31. Answer: D

Explanation:

A standard glass convex lens ($\mu \approx 1.5$) has a higher refractive index than water ($\mu \approx 1.33$). Since the refractive index of the lens is still greater than that of the surrounding medium, the lens continues to behave as a converging (convex) lens when immersed in water. Therefore, the Assertion is false.

The Reason is true because the behavior of any lens depends on the relative refractive index between the lens material and the surrounding medium.

32. Answer: A

Explanation:

According to Cauchy's formula, the refractive index of glass decreases with increasing wavelength. Hence,

$\mu_{\text{violet}} > \mu_{\text{red}}$

Since the angle of deviation produced by a prism is approximately

$\delta = (\mu - 1)A,$

violet light undergoes the maximum deviation, while red light undergoes the minimum deviation.

33. Answer: A

Explanation:

A rainbow is formed due to a combination of:

- Refraction of sunlight,
- Dispersion into different colours, and
- Internal reflection inside water droplets.

After internal reflection, the light emerges from the droplet and reaches the observer. Therefore, the Sun must always be behind the observer, while the rainbow appears in the opposite direction.

34. Answer: A

Explanation:

According to Rayleigh's scattering law, the intensity of scattered light is inversely proportional to the fourth power of its wavelength.

$I \propto \frac{1}{\lambda^4}$

Blue light has a much shorter wavelength than red light and is therefore scattered much more strongly. This is why the sky appears blue during the daytime.

MATRIX MATCHING TYPE

35. Answers

Explanation

A → P:

When an object is placed at the principal focus (F) of a convex lens, the refracted rays emerge parallel to each other. Hence, the image is formed at infinity.

B → Q:

When an object is placed between the optical centre and the principal focus of a convex lens, the image formed is virtual, erect, and magnified on the same side of the lens. This is the principle of a magnifying glass.

C → R:

Parallel rays coming from an object at infinity incident on a concave lens diverge after refraction but appear to originate from the principal focus (F).

D → S:

A concave lens always forms a virtual, erect, and diminished image for any real object placed at a finite distance.

36. Answers

Explanation

A → R: Power of a Lens

$P = \frac{1}{f}$

where $P$is in dioptres and $f$is in metres.

B → P: Magnification of a Lens

$m = \frac{v}{u}$

where $v$is the image distance and $u$is the object distance.

C → Q: Deviation Produced by a Thin Prism

$\delta = (\mu - 1)A$

where $\delta$is the angle of deviation, $\mu$is the refractive index, and $A$is the prism angle.

D → S: Lens Formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

This equation relates the object distance, image distance, and focal length of a lens.

PASSAGE–I S

37. Answer: B

Explanation:

A rainbow is formed due to three successive phenomena occurring inside water droplets:

- Refraction,
- Dispersion of white light into its constituent colours, and
- Internal reflection.

The combined effect of these processes produces the colourful arc seen in the sky.

38. Answer: D

Explanation:

Among all visible colours, violet light has the shortest wavelength and therefore experiences the highest refractive index in glass.

Since

$\delta \propto (\mu - 1)$,

violet light undergoes the maximum deviation.

39. Answer: D

Explanation:

According to Rayleigh's scattering law,

$I \propto \frac{1}{\lambda^4}$

Therefore,

$\frac{I_{\text{blue}}}{I_{\text{red}}} = \left(\frac{\lambda_{\text{red}}}{\lambda_{\text{blue}}}\right)^4$

Given,

$\lambda_{\text{red}} = 2\lambda_{\text{blue}}$

Hence,

$\frac{I_{\text{blue}}}{I_{\text{red}}} = 2^4 = 16$

Thus, blue light is scattered 16 times more strongly than red light.

PASSAGE–II S

40. Answer: B

Explanation:

In a compound microscope, the objective lens forms a real, inverted, and magnified intermediate image of the object inside the microscope tube.

41. Answer: B or C

Explanation:

The eyepiece acts as a simple magnifying glass. It magnifies the intermediate image

formed by the objective lens and produces a virtual, magnified image. Relative to the intermediate image, the final image is erect.
Hence, both options B and C are correct.

42. Answer: C

Explanation:

The total magnifying power of a compound microscope is the product of the magnifications of the objective and eyepiece.

$M = m_o \times m_e$

Given,

$m_o = 15, m_e = 10$

Therefore,

$M = 15 \times 10 = 150$

Thus, the total magnification is 150.

INTEGER TYPE

43. Answer: 50

Explanation:

Given,

$f = +25$ cm, $u = -50$ cm

Since the object is placed at

$2f = 2 \times 25 = 50$ cm,

a convex lens forms a real, inverted image of the same size at a distance of $2f$ on the opposite side.

Therefore,

$v = +50$ cm

Hence, the required answer is

$\boxed{50}$

44. Answer: 20

Explanation:

The power of a lens is related to its focal length by

$P = \frac{100}{f}$

where f is measured in centimetres.

Given,

$P = 5$ D

Therefore,

$f = \frac{100}{5} = 20$ cm

Hence, the answer is

$\boxed{20}$

45. Answer: 4

Explanation:

For a thin prism,

$\delta = (\mu - 1)A$

Given,

$\mu = 1.4, A = 10^\circ$

Therefore,

$\delta = (1.4 - 1) \times 10^\circ$
$\delta = 0.4 \times 10^\circ = 4^\circ$

PREVIOUS YEAR S

1. Correct Option: A. Virtual, Upright, Magnified

Explanation:

The object is placed at position P, which lies between the principal focus (F) and the optical center (O) of the convex (converging) lens.

When an object is placed within the focal length of a convex lens:

- The refracted rays diverge after passing through the lens.
- Their backward extensions meet on the same side of the lens as the object.
- Hence, the image formed is:
  - Virtual
  - Upright (erect)
  - Magnified

2. Correct Option: A. 79

Explanation:

Step 1: Angle of incidence in air

The orifice is 45 cm above the bottom on the left wall, while the observer's eye is 15 cm above the bottom on the right wall.

- Vertical difference = $45 - 15 = 30$ cm
- Horizontal distance (diameter of the container) = $2 \times 15 = 30$ cm

Hence,

$\tan i = \frac{30}{30} = 1$

$i = 45^\circ$

Step 2: Determine the angle of refraction

The liquid level is 30 cm above the bottom.

From the orifice (45 cm) to the liquid surface (30 cm), the ray travels through air.

- Vertical distance in air = $45 - 30 = 15$ cm

Since the ray makes an angle of 45°,

- Horizontal distance covered = 15 cm

Thus, the ray enters the liquid at the midpoint of the container.

To reach the bottom corner of the opposite wall, the refracted ray travels:

- Horizontal distance = 15 cm
- Vertical distance = 30 cm

Therefore,

$\tan r = \frac{15}{30} = 0.5$

Using the right triangle,

$\sin r = \frac{1}{\sqrt{5}}$

Step 3: Apply Snell's Law

$\mu = \frac{\sin i}{\sin r}$

Substituting,

$\mu = \frac{\sin 45^\circ}{1/\sqrt{5}}$

$= \frac{1/\sqrt{2}}{1/\sqrt{5}} = \sqrt{\frac{5}{2}} \approx 1.581$

Given,

$\mu = \frac{x}{50}$

$\frac{x}{50} = 1.581$

$x = 50 \times 1.581 \approx 79.05$

$\boxed{x = 79}$

3. Correct Option: B. (ii) only

Explanation:

The colour of the sky is blue.

When white light passes through a prism, it disperses into its constituent colours because different colours undergo different amounts of deviation.

Prism in orientation (i)

In the normal upright position:

- Red is deviated the least.
- Violet is deviated the most.

Hence, the spectrum from top to bottom is:

Red → Orange → Yellow → Green → Blue → Indigo → Violet

The third colour from the top is Yellow, not Blue.

Prism in orientation (ii)

When the prism is inverted, the direction of deviation reverses.

The spectrum from top to bottom becomes:

Violet → Indigo → Blue → Green → Yellow → Orange → Red

Here, the third colour from the top is Blue, which matches the colour of the sky.

Therefore, only diagram (ii) satisfies the given condition.

4. Correct Answer: A) 16 cm

Explanation:

Let the actual height of water in the container be $h$cm.

- Total height of the container = 28cm.
- Hence, the actual height of the empty space above the water is:

$28 - h$

When the container is viewed from above, the water appears shallower due to refraction.

Given the refractive index of water,

$\mu = \frac{4}{3}$

The apparent depth of the water is:

Apparent Depth $= \frac{\text{Actual Depth}}{\mu} = \frac{h}{4/3} = \frac{3h}{4}$

The student stops pouring when the container appears to be half filled and half empty. Therefore,

Apparent depth of water = Actual height of empty space

$\frac{3h}{4} = 28 - h$

Multiplying both sides by 4,

$3h = 112 - 4h$
$7h = 112$
$h = 16$ cm
Hence, the actual height of the water is
16 cm

5. Correct Answer: C) Continuously decreases

Explanation:

As a light source is moved farther away from a convex lens along its principal axis, the light reaching the lens becomes progressively less intense.
This is because:

- The light spreads over a larger area as the distance from the source increases.
- Consequently, a smaller fraction of the emitted light is intercepted by the lens.
- Therefore, the luminous flux (amount of light energy) passing through the lens decreases continuously.

Although the rays become more nearly parallel as the source moves farther away, the reduction in the amount of light reaching the lens dominates.
Hence, the intensity of the light passing through the lens continuously decreases.
Answer: C) Continuously decreases

6. Correct Answer: C) III

Explanation:

When white light passes through a prism, it undergoes dispersion, splitting into its constituent colours because different wavelengths are refracted by different amounts.
To recombine these colours into white light, a second identical prism must be placed:

- with its refracting edge opposite to that of the first prism, and
- in an inverted orientation.

This second prism produces dispersion in the opposite direction, exactly cancelling the dispersion produced by the first prism and recombining the spectrum into a single beam of white light.
Since the vertex $P$of the first prism points upward, the second prism must have its vertex P'pointing downward. Among the given options, this corresponds to orientation III.
Answer: C) III

8. Electricity

TOPIC-1:
Static Electricity, Current and Electric Cells

1. Answer: B

Explanation:

When two identical conducting spheres are brought into contact, the total charge is shared equally between them.

$Q_{\text{total}} = (+9.6 \times 10^{-18}) + (-3.2 \times 10^{-18}) = +6.4 \times 10^{-18}$ C

Each sphere gets

$$q = \frac{6.4 \times 10^{-18}}{2} = +3.2 \times 10^{-18} \text{ C}$$

2. Answer: B

Explanation:

Total charge transferred:

$Q = ne = (5 \times 10^{18})(1.6 \times 10^{-19}) = 0.8$ C

Current is

$$I = \frac{Q}{t} = \frac{0.8}{40} = 0.02 \text{ A}$$

3. Answer: D

Explanation:

The cells are connected in series, so their EMFs add.

$E = 1.5 + 1.1 + 2.0 = 4.6$ V

Electrical energy supplied is

$W = QE = 240 \times 4.6 = 1104$ J

4. Answer: A

Explanation:

Net loss of electrons:

$2.5 \times 10^{13} - 1.5 \times 10^{13} = 1.0 \times 10^{13}$

Since electrons are lost, the body becomes positively charged.

$Q = (1.0 \times 10^{13})(1.6 \times 10^{-19}) = +1.6 \times 10^{-6}$ C

5. Answer: A

Explanation:

Time:

20 min = 1200 s

Total charge passed:

$Q = It = 2.5 \times 1200 = 3000$ C

Total electrons:

$$n = \frac{3000}{1.6 \times 10^{-19}} = 1.875 \times 10^{22}$$

Only 80% contribute:

$n_{\text{effective}} = 0.8 \times 1.875 \times 10^{22} = 1.5 \times 10^{22}$

6. Answer: B

Explanation:

Total charge:

$Q = +8 - 2 + 6 - 4 = +8$ C

Since the four spheres are identical,

$q = \frac{8}{4} = 2$ C

7. Answer: B

Explanation:

Total charge:

$120 + 480 = 600$ C

Total time:

$2 + 8 = 10$ min $= 600$ s

Average current:

$I_{\text{avg}} = \frac{600}{600} = 1$ A

8. Answer: A

Explanation:

First body has a deficiency of electrons:

$q_1 = +4 \times 10^{12}e$

Second body has an excess of electrons:

$q_2 = -6 \times 10^{12}e$

Net charge:

$Q = -2 \times 10^{12} \times 1.6 \times 10^{-19} = -3.2 \times 10^{-7}$ C

After contact,

$q = \frac{-3.2 \times 10^{-7}}{2} = -1.6 \times 10^{-7}$ C

9. Answer: C

Explanation:

Charge delivered:

$Q = (3 \times 10^{20})(1.6 \times 10^{-19}) = 48$ C

Energy supplied:

$W = QV = 48 \times 6 = 288$ J

10. Answer: A

Explanation:

Current is

$I = \frac{ne}{t}$

If current and time remain constant, then nemust remain constant. Therefore, if the charge on one electron (e) is doubled, the number of electrons (n) must become half.

11. Answer: A

Explanation:

Total EMF:

$E = 1.5 + 1.5 + 1.1 = 4.1$ V

Time:

10 min $= 600$ s

Energy:

$W = EIt = 4.1 \times 2 \times 600 = 4920$ J

12. Answer: 7.5

Explanation:

Charge is quantized:

$Q = ne$

Given charge:
12×10^{-19} C
Number of electrons:
$n = \frac{12 \times 10^{-19}}{1.6 \times 10^{-19}} = 7.5$

13. Answer: B

Explanation:
Time:
2.5 h = 9000 s
Charge:
$Q = It = 0.8 \times 9000 = 7200$ C
Electrons:
$n = \frac{7200}{1.6 \times 10^{-19}} = 4.5 \times 10^{22}$

14. Answer: B

Explanation:
Let the charges be $3x$and $5x$.
$3x + 5x = 1.28 \times 10^{-17}$
$8x = 1.28 \times 10^{-17}$
$x = 1.6 \times 10^{-18}$ C
Smaller charge:
$3x = 4.8 \times 10^{-18}$ C

15. Answer: B

Explanation:
Charge transferred:
$Q = (2.4 \times 10^{21})(1.6 \times 10^{-19}) = 384$ C
Time:
8 min = 480 s
Current:
$I = \frac{384}{480} = 0.8$ A

16. Answer: C

Explanation:
Number of electrons required:
$n = \frac{Q}{e} = \frac{4.8 \times 10^{-17}}{1.6 \times 10^{-19}} = 300$
Therefore, 300 electrons must be added.

17. Answer: A

Explanation:
First contact:
$\frac{12 + (-6)}{2} = +3$ C
The second sphere now has +3 C.
Second contact:
$\frac{3 + 0}{2} = +1.5$ C
Hence, the third sphere finally has
$\boxed{+1.5 \text{ C}}$

18. Answer: D

Explanation:

Potential difference is

$V = \frac{W}{Q}$

$V = \frac{7200}{1200} = 6$ V

19. Answer: B or C

Explanation:

Electrons removed:

$0.25 \times 8 \times 10^{20} = 2 \times 10^{20}$

Charge produced:

$Q = (2 \times 10^{20})(1.6 \times 10^{-19}) = +32$ C

20. Answer: B

Explanation:

Initial charge:

$Q = It = 1.5 \times 40 = 60$ C

Since 20% leaks away,

$Q_{\text{effective}} = 0.80 \times 60 = 48$ C

Therefore, the charge reaching the other end is

48 C

LEVEL-II

SINGLE CORRECT ANSWER TYPE

1. Correct Option: C

Explanation

Since there is a complete transfer of charge each time the conductor is touched, the final net charge is the algebraic sum of all transferred charges.

Final Charge $= 0 + 4 - 7 + 9 = +6$ C

Hence, the correct answer is Option C.

2. Correct Option: 0.5 min

Explanation

First, calculate the charge passing through the circuit per minute.

$Q = (3.6 \times 10^{22})(1.6 \times 10^{-19})$

$Q = 5760$ C/min

Now, determine the time required for a charge of 2880 C.

$t = \frac{Q}{\text{Rate of charge flow}} = \frac{2880}{5760} = 0.5$ min

3. Correct Option: C

Explanation

Time:

$t = 20$ min $= 20 \times 60 = 1200$ s

Charge transferred:

$Q = It = 5 \times 1200 = 6000$ C

Using,

$E = \frac{W}{Q}$

$E = \frac{7.2 \times 10^4}{6000} = 12$ V

Hence, the correct answer is Option C.

4. Correct Option: A

Explanation

Let the initial charges be $5x$and $3x$.

After connection, the total charge is equally shared.

Final charge on each $= \frac{5x + 3x}{2} = 4x$

Given,

$4x = 16$ C

$x = 4$ C

Therefore, the larger initial charge is

$5x = 5 \times 4 = 20$ C

Hence, the correct answer is Option A.

5. Correct Option: C

Explanation

Since the current changes uniformly,

$I_{\text{avg}} = \frac{I_1 + I_2}{2} = \frac{2 + 10}{2} = 6$ A

Total charge transferred:

$Q = I_{\text{avg}} t = 6 \times 40 = 240$ C

Hence, the correct answer is Option C.

6. Correct Option: C

Explanation

Total energy supplied:

$E = VQ = 9 \times 400 = 3600$ J

Since 25% is lost internally,

Useful energy

$= 75\% \times 3600 = 0.75 \times 3600 = 2700$ J

Hence, the correct answer is Option C.

7. Correct Option: B

Explanation

Current is given by

$I = \frac{q}{t}$

$I = \frac{1.6 \times 10^{-19}}{2 \times 10^{-20}} = 8$ A

Hence, the correct answer is Option B.

8. Correct Option: 4.5 C

Explanation

When the first and second spheres are touched,

$Q_1' = Q_2' = \frac{18 + (-12)}{2} = 3$ C

Now the second sphere (3 C) touches the third sphere (6 C).

$Q_2'' = Q_3' = \frac{3 + 6}{2} = 4.5$ C

9. Correct Option: A

Explanation

Forward charge:

$Q_1 = 4 \times (5 \times 60) = 1200$ C

Reverse charge:

$Q_2 = -1 \times (10 \times 60) = -600$ C

Net charge:
$Q = 1200 - 600 = 600$ C
Hence, the correct answer is Option A.

10. Correct Option: B

Explanation

Charge carried:
$Q = (5 \times 10^{21})(1.6 \times 10^{-19}) = 800$ C
EMF:
$E = \frac{W}{Q} = \frac{2400}{800} = 3$ V
Hence, the correct answer is Option B.

11. Correct Option: B

Explanation

Net EMF in series opposition:
$V_{\text{net}} = 12 - 6 = 6$ V
Energy supplied:
$E = VQ = 6 \times 200 = 1200$ J
Hence, the correct answer is Option B.

12. Correct Option: B

Explanation

Net excess electrons:
$-4 \times 10^{13} + 7 \times 10^{13} = 3 \times 10^{13}$
Charge:
$Q = -(3 \times 10^{13})(1.6 \times 10^{-19}) = -4.8 \times 10^{-6}$ C
Hence, the correct answer is Option B.

13. Correct Option: C

Explanation

Total charge:
$Q = (2 \times 100) + (4 \times 200) + (6 \times 300) = 2800$ C
Total time:
$T = 100 + 200 + 300 = 600$ s
Average current:
$I_{\text{avg}} = \frac{2800}{600} = 4.67$ A
Hence, the correct answer is Option C.

14. Correct Option: B

Explanation

Additional charge required:
$\Delta Q = 0.8$ C
Number of electrons removed:
$n = \frac{\Delta Q}{e} = \frac{0.8}{1.6 \times 10^{-19}} = 5 \times 10^{18}$
Hence, the correct answer is Option B.

15. Correct Option: C

Explanation

Total charge supplied:
$Q = It = 3 \times 100 = 300$ C
Charge lost:
$= 25\% \times 300 = 75$ C

Hence, the correct answer is Option C.

16. Correct Option: A

Explanation

Two cells aiding:

$+1.5 + 1.5 = 3.0$ V

Two cells opposing:

$-1.5 - 1.5 = -3.0$ V

Net EMF:

$3.0 - 3.0 = 0$ V

Hence, the correct answer is Option A.

17. Correct Option: C

Explanation

Rate of positive charge gained:

$(2 \times 10^{10})(1.6 \times 10^{-19}) = 3.2 \times 10^{-9}$ C/s

Time required:

$t = \dfrac{1.6 \times 10^{-5}}{3.2 \times 10^{-9}} = 5000$ s

Hence, the correct answer is Option C.

18. Correct Option: C

Explanation

Using

$P = VI$

$I = \dfrac{P}{V} = \dfrac{54}{18} = 3$ A

Hence, the correct answer is Option C.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

19

Correct Options: A, B, C

Explanation:

When two identical conducting spheres carrying charges +8 C and −2 C are brought into contact, charge redistributes equally between them.

Option A (Correct): Conservation of charge

The total charge before and after contact remains the same.

Initial Total Charge = +8 C + (−2 C) = +6 C

Final Total Charge = +3 C + (+3 C) = +6 C

Hence, the law of conservation of charge is satisfied.

Option B (Correct): Equal distribution of charge

Since the spheres are identical, the total charge is shared equally.

Charge on each sphere $= \dfrac{(+8) + (-2)}{2} = +3$ C

Therefore, each sphere finally has +3 C.

Option C (Correct): Direction of electron flow

The negatively charged sphere has excess electrons, while the positively charged sphere has a deficiency of electrons. Therefore, electrons move from the negatively charged sphere to the positively charged sphere until both spheres reach electrostatic equilibrium.

20

Correct Options: A, B, C

Given:

- Current, $I = 2\,\text{A}$
- Time, $t = 15\,\text{min} = 900\,\text{s}$

Explanation:

Option A (Correct): Charge transferred

Using

$Q = It$

$Q = 2 \times 900 = 1800\ \text{C}$

Hence, 1800 C of charge flows through the conductor.

Option B (Correct): Number of electrons transferred

Using

$Q = ne$

where

$e = 1.6 \times 10^{-19}\ \text{C}$

$n = \frac{Q}{e} = \frac{1800}{1.6 \times 10^{-19}} = 1.125 \times 10^{22}$ electrons

Option C (Correct): Definition of electric current

Electric current is defined as the rate of flow of charge.

$I = \frac{Q}{t}$

This is the fundamental relation for steady current.

21

Correct Options: A, B, C

Given:

- Voltage, $V = 4.5\,\text{V}$
- Charge, $Q = 500\,\text{C}$

Explanation:

Option A (Correct): Electrical energy supplied

The electrical energy (or work done) is

$E = VQ$

$E = 4.5 \times 500 = 2250\ \text{J}$

Option B (Correct): Correct formula

Voltage is defined as

$V = \frac{E}{Q}$

Rearranging,

$E = VQ$

Hence, the expression is correct.

Option C (Correct): Energy is directly proportional to charge

Since

$E = VQ$

for a constant voltage,

$E \propto Q$

Therefore, doubling the charge doubles the energy supplied.

22

Correct Options: A, B, D

Given:

Number of electrons lost

$n = 5 \times 10^{13}$

Option A (Correct): Nature of charge

A neutral object that loses electrons is left with an excess of protons and therefore becomes positively charged.

Option B (Correct):

Magnitude of charge

Using

$Q = ne$

$Q = (5 \times 10^{13})(1.6 \times 10^{-19})$

$Q = 8 \times 10^{-6}$ C

Option D (Correct):

Conservation of charge

The law of conservation of charge states that the total electric charge of an isolated system remains constant. Charge can only be transferred from one body to another.

23

Correct Options: A, B, C

Given:

- Three identical cells of 1.5 V each connected in series
- Current, $I = 2$ A
- Time, $t = 5$ min $= 300$ s

Explanation:

Option A (Correct): Total emf

In a series combination, the voltages add.

Total emf = 1.5 + 1.5 + 1.5 = 4.5 V

Option B (Correct):

Charge transferred

Using

$Q = It$

$Q = 2 \times 300 = 600$ C

Option C (Correct):

Energy supplied

Using

$E = VQ$

$E = 4.5 \times 600 = 2700$ J

24. Correct Option: A

Explanation

Assertion (A): True.

When a positively charged rod touches a neutral metal object, electrons flow from the neutral object to the positively charged rod. As a result, the metal object acquires a net positive charge by conduction.

Reason (R): True.

A positively charged rod is electron-deficient and attracts free electrons from the neutral conductor, which is the mechanism responsible for charging by conduction.

Since the Reason correctly explains the Assertion, the correct answer is Option A.

25. Correct Option: C

Explanation

Assertion (A): True.

Using

$Q = It$

$Q = 2 \times (5 \times 60) = 600$ C

Therefore, the assertion is true.

Reason (R): False.

From

$I = \frac{Q}{t}$

for a fixed charge Q, current is inversely proportional to time, not directly proportional.

Hence, the assertion is true, but the reason is false.

Therefore, the correct answer is Option C.

26. Answer: A (Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.)

Explanation:

Zinc is more reactive than carbon and undergoes oxidation more readily. During the chemical reaction inside a dry cell, zinc loses electrons**:**

$Zn \rightarrow Zn^{2+} + 2e^{-}$

These electrons accumulate at the zinc container, making it the negative terminal (anode) of the dry cell. Therefore, both statements are true, and the reason correctly explains the assertion.

27. Answer: A (Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.)

Explanation:

When cells are connected positive terminal to negative terminal, their emfs add algebraically.

For four identical cells of 1.5 V each,

Total emf = 1.5 + 1.5 + 1.5 + 1.5 = 6 V

Hence, the total potential difference becomes 6 V.

28. Matrix Matching (Methods of Charging)

Column I	Column II	Explanation
(A) Charging by friction	(Q) Transfer of electrons by rubbing	Electrons are transferred from one material to another due to rubbing.
(B) Charging by conduction	(R) Charge transfer by direct contact	Charges flow through direct physical contact between objects.
(C) Charging by induction	(P) No physical contact required	Charges are redistributed due to the electric field of a nearby charged object.
(D) Conservation of charge	(S) Total charge remains constant	Charge can neither be created nor destroyed in an isolated system.

Answer:

$\boxed{A \rightarrow Q, B \rightarrow R, C \rightarrow P, D \rightarrow S}$

29. Matrix Matching (Electrical Devices)

Column I	Column II	Explanation
(A) Ammeter	(Q) Measures current	Measures electric current flowing through a circuit.
(B) Gold leaf electroscope	(P) Detects electric charge	Detects the presence and nature of electric charge.
(C) Electric cell	(R) Maintains potential difference	Converts chemical energy into electrical energy and provides a steady potential difference.
(D) Galvanometer	(S) Detects weak current	Sensitive instrument used to detect very small electric currents.

Answer: $\boxed{A \rightarrow Q, B \rightarrow P, C \rightarrow R, D \rightarrow S}$

30. Answer: D

Explanation

When a negatively charged rod is brought near a neutral metallic sphere, the free electrons inside the sphere are repelled to the far side.

While the rod is still nearby, the sphere is earthed, allowing these excess electrons to flow into the ground. After removing the earth connection and then removing the rod, the sphere is left with a net positive charge.

31. Answer: C

Explanation

If the charged rod is removed before disconnecting the earth wire, the induced separation of charges disappears. Electrons immediately flow back from the earth into the sphere, neutralizing it.

Thus, the sphere remains uncharged.

32. Answer: B

Explanation

Since like charges repel, the negatively charged rod repels the free electrons inside the metallic sphere.

Therefore, the electrons move away from the rod toward the opposite side of the sphere.

33. Answer: B

Explanation

One of the three cells is connected in the opposite direction.

Hence,

Effective emf = $1.5 + 1.5 - 1.5 = 1.5$ V

Therefore, the net emf is

$\boxed{1.5 \text{ V}}$

34. Answer: C

Explanation

Initially,

$E_1 = 4.5$ V

After reversing one cell,

$E_2 = 1.5$ V

According to Ohm's Law,

$I = \frac{E}{R}$

Since the resistance remains constant,

$\frac{I_2}{I_1} = \frac{1.5}{4.5} = \frac{1}{3}$

Hence, the current becomes one-third of its original value.

35. Answer: A

Explanation

Electrical energy supplied is

$U = \frac{E^2}{R}\, t$

Initially,

$E_1 = 4.5$ V

Finally,

$E_2 = 1.5$ V

Therefore,

$\frac{U_2}{U_1} = \left(\frac{E_2}{E_1}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

Given,

$U_1 = 270$ J

Hence,

$U_2 = \frac{270}{9} = 30$ J

Therefore,

$\boxed{30\text{ J}}$

36. Answer: 1200

Explanation

Given,

- Current,

  $I = 2.5$ A
- Time,

  $t = 8$ minutes $= 8 \times 60 = 480$ s

Using,

$Q = It$

$Q = 2.5 \times 480 = 1200$ C

Therefore,

$\boxed{1200}$

37. Answer: 8

Explanation

Given,

$n = 5 \times 10^{13}$

Elementary charge,

$e = 1.6 \times 10^{-19}$ C

Using,

$Q = ne$

$Q = (5 \times 10^{13})(1.6 \times 10^{-19}) = 8 \times 10^{-6}$ C

Since,

8×10^{-6} C $= 8\,\mu$C

The required integer answer is

$\boxed{8}$

38. Answer: 3

Explanation

Three cells aid the current, while one opposes it.

Hence,

Effective emf = 1.5 + 1.5 + 1.5 − 1.5 = 3 V

Therefore, the required integer answer is

3

TOPIC-2:

Effects of Electric Current & Chemical Effects

LEVEL-I

SINGLE CORRECT ANSWER TYPE

1

Answer: C) 500 J

Explanation:

The ratio of heat produced in nichrome to copper is 5 : 2. Since copper produces 200 J of heat,

$\frac{H_{\text{nichrome}}}{200} = \frac{5}{2}$

$H_{\text{nichrome}} = 200 \times \frac{5}{2} = 500 \text{ J}$

Therefore, the heat produced in the nichrome wire is 500 J.

2

Answer: C) 8 W

Explanation:

The rate of heat production is equal to power.

Given:

- Heat produced = 960 J
- Time = 2 minutes = 120 s

Using,

$P = \frac{\text{Energy}}{\text{Time}}$

$P = \frac{960}{120} = 8 \text{ W}$

Therefore, the power of the heater is 8 W.

3

Answer: B) Resistance of B is greater

Explanation:

According to Joule's Law of Heating,

$H = I^2Rt$

If the current (I) and time (t) are the same for both wires, then

$H \propto R$

Since wire B produces 500 J of heat while wire A produces 300 J, wire B has the greater resistance.

4

Answer: C) Magnetic field strength increases

Explanation:

The strength of an electromagnet depends on:

- the number of turns of the coil, and
- the current flowing through it.

Doubling both the current and the number of turns makes the electromagnet much stronger, enabling it to attract more iron objects.

5

Answer: D) 16 C

Explanation:

Electric charge is calculated using

$Q = It$

Given:

- Current $I = 8$ A
- Time $t = 2$ s

$Q = 8 \times 2 = 16$ C

Therefore, the charge transferred is 16 C.

6

Answer: B) 20

Explanation:

During electrolysis of water,

$2H_2O \rightarrow 2H_2 + O_2$

Hydrogen and oxygen are produced in the ratio

$H_2 : O_2 = 2 : 1$

If 40 bubbles of hydrogen are produced, then

$\frac{40}{2} = 20$

oxygen bubbles are formed.

Therefore, the correct answer is 20.

7

Answer: C) 4 g

Explanation:

In copper electroplating, the copper anode dissolves into the electrolyte, and an equal mass of copper is deposited on the cathode.

Hence,

- Copper lost by anode = 4 g
- Copper deposited on iron key = 4 g

Therefore, the mass deposited is 4 g.

8

Answer: C) Returns to original position

Explanation:

A compass needle deflects because of the magnetic field produced by the electric current.

When the current is switched off, the magnetic field disappears, and the compass needle aligns once again with the Earth's magnetic field.

Therefore, it returns to its original position.

9

Answer: C) 4 : 1

Explanation:

According to Joule's Law,

$H = I^2Rt$

When current and time are the same,

$H \propto R$

Given:

- $R_1 = 8\,\Omega$
- $R_2 = 2\,\Omega$

$H_1 : H_2 = 8 : 2 = 4 : 1$

Therefore, the ratio of heat produced is 4 : 1.

10

Answer: C) Positive electrode

Explanation:

Hydroxide ions ($0H^-$) carry a negative charge.

During electrolysis, negative ions move towards the positive electrode (anode) because opposite charges attract.

Therefore, hydroxide ions migrate to the positive electrode.

11

Answer: D) 7200 J

Explanation:

Electrical energy converted into heat is calculated using

$E = VIt$

Given:

- Voltage $V = 12$ V
- Current $I = 2$ A
- Time $t = 5$ minutes = 300 s

$E = 12 \times 2 \times 300 = 7200$ J

Therefore, the electrical energy converted into heat is 7200 J.

12

Answer: B) Chemical energy converts into electrical energy

Explanation:

An electric cell produces electricity through chemical reactions occurring inside it.

Thus,

Chemical Energy → Electrical Energy

The electrical energy then powers the LED, causing it to glow.

Therefore, the correct answer is Chemical energy converts into electrical energy.

13

Answer: C) It can be reset after tripping

Explanation:

Unlike a traditional fuse wire, which melts when excessive current flows and must be replaced, a Miniature Circuit Breaker (MCB) works as an automatic switch. During an overload or short circuit, it trips and disconnects the circuit. Once the fault is corrected, the MCB can simply be reset manually and used again.

14

Answer: B) Act as anode

Explanation:

In an electrolytic cell, the electrode connected to the positive terminal of the battery is called the anode, while the electrode connected to the negative terminal is called the cathode. If the battery connections are reversed, the iron key becomes connected to the positive terminal and therefore acts as the anode.

15

Answer: C) Magnetic field

Explanation:

A magnetic field is the region around a magnet or a current-carrying conductor

where magnetic forces can be detected. Any magnetic material or moving charge placed within this region experiences the effect of the magnetic force.

16

Answer: B) Ionisation increases

Explanation:

Pure distilled water is a poor conductor because it contains very few free ions. When sulphuric acid is added, it ionizes into H^+ and SO_4^{2-} ions. The increased number of free ions allows electric current to pass through the solution more easily, thereby increasing its conductivity.

17

Answer: C) Heating effect

Explanation:

A safety fuse works on the heating effect of electric current (Joule heating). When the current exceeds the safe limit, the fuse wire becomes excessively hot and melts, breaking the circuit and protecting electrical appliances from damage.

18

Answer: B) Current stops flowing

Explanation:

An electromagnet produces magnetism only when electric current flows through its coil. As soon as the current stops flowing, the magnetic field disappears, and the electromagnet immediately loses its magnetism.

19

Answer: B) 2 Ω

Explanation:

Using Joule's law of heating,

$H = I^2Rt$

Given:

$H = 1800\ \text{J}, I = 3\ \text{A}, t = 100\ \text{s}$

Substituting the values,

$1800 = (3)^2 \times R \times 100$

$1800 = 9 \times 100 \times R$

$1800 = 900R$

$R = \frac{1800}{900} = 2\ \Omega$

Therefore, the resistance is

$\boxed{2\ \Omega}$

20

Answer: C) Anode

Explanation:

In copper electrolysis, the electrode connected to the positive terminal of the battery is called the anode. Copper atoms at the anode lose electrons (oxidation) and dissolve into the electrolyte as Cu^{2+} ions. As a result, the copper anode gradually loses mass and becomes thinner.

LEVEL-II

SINGLE CORRECT ANSWER TYPE

1. Answer: B) Mineral salts

Explanation:

Distilled water is nearly pure and contains very few ions, so it is a poor conductor

of electricity. Tap water contains dissolved mineral salts that dissociate into free ions, which carry electric current and make it a good conductor.

2. Answer: C) 1800 J

Explanation:

Useful heat produced = Efficiency × Input energy

$= 0.90 \times 2000 \text{ J} = 1800 \text{ J}$

3. Answer: A) Cathode

Explanation:

The object to be plated is connected to the negative terminal (cathode). Positively charged metal ions (cations) are attracted to the cathode, where they gain electrons and form a thin coating on the object.

4. Answer: B) Current is switched off

Explanation:

An electromagnet produces a magnetic field only while electric current flows through its coil. When the current is switched off, the magnetic field disappears, and the crane releases the iron scrap.

5. Answer: D) 1200 C

Explanation:

Using the formula:

$Q = I \times t$

Given,

- $I = 5$ A
- $t = 4$ minutes $= 240$ s

$Q = 5 \times 240 = 1200$ C

6. Answer: B) Free ions

Explanation:

Electric current in solutions is carried by free-moving ions, not electrons. These ions move toward opposite electrodes and complete the electric circuit.

7. Answer: C) Decreases

Explanation:

The strength of an electromagnet is directly proportional to the current flowing through the coil. Therefore, reducing the current reduces the magnetic field strength.

8. Answer: C) Gas bubbles at electrodes

Explanation:

The formation of gas bubbles at the electrodes indicates that a chemical reaction is taking place due to the passage of electric current. Other chemical effects include metal deposition and colour change.

9. Answer: C) 6.67 W

Explanation:

Power is given by:

$$P = \frac{\text{Energy}}{\text{Time}}$$

Given,

- Energy = 1200 J
- Time = 3 minutes = 180 s

$$P = \frac{1200}{180} = 6.67 \text{ W}$$

10. Answer: C) Cathode

Explanation:

Hydrogen ions (H^+) are positively charged. They move towards the negatively charged cathode, where they gain electrons to form hydrogen gas.

11. Answer: B) Stronger

Explanation:

The magnetic field produced by an electromagnet increases with the number of turns in the coil. Increasing the turns makes the electromagnet stronger.

12. Answer: A) Copper ions

Explanation:

The characteristic blue colour of copper sulphate solution is due to hydrated copper (Cu^{2+}) ions present in the solution.

13. Answer: C) 1800 J

Explanation:

According to Joule's law, when current and resistance remain constant, heat produced is directly proportional to time.

Since the time doubles from 3 minutes to 6 minutes,

$H = 900 \times 2 = 1800$ J

14. Answer: B) Thicker

Explanation:

During electrolytic refining, copper atoms from the impure anode dissolve into the solution as copper ions. These ions are deposited as pure copper on the cathode, making it thicker over time.

15. Answer: D) 48 W

Explanation:

Power is calculated using:

$P = V \times I$

Given,

- $V = 24$ V
- $I = 2$ A

$P = 24 \times 2 = 48$ W

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. Correct Options: A, B, C

Explanation:

The strength of an electromagnet depends on the magnetic field produced by the coil. For a solenoid,

$B = \mu nI$

where:

- B= Magnetic field strength
- μ= Magnetic permeability of the core material
- n= Number of turns per unit length
- I= Current through the coil
- **A) Increasing current**

  Increasing the current increases the magnetic field strength because $B \propto I$.
- **B) Increasing the number of turns**

  More turns per unit length produce a stronger magnetic field.
- **C) Using a soft iron core**

  Soft iron has high magnetic permeability, greatly increasing the magnetic field strength.

17. Correct Options: A, B, C

Explanation:

When electric current passes through an electrolyte, it causes **electrolysis**, producing chemical changes.

- **A) Formation of gas bubbles**
  Gases such as hydrogen or oxygen are released at the electrodes during electrolysis.
- **B) Change in the colour of the solution**
  Chemical reactions alter the composition of the solution, leading to a colour change.
- **C) Deposition of metal on the electrodes**
  Metal ions gain electrons and get deposited as a solid metal on the cathode.

18. Correct Options: A, B, C

Explanation:

Electrolysis uses electrical energy to bring about non-spontaneous chemical reactions.

- **A) Electroplating**
  A thin layer of one metal is deposited on another metal to improve appearance or prevent corrosion.
- **B) Purification of metals**
  Electrolysis is used to refine impure metals such as copper into pure metals.
- **C) Extraction of metals**
  Highly reactive metals like aluminium, sodium, and magnesium are extracted from their molten ores by electrolysis.

19. Correct Options: A, B, C

Explanation:

Liquids conduct electricity only if they contain **free-moving ions**.

- **A) Lemon juice**
  Lemon juice contains citric acid, which dissociates into ions that conduct electricity.
- **B) Tap water**
  Tap water contains dissolved salts and minerals that provide ions for electrical conduction.
- **C) Distilled water with salt**
  Pure distilled water is a poor conductor, but adding salt produces sodium (Na^+) and chloride (Cl^-) ions, making it a good conductor.

20. Correct Options: A, B, D

Explanation:

Safety devices protect electrical circuits from excessive current due to overloads or short circuits.

- **A) A fuse melts due to the heating effect of electric current.**
  A fuse contains a low-melting-point wire. Excess current produces heat according to
  $H = I^2Rt$
  causing the fuse wire to melt and break the circuit.
- **B) An MCB can be reset after tripping.**
  Unlike a fuse, an MCB does not melt. It can be switched back on after the fault is removed.

- **D) An MCB automatically breaks the circuit during overload or short circuit.**
  It trips automatically to interrupt the current and protect the circuit and appliances.

Q21. Answer Key: A

Explanation:

An electromagnet exhibits magnetic properties only when electric current flows through its coil. As soon as the current is switched off, the magnetic field disappears instantly. Therefore, an electromagnetic crane can easily release the iron scrap by simply cutting off the current.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Q22. Answer Key: A

Explanation:

A fuse wire is made of a special alloy (commonly tin and lead) that has a low melting point and high resistance. When excessive current flows, the fuse wire heats up rapidly and melts, breaking the circuit before electrical appliances are damaged.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Q23. Answer Key: A

Explanation:

During electroplating, the object to be coated (such as an iron key) is connected to the negative terminal (cathode). Positively charged copper ions (Cu^{2+})move towards the cathode, gain electrons, and get deposited as a thin layer of copper on its surface.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Q24. Answer Key: A

Explanation:

Whenever electric current flows through a conductor, it produces a magnetic field around it (discovered by Hans Christian Oersted). Because of this magnetic field, the conductor behaves like a temporary magnet.

Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

Q25. Answer Key:

P – 2, Q – 1, R – 3, S – 4

Explanation:

- Fuse (P): Melts when excessive current flows, protecting the circuit.
- MCB (Q): Trips automatically during overload or short circuit and can be reset.
- Electroplating (R): Uses electric current to coat an object with another metal.
- Electromagnet (S): Becomes magnetic only while electric current flows.

Q26. Answer Key:

P – 1, Q – 2, R – 3, S – 4

Explanation:

- Nichrome wire (P): Has high resistance and is commonly used in heaters.
- Copper wire (Q): Has low resistivity, making it ideal for electrical wiring.
- Distilled water (R): Contains almost no dissolved ions and is therefore a poor conductor.

- Lemon juice (S): Contains citric acid that dissociates into ions, allowing it to conduct electricity.

Q27. Answer Key: C

Explanation:

The formation of gas bubbles at the electrodes when electric current passes through a liquid indicates that electrolysis is taking place. This is evidence of the chemical effect of electric current, where a chemical reaction occurs due to the passage of electricity.

Q28. Answer Key: B

Explanation:

A compass needle deflects only when a current-carrying wire produces a magnetic field around it. Once the current is switched off, the magnetic field disappears, and the compass needle returns to align with Earth's magnetic field.

Q29. Answer Key: C

Explanation:

Acidulated water contains free ions that conduct electricity effectively, resulting in vigorous electrolysis and the formation of gas bubbles. Distilled water contains very few ions and is therefore a poor conductor, producing little or no chemical effect.

Q30. Answer Key: C

Explanation:

The immediate release of iron scrap when the current is switched off demonstrates that an electromagnet is a temporary magnet. It loses its magnetic property as soon as the electric current stops flowing.

Q31. Answer Key: B

Explanation:

The strength of an electromagnet depends directly on the current flowing through its coil. Increasing the current increases the magnetic field, making the electromagnet capable of lifting heavier objects.

Q32. Answer Key: B

Explanation:

Electromagnets are preferred in industries because their magnetic effect is controllable. They can be switched ON or OFF whenever required, and their strength can be adjusted by varying the electric current.

Q33. Answer Key: 20

Explanation:

Power is calculated using:

$P = \frac{E}{t}$

Given:

- Energy, $E = 3600$ J
- Time = 3 minutes = 180 s

$P = \frac{3600}{180} = 20$ W

Q34. Answer Key: 30

Explanation:

During electrolysis of water,

$2H_2O \rightarrow 2H_2 + O_2$

Hydrogen and oxygen are produced in the ratio:

$H_2 : O_2 = 2 : 1$

If 60 bubbles of hydrogen are formed,

Oxygen bubbles $= \frac{60}{2} = 30$

Q35. Answer Key: 200

Explanation:

According to Joule's Law of Heating,

$H = I^2Rt$

Given:

- $I = 2$ A
- $R = 5\ \Omega$
- $t = 10$ s

$H = (2)^2 \times 5 \times 10$

$H = 4 \times 5 \times 10 = 200$ J

PREVIOUS YEAR QUESTIONS

1. Answer: C

Explanation:

Sodium chloride, copper sulphate, and silver nitrate are ionic compounds. When dissolved in water, they dissociate into free-moving ions that conduct electricity. Sugar solution contains only neutral molecules and does not produce ions, so it does not conduct electricity.

Therefore, solutions I, II, and III complete the circuit and cause the bulb to glow.

2. Answer: A

Explanation:

The correct matching is:

- P (Anode) → 2 (Positive terminal)
- Q (Cathode) → 3 (Negative terminal)
- R (Ions) → 1 (Positively or negatively charged particles)
- S (Electrolyte) → 4 (Conducting solution)

Hence, the correct answer is P-2, Q-3, R-1, S-4.

3. Answer: C

Explanation:

- Statement (i): Incorrect. According to Faraday's law, the mass of substance deposited is directly proportional to the quantity of electricity passed through the electrolyte ($m \propto Q$).
- Statement (ii): Incorrect. The object to be electroplated must be connected to the negative terminal (cathode) so that metal ions are deposited on its surface.
- Statement (iii): Correct. Positively charged copper ions (Cu^{2+})move towards the cathode (negative electrode), where they gain electrons and are deposited as copper metal.

Therefore, only Statement (iii) is correct.

4. Answer: B

Explanation:

From the given diagram:

- Q represents the electrolyte.
- P is connected to the positive terminal, so it is the anode.
- R is connected to the negative terminal, so it is the cathode.

Hence, the correct order is Q, P, R.

5. Answer: B

Explanation:

A fuse is a safety device that protects an electrical circuit by melting and breaking the circuit when the current exceeds the safe limit.

- Statement II is correct.
- Statements I, III, and IV are incorrect because a fuse does not block the live or neutral wire, nor does it discharge current to the Earth.

Therefore, only Statement II is correct.

6. Answer: D

Explanation:

During the electrolysis of water:

- Carbon (graphite) rods are commonly used as electrodes.
- Electrode A is connected to the positive terminal and acts as the anode.
- Electrode B is connected to the negative terminal and acts as the cathode.
- Oxygen gas is released at the anode.
- Hydrogen gas is released at the cathode.

Hence, all the given statements are correct.

7. Answer: A

Explanation:

- Statement I: Correct. The electrical conductivity of a liquid depends on the presence of free-moving ions.
- Statement II: Incorrect. By convention, the direction of electric current is taken as the direction of the flow of positive charges, which is opposite to the movement of electrons or negative ions.

Therefore, Statement I is correct, but Statement II is incorrect.

8. Answer: C

Explanation:

A circuit breaker is an automatic safety device that protects an electrical circuit by interrupting the flow of current whenever a short circuit or overload occurs. Once the fault is removed, it can usually be reset and used again without replacement.
Hence, the correct answer is C.

9. Correct Option: C) Statement 1 is true but Statement 2 is false.

Explanation:

- Statement 1 is True: The chemical effect of electric current refers to the chemical changes that occur when an electric current passes through a conducting solution. These changes may include the decomposition of compounds, the deposition of metals, or the evolution of gases.
- Statement 2 is False: Cane sugar (sucrose) is a covalent compound. When dissolved in water, it remains as neutral molecules and does not dissociate into ions. Since free ions are required to carry electric current, a cane sugar solution does not conduct electricity.

10. Correct Option: C) Electroplating

Explanation:

The table classifies applications based on the effects of electric current.

- Heating Effect: Devices such as geysers, room heaters, hot plates, and hair dryers operate by converting electrical energy into heat using high-resistance heating elements. Hence, they are correctly placed under the heating effect.
- Magnetic Effect: Electric bells, electromagnets, and magnetic cranes work because an electric current produces a magnetic field. Therefore, they are correctly classified under the magnetic effect.

- Incorrect Entry: Electroplating does not depend on the magnetic effect of electric current. It is based on the chemical effect of electric current, in which a thin layer of one metal is deposited onto another object through electrolysis. Therefore, Electroplating is the incorrectly categorized item in the table.